

ANNEXURE - E							
INDICATIVE MINIMUM MACHINING RATE BASED ON MFP (BILL VALUE OF NEW ITEMS)							
Sl no	Item	BR (MFP)-Kg	BW -Kg	FW-Kg	Swarf-Kg	Qty	Intr
1	Wheel for 80 T/100 T Bogie	52.53	1290	442	848	2	0.5
2	Axle for 80 T/100 T Bogie	52.53	1078	729	349	1	1
3	Wheel and Axle heat treatment , shrink fitting and assembly factor (2 wheels + 1 Axle)						0.25
							One Set Wheel Axle
4	Sinter Frame	52.53	1272	1172	100	1	1.4
5	Sinter Frame Shaft	52.53	125	43	82	4	1.4
6	Sinter frame Assembly						0.6
							One Set Sinter
7	Side Beam Fitted with 2 covers 80T/100 T (fitted with covers- 80 T)	52.53	1060	990	70	2	4
8	Cross Beam 80T/100T	52.53	1250	1100	150	1	2
9	Axle Box	52.53	140	75	65	4	1.4
10	Collar	52.53	25	11	14	4	1.4
11	Front Cover	52.53	65	47	18	4	1.4
12	Rear Cover	52.53	70	47	23	4	1.4
	Bogies Assembly factor (including all hardware /machined items)						0.3
							One Set Bogie
13	(FORGINGS) FOR ROLL HEADS/WRM COOLING ELEMENTS						
A	SWARF up to 100 Kg	52.5	37.9	8.2	29.8	1.0	1.0
B	SWARF (100-200 Kg)	52.5	166.0	30.3	135.6	1.0	0.90
C	SWARF (200 -300Kg)	52.5	237.3	61.0	176.3	1.0	0.70
D	SWARF (300-400 Kg)	52.5	384.7	30.0	354.7	1.0	0.50

INTRICACY TABLE FOR OPERATION CONTRACT FOR MACHINE

OPERATIONS		INTRICACY LEVELS															
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
NO. OF MACHINES USED FOR MACHINING OF JOB	1	BR															
	2		1.4 BR														
	3			1.6 BR													
	4				1.8 BR												
	5					2 BR											
	6						3 BR										
	More than 6							3.5 BR									
	*****"X"*****																
N6 – N4(THREE DELTA)-FINE FINISH AND SUPER FINE								1.4 X									
*****"Y"*****																	
HARDNE SS	UP to 250 BHN	NIL															
	MORE THAN 250 BHN										1.8 Y						
	*****"Z"*****																
HYDRAULIC TESTING	1 TIME												1.1 Z				
	MORE THAN 1 TIME													1.3 Z			
	*****"P"*****																
ASSEMBLY	UP TO 2 MAJOR PARTS	*****"P"*****															
	3 MAJOR PARTS														1.2 P		
	4 MAJOR PARTS															1.4 P	
	MORE THAN 4 MAJOR PARTS																1.6 P
	Note: If the finished weight of the assembly is more than 1000 kg, then the assembly factor will be 1.10 P For items other than Bogie assly (80T & 100 T), Sinter frame.																
*****"Q"*****																	

Table shows Intricacy factors in terms of Base Rate (Rate quoted by the tenderer in the "schedule of rates") at various Intricacy levels & for different operations and activities to be performed. The Table shall be applicable for calculation of Machining Rate (MR) **The Table shall be applicable for calculation of Machining Rate (MR) for those items which are not given in the fixed intricacy table..** In order to calculate the Machining Rate (MR) for a job, different operations and activities required to be performed shall be first worked out by the help of drawings and /or Customer's Work Order and then shall be mutually decided and agreed upon as per the above table. Such Machining Rate for an Item shall be calculated only once for each Item and shall be applicable till the end of the Contract for that Item. Based on the Intricacy factors provided at various Intricacy levels, the 'Machining Rate' shall be worked out in terms of 'Base Rate'.

Notations:

(X) = Factor w. r. t BR for No. of Machines used and Intricacy factor assigned to them for a particular Item machined

(Y) = Factor w. r. t (X) for Intricacy assigned for Surface Finish to a particular Item machined

(Z) = Factor w. r. t (Y) for Intricacy assigned for Hardness of Material to be machined for a particular Item

(P) = Factor w. r. t (Z) for Intricacy assigned for Hydraulic testing of various Items.

(Q) = Factor w. r. t (P) for Intricacy assigned for Assembly of various parts of a particular Item.

Annex: C

Special intricacies (Applicable in conjunction with the intricacy table only):

NOTE:

- 1 (a) In case of any product having a black weight up to 5 Kg which requires machining, then the Intricacy factor as applicable as per the Intricacy Table will be multiplied by 3. This will include even threading requirement.
- 1(b) In case of any product having a black weight more than 5 Kg which requires machining and the Swarf generated is less than 3 Kg then the Intricacy factor as per the Intricacy table will be multiplied by 3 as per 1(a).
- 1 (c) In case of threading, where the black weight is more than 5 Kg and the Swarf generated is more than 3 Kg, every start of the thread will be considered as 10 % more than the actual calculation of the Intricacy on linear basis.
2. Intricacy of 1.1 will be given for the gear cutting jobs.
3. One time Machining of gauges, fixtures would be considered as another job.
4. The weight of the Cut pieces /Risers will be deducted from the black weight of the Castings.
5. Castings requiring marking only but not machining will be paid @ Rs 50/- per piece.
6. In-process marking or otherwise, being the requirement for any machining job, will not be addressed separately.
7. Heat treatment activities will call for 10 % more than the actual amount.
8. Hand grooving will be considered as another machine used.
9. No intricacy will be given for operations which are required to facilitate the machining process, like drilling for dowel pin holes, making centering holes on shafts, clamping etc.
10. Grooving by hand Chipping /hand operated electrical/pneumatic machine will be considered as another machine used.
11. The Machining rate will be multiplied by factor of **0.90** for those **ferrous Casting items where the swarf generated from single ferrous casting is 100 kg or more than 100 kg.(Applicable for Castings only).**

ANNEXURE-D - Pg-1

FIXED INTRICACIES

Sl no	Name of Item	Black weight range in (Kg)	Applicable intricacy (Fixed Intricacy)
1	Copper Tuyeres and Coolers	Any weight	2.85
2	All types Split nuts of BSL & DSP including Leading nuts of BSP	Any weight	2.40
3	All types of Zn Alloy Nuts	Any weight	1.8
4	Flat Liners, Wearing strips , Wearing plates (excluding right angled liners, slipper brackets , round wearing plates of DSP.)	Any weight	2.2
5	Very Small size slipper pads/inserts	Up to 10 kg	9.0
6	Large & Medium size Slipper pads/Bushing	Above 10 Kg	4.5
7	Very small bearings/ bushes like mould foot roller bushes of DSP	Up to 1.00 Kg	3.0
8	Small Size bearings / Bushes	above 1 kg to 10 kg	5.0
9	Medium size bearings/ Bushes including labyrinth ring of RSP	above 10 kg to 150 kg	1.8
10	Heavy bearings like Roll neck bearing of ISP, Bearing shell of RSP, Carriers bearings of ISP, Carrier bearing liner of ISP AND RING THRUST BRGS OF DSP	above 150 kg	3.0
11	Plain cylindrical jobs like PB rounds of BSP, ferrules etc	Any weight	1.0
12	Crane Bushes of RSP	Any weight	1.50
13	Swivel Blocks of RSP	-	4.50
14	Manufacturing of HSS Tools for CNC machines, End mills , Profile tools etc.	Any weight	10
15	Fire grate plates for Self balancing screen of BSP	-	7
16	Pressure Roller Bushes of BSP (fitted with Bush)	-	2.376
			Annexure- D- Pg-2 Applicable intricacy (Fixed Intricacy)

17	Wheel for 80 T/100 T Bogie	Refer Drawings	0.5
18	Axle for 80 T/100 T Bogie	Do	1.0
19	Wheel and Axle heat treatment , shrink fitting and assembly factor (2 wheels + 1 Axle)	Do	0.25
20	Sinter Frame	Do	1.40
21	Sinter Frame Shaft	Do	1.40
22	Sinter frame Assembly	Do	0.60
23	- Side Beam 80T fitted with covers - Side Beam 100 T	Do	5.0
24	Cross Beam 80T/100T	Do	2.0
25	Axle Box , Collar , Front and Rear Cover	Do	1.40
26	Bogies Assembly factor (including supply of all hardware items like fasteners, packings, seals , wire , nut of withdrawal sleeve AH 2336 /machined items etc.)	Do	0.3
27	FORGINGS (ROLL HEADS/COOLING ELEMENTS ETC. OF ISP	Do	
A	SWARF up to 100 Kg	Do	1.0
B	SWARF (100-200 Kg)	Do	0.90
C	SWARF (200 -300Kg)	Do	0.70
D	SWARF (300-400 Kg)	Do	0.50
28	Special Jobs which require high degree of skill , accuracy and safety (Such as sleeve and arbor fitting , roll chock refurbishment etc.)- Pls refer clause (ii) of scope of work.	Do	intricacy factor for Machining Rate (MR) will be fixed and decided by EA/OA which will be binding on the Contractor.
	NOTE: FOR ESTIMATION OF MINIMUM BILL VALUES FOR NEW ITEMS (SL NOS- 17 TO 28) BASED ON MFP , PLS REFER ANNEXURE- E		

ANNEXURE-F

MACHINE SHOP NO: 1 : MAIN PRODUCTION MACHINES								
SL NO.	Name of M/c:	M/C NO.	model/ma ke	Date of commissi on.	Capacity:	Remarks	Location	Deputatio n of machine operator
1	Niles double column vertical boring	113	DKZ	1965	Max. turnibg dia:1350 mm,Max. bore dia:1250 mm,Max. height of job: 2000 mm,Swivel angle of beam: 30 deg.	The machine is used for machining of heavy ferrous jobs like Roll liners, grinding rings, wheel etc.	MS-1	1
2	Slotting Machine	28	Butler	1954	Max. Stroke: 14", Table dia: 3'-3"	Roll liner key ways, wheel key ways etc.	MS-1	1
3	Lathe	27	D.S.G.	1951	Swing over bed:23",Admit between centre: 10'-4",Longitudinal traverse: 4'-6"	The machine is used for machining of Non ferrous split nuts of DSP-Tilting Car and other ferrous jobs	MS-1	1
4	Vertical Milling	1	Batliboi	,LCD	Table size : 1000 mm x 700 mm, table swivel in both directions up to 30 deg , vertical head swivel 30 degree		MS-1	1
5	Horizontal boring	96	Kearns	1958	Spindle dia:4",table size:4'-0"x4'-0",Vertical traverse: 2'-0", Longitudinal Traverse- 4'-6"	The machine is used for machining of skids castings, dummy bars and ferrous Jobs	MS-1	1

6	Horizontal boring	95	Kearns	1958	Spindle dia:3",table size:3'-0"x3'-0",Vertical traverse: 20", Longitudinal traverse-3'-6"	The machine is used for machining of skids castings, dummy bars and ferrous Jobs	MS-1	1
7	Horizontal boring	24	Kearns	1958	Face plate dia: 3'-0", Table size: 33"x33", Longitudinal traverse: 5'-0", Vertical traverse: 2'-0"	The machine is used for machining of skids castings, dummy bars and ferrous Jobs	MS-1	1
8	Radial drill M/c	22	Asquith	1949	Radius of arm:5'-6",max drill: 2"	The machine is used for drilling and tapping of copper tyrs and other non ferrous jobs	MS-1	1
9	Radial drill M/c	21	Asquith	1949	Radius of arm:5'-6",max drill: 2"	The machine is used for drilling and tapping of copper tyrs and other non ferrous jobs	MS-1	1
10	Boring lathe	100	Swift	1958	Centre height:16", length of bed:7'	Machine is used for machining of copper tyrs only	MS-1	1
11	Heavy duty lathe	19	Swift-L-1021	1951	Centre height: 27",centre to centre ,distance:20'-0"	Machine is used for Long size jobs such as Long bushes, spindles, slipper pads etc.	MS-1	1
12	Heavy duty lathe	20	Swift-L-1027	1951	Centre height: 21",centre to centre distance:18'-0"	Machine is used for Long size jobs such as Long bushes, spindles, slipper pads etc.	MS-1	1

13	CNC Lathe	124	Retrofitted	2009	Ht. of center: 400 mm, swing over bed: 800 mm , distance between center: 4000 mm		MS-1	1
14	Planomiller - DOUBLE COLUMN	15	Maclellan	1950	Width of table: 5'-0", max planing width: 3 '	Machine is used for machining of pig moulds, skids, dummy bars, inclined grate bars, Mn Steel armour plates ,axle boxes	MS-1	1
15	Heavy radial drill	117	HEC-DR-8	1977	Max. drill cap:80 mm, radius of arm:2790	Used for drilling of above mentioned heavy castings	MS-1	1
16	Plano-milling m/c, open side	11	Butler	1950	Max stroke: 18'-0",width of table: 5'-0", max planing width: 6'-0"	Machine is used for machining of pig moulds, skids, dummy bars, inclined grate bars, Mn Steel armour plates , axle boxes	MS-1	1
17	Heavy duty floor type horizontal Boring	10	Asquith-MEB:5	1952	Spindle dia: 6" max. boring cap: 3'-0",Max. horizontal traverse: 19'-0"	Used for milling heavy Jobs	MS-1	1
18	Draw Cut Shaper	9	Butler	1952	Max vertical travel : 4'-0", Logitudinal traverse: 19'-0"	Machining of heavy castings like dummy bars, bogie items	MS-1	1

19	Heavy duty centre Lathe	8	Shanks	1949	Swing over bed: 9'-0", length of bed: 21'-0"	used for machining of Bowl Liners and Lower armours of Bhilai- Rajhara Mines	MS-1	1
20	Horizontal drill	103	Asquith	1959	Maximum drill: 2",Max. longitudinal traverse: 15	used for drilling of Light Jobs	MS-1	1
21	CNC Retrofitted Horizontal Boring machine	123	2009		Max boring dia: 450 mm, Table size: 1000 mm x 1500 mm		MS-1	1
22	Vertical boring	112	Webster & Bennet model DH	1965	Table dia: 72",max ht. Of job: 30"	Machining jobs - Slipper pads, Liners -All Non Ferrous Jobs	MS-1	1
23	Horizontal boring	111	Richards	1965	Max. Boring capacity: 1220 mm, Longitudinal traverse of table: 2400 mm, Vertical traverse of tool head: 2150 mm	Machining of heavy castings like dummy bars, bogie items	MS-1	1
24	Vertical Milling	64		1951	28 " Table travel in long direction		MS-1- NF SECTION	1
25	Lathe	98	HMT: H-22	1958	Swing over bed: 10", length of bed: 5'	Used for machining of Light weight Jobs	MS-1- SIDE BAY	1
26	Lathe	97	HMT: H-22	1958	Swing over bed: 10", length of bed: 5'	Used for machining of Light weight Jobs	MS-1- SIDE BAY	1

27	Swift Lathe	46	Swift	1950	Center height: 8.5 " Long bed	Used for machining of Light weight Jobs	MS-1-SIDE BAY	1
28	Lathe	47	Swift	1950	Center height: 8.5 " Long bed	Used for machining of Light weight Jobs	MS-1-SIDE BAY	1
29	Horizontal Milling	65	Edgwick	1951	Long travel: 18 "		MS-1-SIDE BAY	1
30	Heavy duty lathe	14/2	Waldrich	1958	Centre height:27", dist. Between centres: 21'	Used for machining of Heavy Jobs	MS-2	1
31	Vertical borer	S-4	SK-12	1964	Table dia: 1108 mm, Height of Job: 1000 mm	Used for machining of Roll liners, sprockets, bearing housings , wheels	SF MS	1
32	Vertical borer	S-5	Model DH-5	1957	Table dia:60", max ht of job: 27",Horizontal travel:1250 mm ,cross traverse:335 mm	Used for machining of Roll liners, sprockets, bearing housings , wheels	SF MS	1
33	Vertical borer	S-14	Model 13 DK/12	1967	Max Turning Dia: 1250 mm, Turning Height: 1400 mm	Used for machining of Roll liners, sprockets, bearing housings , wheels	SF MS	1
34	Lathe	S-45					SF MS	1
35	HMT Lathe	114	HMT	1975	12" Centre Height	Used for machining of Non ferrous and Steel Jobs	MS-1-T/ROOM	1

36	Lathe	Stanley-10/2	Stanley-10/2	1958	Swing over bed dia 22", admit between , Centre: 12'-5", adjustable bed	Machining of medium type NF and Ferrous Jobs such as bearings and shafts	MS-2	1
37	Slotting	26/2	Cooper	1954	Max stroke: 14", table dia: 32"	Used for machining of Sprockets	MS-2	1
								37
SUPPORTING (SECONDARY MACHINES)								
1	Planomiller	16			6 feet bed length, Bed Width: 2'-0"	Small size non ferrous Liners	MS-1	1
2	Crank Planning	14					MS1	1
3	Horizontal drill PMS	D-5	Asquith	1954	Max Drill- 2"	-	MS-1	1
4	Horizontal drill PMS	D-6	Asquith	1954	Max Drill- 2"	-	MS-1	1
5	Vertical surface grinder	2	LUMSDE N model 92 E	1942	Table dia: 48", max ht: 18 "	Lying Idle for want of Jobs such as Grizzly Discs- ISP	MS-1	1
6	Shaping	59	Butler	1949	Max stroke : 18", max ht. Of job: 17"	Non ferrous Jobs	MS-1- NF SECTION	1
7	Shaping	60	Butler	1949	Max stroke : 18", max ht. Of job: 17"	Non ferrous Jobs	MS-1- NF SECTION	1
8	Pillar Type Drilling		Asquith		Max drill- 2" dia		MS-1- NF SECTION	1
9	Lathe	92	Lang	1958	Center height: 12 " CTCD: 36 "		MS-1- NF SECTION	1
10	Radial drill	78	-	1949	Max 1" dia drill	Very precise small drill holes	MS-1- T/ROOM	1

11	Horizontal Milling	63		1949	Table size: 38 "x 19 "		MS-1-T/ROOM	1
12	Universal Milling	118		1977	Table Travel: 950 mm		MS-1-T/ROOM	1
13	HMT Lathe	102					MS-1-T/ROOM	1
14	Turret Lathe	11/2	Lang	1958	Turret lathe		MS2	1
15	Heavy duty lathe	14/2	Waldrich	1958	Centre height:27", dist. Between centres: 21'	Used for machining of Heavy Jobs	MS-2	1
16	Lathe	22/2	HMT	1977	Max Center height: 260 mm, Swing Over bed: 530 mm dia, Admit between center: 3 meter	Used for machining of Long Shafts	MS-2	1

16

NOT MINIMUM 25 MACHINES SHOULD RUN OUT OF THESE MACHINES IN A MONTH.

Total

53

Annexure -A LIST OF JOBS DONE

SL NO.	Item	Customer	Qty	Black Wt(Each/ In Kg)	T/B	Wt(Each/	T/F	Total Swarf (Kg)
1	Modified Rope Guide Bush for OBC Grab Bucket, Drg: BF-97-020 (3% JOB DONE)	BSP	1	41	41	30	30	11.00
2	Modified Rope Guide Bush for OBC Grab Bucket, Drg: BF-97-020 (REJECTED DUE TO BLOWN) (3% JOB DONE)	BSP	1	41	41	30	30	11.00
3	Modified Cast Tuyere Assly, Drg: BF-2011-013 (5% JOB DONE)	BSP	4	145	580	130.1	520.4	59.60
4	Modified Cast Tuyere Assly, Drg: BF-2011-013 (20% JOB DONE)	BSP	3	145	435	130.1	390.3	44.70
5	Modified Tuyere Cooler, Drg: BF-2011-012 (5% JOB DONE)	BSP	1	165	165	160	160	5.00
6	Modified Tuyere Cooler, Drg: BF-2011-012 (20% JOB DONE)	BSP	1	165	165	160	160	5.00
7	Copper Tuyere, Drg: 5017165-0384407 (5% JOB DONE)	BSL	3	301	903	287	861	42.00
8	Copper Tuyere, Drg: 5017165-0384407 (20% JOB DONE)	BSL	2	301	602	287	574	28.00
9	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098 (20% JOB DONE)	BSL	02	177	354	160	320	34.00
10	SLIPPER PAD, DWN: CDB/R-26095 R4, POS: 3 FOR HSM R-1 ROLL SIDE, MATERIAL AS PER rsn: 31.00.03.06, DRG: CDB/R-26095 R4 POSITION NO: 3 (55% JOB DONE)	RSP	02	165	330	112.4	224.8	105.20
11	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2 (50% JOB DONE)	DSP	04	05	20	03.5	14	6.00

12	Modi Wearing Plates for Bottom work Roll with Chock Assembly of Four High Roughing Stand, Drg: PM-18-104 Serial-2 (34mm Thickness) (3% JOB DONE)	BSP	01	200	200	137.4	137.4	62.60
13	WEARING PLATE FOR WORK ROLL WITH CHOCK ASSLY DRG: PMM-177 R2 (3% JOB DONE)	BSP	01	202	202	133	133	69.00
14	Modified Rope Guide Bush for OBC Grab Bucket, Drg: BF-97-020 (100% JOB DONE)	BSP	02	41	82	30	60	22.00
15	Copper Tuyere, Drg: 5017165-0384407 (100% JOB DONE)	BSL	01	301	301	287	287	14.00
16	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair) (30% JOB DONE)	DSP	01	70	70	39.5	39.5	30.50
17	Bush, Bronze, Drg: PMM-1617-1 R1 Position No. 2 (5% JOB DONE)	BSP	15	01.5	22.5	0.1	1.5	21.00
18	BUSH, PN: 3460331601100/08 FOR C/C SHEAR BLADE HOLDER, DIA: 100/70X33MM DRG: 3460331601100/08 DRAWING NO: 346033160110007 ITEM NO: 08 (10% JOB DONE)	DSP	03	03	9	1.4	4.2	4.80
19	BUSH, PN: 3460331601100/09 FOR C/C SHEAR BLADE HOLDER, DIA: 150/120X60MM, DRG: 346033160100/09 DRAWING NO: 346033160110007 ITEM NO: 09 (5% JOB DONE)	DSP	03	6.0	18	2.0	6	12.00
20	BUSH, PN: 3460331601100/25 FOR C/C SHEAR BLADE LEVER, DIA: 220/179.4X95MM DRG: 3460331601100/25 DRAWING NO: 346033160110005 ITEM NO: 25 (3% JOB DONE)	DSP	01	15	15	6.2	6.2	8.80
21	Modified Tuyere Cooler, Drg: BF-2011-012 (100% JOB DONE)	BSP	01	165	165	160	160	5.00
22	Lining Plate, 13 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00021 POSITION NO - 13 (100% JOB DONE)	ISP	01	75	75	74.7	74.7	0.30

23	Lining Plate, 12 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00020 POSITION NO - 12 (100% JOB DONE)	ISP	01	55	55	54.7	54.7	0.30
24	Lining Plate, 10 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00018 POSITION NO - 10 (100% JOB DONE)	ISP	01	80	80	79.7	79.7	0.30
25	Lining Plate, 20 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00025 POSITION NO - 20 (100% JOB DONE)	ISP	01	65	65	64.7	64.7	0.30
26	Lining Plate, 14 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00022 POSITION NO - 14 (100% JOB DONE)	ISP	01	60	60	59.7	59.7	0.30
27	Lining Plate, 26A for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00026A POSITION NO - 23 (100% JOB DONE)	ISP	01	55	55	54.7	54.7	0.30
28	Lining Plate, 26A for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00026A POSITION NO - 24 (100% JOB DONE)	ISP	01	55	55	54.7	54.7	0.30
29	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B (5% JOB DONE)	BSL	01	1600	1600	1300	1300	300.00
30	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing (100% JOB DONE)	F/O	10	26.8	268	20.4	204	64.00
31	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K (90% JOB DONE)	DSP	1	580	580	560	560	20.00
32	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K (100% JOB DONE)	DSP	1	580	580	560	560	20.00
33	Machining of Hub (Proof Machined Forged Round to 400mm Dia x 400L) for 450T HMTC, Drg: As per reference drawing (60% JOB DONE)	BSL	01	414.103	414.1032	236.2	236.2	177.90
34	Skid Casting Drg: A2/004.07.002.90252 (100% JOB DONE)	DSP	02	140	280	138	276	4.00

35	BASKET FOR MILL SHEAR CROP COLLECTION. SHEAR#1, 2, CVR AND CVRD. PART No. 1.11 TRUNION, DRG: MOD-20228 REV. No. 0 (100% JOB DONE)	ISP	02	35.618	71.23561	06.9	13.8	57.44
36	450T HMTTC Floating Shaft as per drg no. 038.007.SM.025 (5% JOB DONE)	BSL	1	262.512	262.5119	134.1	134.1	128.41
37	450T HMTTC Floating Shaft as per drg no. 038.007.SM.025 (10% JOB DONE)	BSL	2	262.512	525.0237	134.1	268.2	256.82
38	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A (60% JOB DONE)	BSL	02	390	780	365	730	50.00
39	Wheel Track DIA: 800MM, Track Wheel for charging car and coke pusher, Drg: COB-1-882 R1 (30% JOB DONE)	BSP	01	460	460	315	315	145.00
40	Housing Drg: 036.182.00.050A (100% JOB DONE)	BSL	01	140.0	140	72.3	72.3	67.70
41	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D (90% JOB Done)	DSP	01	1780	1780	1500	1500	280.00
42	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B (100% JOB DONE)	BSL	03	1600	4800	1300	3900	900.00
1	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2 (50% Done)	DSP	08	05	40	03.5	28	12.00
2	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair) (50% Done)	DSP	01	70	70	39.5	39.5	30.50
3	Bush, Pedestal, 180 x 160 x 140 mm Top Roller Pedestal Bush for Apron Feeder, Drg: RM-2010-010	BSP	04	14	56	05.6	22.4	33.60
4	Bush, Pedestal, 180 x 160 x 140 mm Top Roller Pedestal Bush for Apron Feeder, Drg: RM-2010-010 (Due to Blown)	BSP	01	14	14	05.6	5.6	8.40
5	Bush for Chain Support of Inspection Bed DRG: PMM-169 R1 (5% Done)	BSP	30	3.00	90	1.00	30	60.00
6	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	02	165	330	160	320	10.00

7	Slipper Pad for PM Roll Side, Material as per RSN number 31.00.03.06 Drg:- CDB/R 18397 R3 (20% Done)	RSP	02	259	518	187	374	144.00
8	PLATE COOLER, LMS, ROW: 38-40, RS-3.50.048 FOR BLAST FURNACE-1 LOWER MIDDLE STACK ZONE, ROW 38 TO 40, DRG: RS-3.50.048, ROW NO.: 38 TO 40	RSP	02	142	284	129	258	26.00
9	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	50	0.625	31.25	0.1	5	26.25
10	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
11	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
12	Slipper Pad, Drg:-CDB/R-26095 R4 Position No:-3 (95% Done)	RSP	02	165	330	112.4	224.8	105.20
13	Modified Liner-1, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-106	BSP	01	160.00	160	81.10	81.1	78.90
14	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	50	0.625	31.25	0.1	5	26.25
15	Modified Liner-4, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-109	BSP	01	127	127	81.1	81.1	45.90
16	Modified Liner-1, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-106 (35% Done)	BSP	01	160.0	160	81.10	81.1	78.90
17	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107 (40% Done)	BSP	1	160.0	160	81.10	81.1	78.90
18	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107	BSP	2	160.0	320	81.10	162.2	157.80
19	Bush, Bronze, Drg: PMM-1617-1 R1 Position No. 2 (5% Done)	BSP	15	01.5	22.5	0.1	1.5	21.00
20	Bushing For Spindle ASSLY of LEVELLER NO. 2, Drg: PMM-548 R1 (70% Done)	BSP	06	19	114	12	72	42.00
21	Modified Liner-3, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-108	BSP	02	127	254	81.1	162.2	91.80
22	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107	BSP	01	160.0	160	81.10	81.1	78.90

23	Modified Liner-4, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-109	BSP	01	127	127	81.1	81.1	45.90
24	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2 (70% Done)	BSP	01	109	109	80.2	80.2	28.80
25	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
26	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2 (70% Done)	BSP	01	109	109	80.2	80.2	28.80
27	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
28	Bush, Pedestal, 180 x 160 x 140 mm Top Roller Pedestal Bush for Apron Feeder, Drg: RM-2010-010	BSP	01	14	14	05.6	5.6	8.40
29	Modified Cast Tuyere Assly, Drg: BF-2011-013 (15% Done)	BSP	01	145	145	130.1	130.1	14.90
30	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	02	145	290	130.1	260.2	29.80
31	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	01	165	165	160	160	5.00
32	Bush Half for Universal Spindle Dia: 150MM of Leveller 1 & 3 Material: ALBRZ, Drg: PMM-940 R1 (55% Done)	BSP	08	5.5	44	3.1	24.8	19.20
33	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
34	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	01	177	177	160	160	17.00
35	PLATE COOLER, LMS, ROW: -27-37, RS-3.50.048 FOR BLAST FURNACE-1 LOWER STACK ZONE, ROW 27 TO 32 AND LOWER MIDDLE STACK ZONE, ROW 33-37, DRG: RS-3.50.048, ROW NO.: 27 TO 37 (5% Done)	RSP	01	142	142	129	129	13.00
36	PLATE COOLER, LMS, ROW: -27-37, RS-3.50.048 FOR BLAST FURNACE-1 LOWER STACK ZONE, ROW 27 TO 32 AND LOWER MIDDLE STACK ZONE, ROW 33-37, DRG: RS-3.50.048, ROW NO.: 27 TO 37	RSP	02	142	284	129	258	26.00
37	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00

38	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
39	TUYERE, BLAST, 140X600 WITH REFRACTORY LINING MATERIAL: MODIFIED TUYERE COPPER WITH REFRACTORY CASTING AS PER DRAWING, DRG: PD/R-355953, BF4	RSP	08	240	1920	220	1760	160.00
40	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	01	315	315	298	298	17.00
41	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75571, Item No. 1 & 2	DSP	02	05	10	03.5	7	3.00
42	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	50	0.625	31.25	0.1	5	26.25
43	Bush for Mould Foot Roll, Drg: 540.01.07.V/77934	DSP	20	2.0	40	1.0	20	20.00
44	Bushing for foot roll assembly of Size 150mm×150mm Drg: ISP-DES-VI-029-001-55-DE-00016	ISP	45	2.00	90	0.40	18	72.00
45	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
46	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	03	177	531	160	480	51.00
47	TUYERE, BLAST, 140X600 WITH REFRACTORY LINING MATERIAL: MODIFIED TUYERE COPPER WITH REFRACTORY CASTING AS PER DRAWING, DRG: PD/R-355953, BF4	RSP	03	240	720	220	660	60.00
48	Bush, Bronze, Drg: PMM-1617-1 R1 Position No. 2 (5% Done)	BSP	10	01.5	15	0.1	1	14.00
49	Bush, Bronze, Drg: PMM-1617-1 R1 Position No. 2 (10% Done)	BSP	11	01.5	16.5	0.1	1.1	15.40
50	Bush, Bronze, Drg: PMM-1617-1 R1 Position No. 2	BSP	04	01.5	6	0.1	0.4	5.60
51	Bush for Chain Support of Inspection Bed DRG: PMM-169 R1 (5% Done)	BSP	26	3.00	78	1.00	26	52.00
52	Bush for Chain Support of Inspection Bed DRG: PMM-169 R1	BSP	40	3.00	120	1.00	40	80.00

53	Modified Liner-1, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-106	BSP	02	160.0	320	94.10	188.2	131.80
54	Modified Liner-4, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-109	BSP	01	127	127	81.1	81.1	45.90
55	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	09	145	1305	130.1	1170.9	134.10
56	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	01	145	145	130.1	130.1	14.90
57	Modified Liner-3, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-108	BSP	01	127	127	81.1	81.1	45.90
58	Bush for Mould Foot Roll, Drg: 540.01.07.V/77934	DSP	16	2.0	32	1.0	16	16.00
59	Bush for Mould Foot Roll, Drg: 540.01.07.V/77934	DSP	04	2.0	8	1.0	4	4.00
60	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107 (Due to Blown)	BSP	01	160.0	160	94.10	94.1	65.90
61	Modified Liner-3, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-108 (Due to Blown)	BSP	01	127	127	81.1	81.1	45.90
62	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	DSP	02	580	1160	560	1120	40.00
63	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	02	580	1160	560	1120	40.00
64	Skid Casting Drg: A2/004.07.002.90252	DSP	07	140	980	138	966	14.00
65	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A (60% Done)	BSL	02	390	780	365	730	50.00
66	Skid Casting Drg: A2/004.07.002.90252	DSP	05	140	700	138	690	10.00
67	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	03	1600	4800	1300	3900	900.00
68	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A (70% Done)	RSP	03	120	360	65	195	165.00
69	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A (30% Done)	BSL	02	390	780	365	730	50.00

70	Machining of Hub (Proof Machined Forged Round to 400mm Dia x 400L) for 450T HMTc, Drg: As per reference drawing	BSL	01	414.103	414.1032	236.2	236.2	177.90
71	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	02	1600	3200	1300	2600	600.00
72	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A (70% Done)	BSL	01	120	120	65	65	55.00
73	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	01	120	120	65	65	55.00
74	Skid Moving, CS DWI: B Upper Series along with 2 Nos of plates Item A. 1 No plate item F of Drawing 04.06.001.117A and 8 Nos of 3/4" BSW Bolts X 2-1/2" Long, All in Assembled Condition., Drg: 04.06.001/105 Drawing Item No: B (75% Done)	DSP	01	800	800	700	700	100.00
75	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	06	1600	9600	1300	7800	1800.00
76	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	RSP	07	120	840	65	455	385.00
77	CHARGING HOLE COVER, DRG: COB-1558	BSL	50	89.9	4495	59.4	2970	1525.00
78	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	02	18.4	36.8	10.2	20.4	16.40
79	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	38	26.8	1018.4	20.4	775.2	243.20
80	Plate Liner Drg: 03601601200 Pos 1	BSL	04	22	88	20	80	8.00
81	Plate Liner Drg: 03601601200 Pos 2	BSL	05	22	110	20	100	10.00
82	Plate Liner Drg: 03601601200 Pos 4	BSL	04	22	88	20	80	8.00
83	Plate Liner Drg: 03601601200 Pos 5	BSL	05	22	110	20	100	10.00
84	Plate Liner Drg: 03601601201 Pos 2	BSL	03	22	66	20	60	6.00
85	Plate Liner Drg: 03601601201 Pos 5	BSL	03	22	66	20	60	6.00
86	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
87	Pivot for ISP as per Drg: PSL1608 & 1609 (Including Shaft Washer & Plate)	F/O	12	12.21	146.52	5.30	63.6	82.92
88	Pivot for GUA as per Drg: PE/271-R, & PE/560 (Including Shaft Washer & Plate)	F/O	42	10.30	432.6	5.00	210	222.60

89	Guide Roller for ISP as per Drg: PSL-1608	F/O	12	4.75	57	3.21	38.52	18.48
90	Guide Roller for ISP as per Drg: PSL-1609	F/O	12	6.75	81	4.5	54	27.00
91	Guide Roller for GUA as per Drg: PE/271	F/O	84	5.8	483	3.8	315.084	167.92
92	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	03	1600	4800	1300	3900	900.00
93	Track Wheel Modified for Door Extractor No.1,2,3,4 Drg: COB-78-75 R1	BSP	02	240	480	183	366	114.00
94	Hub Drg 038.067.00.027	BSL	02	147	294	86	172	122.00
95	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	01	1600	1600	1300	1300	300.00
96	Machining of Shaft as per sample	M/C	01	110	110	41	41	69.00
97	Key way to be made	M/C	01	20	20	13	13	7.00
98	Cu Plug for Tuyere & Cooler	M/C	45	0.85	38.25	0.25	11.25	27.00
99	MS Plug for all Tuyere	M/C	98	1.25	122.5	0.75	73.5	49.00
100	MS Plug for RSP Tuyere 140X600	M/C	11	1.20	13.2	0.60	6.6	6.60
101	MS Cap 2" NPT for RSP Plate Cooler (5X5)	M/C	10	1.75	17.5	0.70	7	10.50
Grand Total					0		0	11083.87
					0		0	
CALCULATION SHEET FOR THE MONTH OF FEBRUARY-2024 (01.02.2024 TO 29.02.2024)					0		0	
1	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107	BSP	01	160.0	160	94.10	94.1	65.90
2	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	02	145	290	130.1	260.2	29.80
3	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	01	315	315	298	298	17.00
4	Bush, Bronze, Drg: PMM-1617-1 R1 Position No. 2	BSP	04	01.5	6	0.1	0.4	5.60
5	Bush for Chain Support of Inspection Bed DRG: PMM-169 R1	BSP	04	3.00	12	1.00	4	8.00
6	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50

7	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
8	Bush, Phosphor Bronze,115/90.8×155 mm for Wheel Dia: 500mm for Crane Drg: PD/R-40705 R8	RSP	02	10.0	20	4.6	9.2	10.80
9	Bush, Phosphor Bronze,135/110.8×190 mm for Wheel Dia: 710mm for Crane Drg: PD/R-40705 R8	RSP	01	15.0	15	6.8	6.8	8.20
10	Intermediate Gear Bush, Pn:3 for Size:130×110×150 mm Long as per Part No.3 10mm Drg: PD/R-52056R2	RSP	03	13.0	39	6.6	19.8	19.20
11	Sector Gear Bush, Pn:2 for 3.5 Cu. Metre Bucket, Size: Dia 130/150×140 mm Long as per Part No.2 10mm Drg: PD/R-19168 R3	RSP	02	16.0	32	7.9	15.8	16.20
12	Bush for Support Block Drg:4.56555.R Close Flap Device 8.339735.Q	ISP	04	20.5	82	7.9	31.6	50.40
13	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
14	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
15	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	100	0.625	62.5	0.1	10	52.50
16	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	DSP	10	05	50	03.5	35	15.00
17	Sector Gear Bush, Pn:2 for 3.5 Cu. Metre Bucket, Size: Dia 130/150×140 mm Long as per Part No.2 10mm Drg: PD/R-19168 R3	RSP	01	16.0	16	7.9	7.9	8.10
18	Brass Bush F/Coke Guide for Top Roller Drg: DWN:CO-20-115	BSP	09	7.5	67.5	3.68	33.12	34.38
19	Brass Bush F/Coke Guide for Top Roller Drg: DWN:CO-20-115 (Due to Blown)	BSP	03	7.5	22.5	3.68	11.04	11.46

20	Brass Bush F/DE Ram Boomring Lever Drg: DWN:CO-20-114	BSP	04	22.5	90	10.86	43.44	46.56
21	Brass Bush F/DE Ram Boomring Lever Drg: DWN:CO-20-114 (Due to Blown)	BSP	02	22.5	45	10.86	21.72	23.28
22	Plate, for Top work Roll Chock of four High Roughing Stand Drg: PMM-182-1 R1 (Due to Blown)	BSP	01	166	166	100	100	66.00
23	Copper Tuyere, Drg: 5017165-0384407	BSL	03	301	903	287	861	42.00
24	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	09	177	1593	160	1440	153.00
25	Main Shaft Step Bearing for Secondary Crusher Drg:IRON/OHP/CR/2/13/72	RSP	02	93.5	187	62.4	124.8	62.20
26	Piston Wearing Plate for Secondary Crusher Drg:IRON/OHP/CR/2/12/72	RSP	02	78.5	157	56.0	112	45.00
27	Slipper Pad, Drg:-CDB/R-26095 R4 Position No:-3 (In Pair)	RSP	02	165	330	112.4	224.8	105.20
28	Slipper Pad for PM Roll Side, Material as per RSN number 31.00.03.06 Drg:- CDB/R 18397 R3	RSP	02	259	518	187	374	144.00
29	Bush, Phosphor Bronze,155/125.8×210 mm for Wheel Dia: 800mm for Crane Drg: PD/R-40705 R8	RSP	01	21.5	21.5	9.5	9.5	12.00
30	Bush, Phosphor Bronze,170/140.8×240 mm for Wheel Dia: 900mm for Crane Drg: PD/R-40705 R8	RSP	06	31.0	186	12.9	77.4	108.60
31	Bush, Phosphor Bronze,135/110.8×190 mm for Wheel Dia: 710mm for Crane Drg: PD/R-40705 R8	RSP	01	15.0	15	6.8	6.8	8.20
32	Bush for Mould Foot Roll, Drg: 540.01.07.V/77934	DSP	35	2.0	70	1.0	35	35.00
33	Brass Bush F/Coke Guide for Top Roller Drg: DWN:CO-20- 116	BSP	11	4.0	44	0.99	10.89	33.11
34	Bush-1 Long Lever For DE RAM, Drg: DWN:CO-20-110	BSP	12	5.5	66	2.01	24.12	41.88
35	Bush-1 Small Lever For DE RAM, Drg: DWN:CO-20-108	BSP	07	12.5	87.5	5.0	35	52.50
36	Bush-1 Small Lever For DE RAM, Drg: DWN:CO-20-108 (Due to Blown)	BSP	01	12.5	12.5	5.0	5	7.50
37	Bush-2 Small Lever For DE RAM, Drg: DWN:CO-20-107	BSP	10	8.0	80	3.4	34	46.00

38	Bush-2 (Side) For DE RAM Swinging Tunion Drg:CO-20-106	BSP	10	2.3	23	0.58	5.8	17.20
39	Bush-2 (Top/Bottom) For DE RAM Swinging Tunion Drg:CO-20-105	BSP	10	2.5	25	0.7	7.38	17.62
40	Bush Half for Universal Spindle Dia: 150MM of Leveller 1 & 3 Material: ALBRZ, Drg: PMM-940 R1	BSP	10	5.5	55	3.1	31	24.00
41	Bushing For Spindle ASSLY of LEVELLER NO. 2, Drg: PMM-548 R1	BSP	02	19	38	11.5	23	15.00
42	Nut for Door Extractor, Drg:COB-1-465 R1	BSP	01	45	45	24.1	24.1	20.90
43	Eccentric Wearing Plate for Secondary Crusher, Drg:IRON/OHP/CR/32/72	RSP	02	81.0	162	46.0	92	70.00
44	Top Insert for Bridge Travel, Drg:BF-22-086 R0	BSP	01	41	41	12.4	12.4	28.60
45	Top Insert for Bridge Travel, Drg:BF-22-086 R0	BSP	01	41	41	12.4	12.4	28.60
46	Bottom Insert for Bridge Travel, Drg:BF-22-085 R0	BSP	01	41	41	11.6	11.6	29.40
47	Bottom Insert for Bridge Travel, Drg:BF-22-085 R0	BSP	01	41	41	11.6	11.6	29.40
48	TUYERE, BLAST, 140X600 WITH REFRACTORY LINING MATERIAL: MODIFIED TUYERE COPPER WITH REFRACTORY CASTING AS PER DRAWING, DRG: PD/R-355953, BF4	RSP	04	240	960	220	880	80.00
49	PLATE COOLER, LMS, ROW: 38-40, RS-3.50.048 FOR BLAST FURNACE-1 LOWER MIDDLE STACK ZONE, ROW 38 TO 40, DRG: RS-3.50.048, ROW NO.: 38 TO 40	RSP	06	142	852	129	774	78.00
50	Slipper Pad, Drg:-CDB/R-26095 R4 Position No:-3 (In Pair)	RSP	02	165	330	112.4	224.8	105.20
51	Bush for OTC Travel Mechanism, Drg:B-001-107 R1 (TOP)	BSP	03	33	99	19	57	42.00
52	Bush for OTC Travel Mechanism, Drg:B-001-107 R1 (Bottom)	BSP	03	25	75	14.5	43.5	31.50
53	Modified Rope Guide Bush for OBC Grab Bucket, Drg: BF-97-020	BSP	02	41	82	30	60	22.00
54	Bush-1 Long Lever For DE RAM, Drg: DWN:CO-20-110	BSP	06	5.5	33	2.01	12.06	20.94
55	Bush-2 Small Lever For DE RAM, Drg: DWN:CO-20-107	BSP	02	8.0	16	3.4	6.8	9.20
56	Bush for Cradle, Drg:CO-20-111	BSP	05	9.5	47.5	5.8	29	18.50

57	Nut, Drg:COB-1-201 R3	BSP	02	24	48	10.9	21.8	26.20
58	Copper Tuyere, Drg: 5017165-0384407	BSL	03	301	903	287	861	42.00
59	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	06	177	1062	160	960	102.00
60	Tuyere, Intermediate, Drg: 036.008.00.049C	BSL	09	181	1629	170	1530	99.00
61	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	60	0.625	37.5	0.1	6	31.50
62	Slipper Pad for PM Roll Side, Material as per RSN number 31.00.03.06 Drg:- CDB/R 18397 R3	RSP	02	259	518	187	374	144.00
63	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107	BSP	02	160	320	94.10	188.2	131.80
64	Modified Liner-4, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-109	BSP	01	127	127	81.1	81.1	45.90
65	MODIFIED LINER, For Chock Assly of Bottom backup roll C/S Exit (L2) Outer Side of 4 High Roughing/Finishing Stand, Drg: PM-17-013	BSP	02	142	284	92.5	185	99.00
66	Bush-1 Long Lever For DE RAM, Drg: DWN:CO-20-110	BSP	04	5.5	22	2.01	8.04	13.96
67	Bush-2 Long Lever For DE RAM, Drg: DWN:CO-20-109	BSP	08	11.0	88	4.83	38.64	49.36
68	Bush for Cradle, Drg:CO-20-111	BSP	03	9.5	28.5	5.8	17.4	11.10
69	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
70	Hub Drg 038.067.00.027	BSL	01	147.31	147.31	86	86	61.31
71	Skid, Stationary, CS,PN:106347-AA# Drg: 004.06.001.0000106# Item No. AB"	DSP	01	560	560	465	465	95.00
72	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	04	1600	6400	1300	5200	1200.00
73	Link 1 st for Dummy Bar Chain Assembly of Slab Caster 250 x 1500 mm, Drg: CC-2004-037	BSP	01	445	445	331	331	114.00
74	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	03	475	1425	435	1305	120.00
75	Hub Drg 038.067.00.027	BSL	01	147.31	147.31	86	86	61.31
76	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	03	475	1425	435	1305	120.00

77	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	04	1600	6400	1300	5200	1200.00
78	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	48	26.8	1286.4	20.4	979.2	307.20
79	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	04	120	480	65	260	220.00
80	Hammer for Coal Crusher. Drg: 001.027.00.057	BSL	36	21	756	18.2	655.2	100.80
81	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	03	1600	4800	1300	3900	900.00
82	Pin as per Drg:038.067.00.028	BSL	02	649.68	1299.36	255.4	510.8	788.56
83	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	08	120.0	960	65.0	520	440.00
84	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	04	18.4	73.6	10.2	40.8	32.80
85	Clay Gun Nozzle Tip with 2 Nos Locking Pin, Drg:ISP-DES-V-007-323-01-DE-004	ISP	05	45.0	225	31.3	156.5	68.50
86	Clay Gun Nozzle Tip with 2 Nos Locking Pin, Drg:ISP-DES-V-007-323-01-DE-004	ISP	03	45.0	135	31.3	93.9	41.10
87	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	03	1600	4800	1300	3900	900.00
88	Pin as per Drg:038.067.00.028	BSL	2	649.68	1299.36	255.4	510.8	788.56
89	Assemble Drive Side Wheel for Transfer Car CRM-III, Drg:- 412.1409.00.002 - Refer sheet A attached	BSL	02		0		0	0.00
90	Split Colar= 2 Nos 100%	BSL	2	5.000	10	4.000	8	2.000
91	Bearing Cover	BSL	1	35.000	35	15.600	15.6	19.400
92	Lock Plate	BSL	4	6.000	24	2.300	9.2	14.800
93	Key for Drive Side	BSL	6	0.750	4.5	0.250	1.5	3.000
94	NFF Crane C.T Wheel Welding	M/C	05	9.8	49	4.9	24.5	24.50
95	Ferrous for Electric Dept.	M/C	04	5.8	23.2	3.00	12	11.20
96	Shaft Key way cutting for Electric Dept.	M/C	01	26.1	26.1	23.9	23.9	2.20
97	PB pads for Shapping M/C	M/C	02	3.0	6	1.0	2	4.00
98	Fibre Bush and Shaft Electric Dept.	M/C	16	3.3	52.8	1.2	19.2	33.60
99	10 Ton Crane Bolt M/c Shop	M/C	12	2.75	33	1.3	15.6	17.40
100	BSL Pin for Laddle	M/C	04	11.00	44	6.50	26	18.00
101	Plate for Milling & Drilling for Structure Shop	M/C	06	41.50	249	28.00	168	81.00
102	Shaft with Thread for Fabrication	M/C	04	1.20	4.8	0.40	1.6	3.20

103	Pin for RSP Remachining	M/C	02	18.48	36.96	10.00	20	16.96
104	Pin for RITES	M/C	02	1.20	2.4	0.40	0.8	1.60
105	Pin for RITES	M/C	02	1.20	2.4	0.40	0.8	1.60
106	Cu Plug for Tuyere & Cooler	M/C	46	0.85	39.1	0.25	11.5	27.60
107	MS Plug for all Tuyere	M/C	102	1.25	127.5	0.75	76.5	51.00
108	MS Plug for RSP Tuyere 140X600	M/C	8	1.20	9.6	0.60	4.8	4.80
109	MS Cap 2" NPT for RSP Plate Cooler (5X5)	M/C	12	1.75	21	0.70	8.4	12.60
Grand Total					0		0	10950.05
					0		0	
CHECKED CALCULATION SHEET FOR THE MONTH OF MARCH-2024 (01.03.2024 TO 31.03.2024)					0		0	
1	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	12	145	1740	130.1	1561.2	178.80
2	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
3	Bush for Mould Foot Roll, Drg: 540.01.07.V/77934	DSP	16	2.0	32	1.0	16	16.00
4	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	140	1.0	140	0.1	14	126.00
5	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75571, Item No. 1 & 2	DSP	42	5.0	210	3.5	147	63.00
6	Bush (Foot Roll Bracket) for 370 Dia Round Mould Assly Drg:540.01.07.V/89961	DSP	04	1.5	6	0.70	2.8	3.20
7	Copper Cooling Plate For Furnace Cooling System 520X550X850L, Drg:ISP-POSCO E&C-V-007-135-40-DE-14941	ISP	01	155	155	138.0	138	17.00
9	Bush Stand for F/V-Rake Drg: 339.71.30.7.13111 (8 Halves)	DSP	04	42	168	18	72	96.00
10	Bush Storage Bank, Drg: 3.39.71.30.9.13240 (4 Halves)	DSP	02	13	26	4.6	9.2	16.80
11	Spindle Support Basr Liner 600X70X10, Drg: MOD-2159 for Stand 11,13,15	ISP	06	19	114	3.3	19.8	94.20

12	Spindle Support Basr Liner 600X28X10, Drg: MOD-2159 for Stand 11,13,15	ISP	10	4.5	45	1.3	13	32.00
13	Spindle Support Basr Liner 600X35X10, Drg: MOD-2159 for Stand 11,13,15	ISP	05	5.0	25	1.6	8	17.00
14	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
15	PLATE COOLER, LMS, ROW: 38-40, RS-3.50.048 FOR BLAST FURNACE-1 LOWER MIDDLE STACK ZONE, ROW 38 TO 40, DRG: RS-3.50.048, ROW NO.: 38 TO 40	RSP	01	142	142	129	129	13.00
16	Slipper Pad, Drg:-CDB/R-26095 R4 Position No:-3 (In Pair)	RSP	02	165	330	112.4	224.8	105.20
17	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.00
18	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.00
19	Bush Storage Bank, Drg: 3.39.71.30.9.13240 (4 Halves)	DSP	02	13	26	4.6	9.2	16.80
20	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	70	1.0	70	0.1	7	63.00
21	Top Insert for Bridge Travel, Drg:BF-22-086 R0	BSP	08	21	168	12.4	99.2	68.80
22	Bottom Insert for Bridge Travel, Drg:BF-22-085 R0	BSP	08	21	168	11.6	92.8	75.20
23	Bush-1 Long Lever For DE RAM, Drg: DWN:CO-20-110	BSP	10	5.5	55	2.01	20.1	34.90
24	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
25	MODIFIED LINER, For Chock Assly of Bottom backup roll D/S Entry (R1) Inner/Barrel Side of 4 High Roughing/Finishing Stand, Drg: PM-17-016	BSP	01	130	130	77.5	77.5	52.50
26	Brass Bush F/Coke Guide for Top Roller Drg: DWN:CO-20-115	BSP	03	7.5	22.5	3.68	11.04	11.46
27	Brass Bush F/DE Ram Boomring Lever Drg: DWN:CO-20-114	BSP	02	22.5	45	10.86	21.72	23.28

28	BUSH, PN: 3460331601100/09 FOR C/C SHEAR BLADE HOLDER, DIA: 150/120X60MM, DRG: 346033160100/09 DRAWING NO: 346033160110007 ITEM NO: 09	DSP	30	6.0	180	2.0	60	120.00
30	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	01	315	315	298	298	17.00
31	Bush, for Unloader Shaft and Lateral Transfer Shaft, Drg: 4.38.01.21.W/81429	DSP	08	9.2	73.6	3.2	25.6	48.00
32	Bush for Shaft W/Lever of Unloader Mechanism, Drg: 540.01.21.W/85463 (In Set)	DSP	04	21	84	07	28	56.00
33	Bush for Shaft W/Lever of Unloader Mechanism, Drg: 540.01.21.W/85463 (In Set)	DSP	08	21	168	07	56	112.00
34	BUSH, PN: 3460331601100/25 FOR C/C SHEAR BLADE LEVER, DIA: 220/179.4X95MM DRG: 3460331601100/25 DRAWING NO: 346033160110005 ITEM NO: 25	DSP	03	15	45	6.2	18.6	26.40
35	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
36	Bush for OTC Travel Mechanism, Drg:B-001-107 R1 (TOP)	BSP	02	33	66	19	38	28.00
37	Bush for OTC Travel Mechanism, Drg:B-001-108 R1 (Bottom)	BSP	02	25	50	14.5	29	21.00
38	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	05	165	825	160	800	25.00
39	MODIFIED LINER, For Chock Assly of Bottom backup roll C/S Exit (L2) Inner/Barrel Side of 4 High Roughing/Finishing Stand, Drg: PM-17-012	BSP	01	140	140	91.0	91	49.00
40	MODIFIED LINER, For Chock Assly of Bottom backup roll D/S Entry (R1) Outer Side of 4 High Roughing/Finishing Stand, Drg: PM-17-017	BSP	01	137	137	82.5	82.5	54.50
41	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	07	177	1239	160	1120	119.00
42	PLATE COOLER, LMS, ROW: 38-40, RS-3.50.048 FOR BLAST FURNACE-1 LOWER MIDDLE STACK ZONE, ROW 38 TO 40, DRG: RS-3.50.048, ROW NO.: 38 TO 40	RSP	01	142	142	129	129	13.00

43	Slipper Pad, Drg:-CDB/R-26095 R4 Position No:-3 (In Pair)	RSP	02	165	330	112.4	224.8	105.20
44	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	01	315	315	298	298	17.00
45	MOD.CU TUYERE W/HARD FACING. Drg:320.01.31.W/89465	DSP	04	170	680	146	584	96.00
46	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	04	390	1560	365	1460	100.00
47	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	01	1600	1600	1300	1300	300.00
48	Clay Gun Nozzle Tip with 2 Nos Locking Pin, Drg:ISP-DES-V- 007-323-01-DE-004	ISP	03	45	135	31.3	93.9	41.10
49	Dipper Casting (Inner) for RHF The Holes shall be drilled to round holes of 16 mm Drg: 358030 Part-1	VISL	12	40	480	23.5	282	198.00
50	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	04	390	1560	365	1460	100.00
51	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	02	1600	3200	1300	2600	600.00
52	Hammer for Coal Crusher. Drg: 001.027.00.057	BSL	36	21	756	18.2	655.2	100.80
53	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	03	18.4	55.2	10.2	30.6	24.60
54	Plate Liner Drg: 03601601200 Pos 1	BSL	02	22	44	20	40	4.00
55	Plate Liner Drg: 03601601200 Pos 2	BSL	02	22	44	20	40	4.00
56	Plate Liner Drg: 03601601200 Pos 3	BSL	02	22	44	20	40	4.00
57	Plate Liner Drg: 03601601200 Pos 4	BSL	02	22	44	20	40	4.00
58	Plate Liner Drg: 03601601200 Pos 5	BSL	02	22	44	20	40	4.00
59	Plate Liner Drg: 03601601200 Pos 6	BSL	02	22	44	20	40	4.00
60	Bushing as per drg no. 038.067.SM.006A	BSL	04	59.59	238.36	21.8	87.2	151.16
61	Bushing as per drg no. 038.067.SM.0064 & 011A	BSL	08	28.55	228.4	7.3	58.4	170.00
62	Bearing Block attachment of Blind Cover Drg: 038.067.SM.013A/ Drg: 038.067.00.026 & 038.067.00.015 (3 Set)	BSL	03	200.0	600	139.7	419.1	180.90
63	Blind Cover as per Drg: 038.067.SM.013A	BSL	08	104.2	833.6	33.5	268	565.60
65	Pig Mould Drg: 036.108.00.002 B	BSL	24	105	2520	102.5	2460	60.00

66	FIRE GRATE , FOR SELF BALANCING SCREEN DRG: SP2-MM-1605-R1	BSP	01	95	95	85	85	10.00
67	Pig Mould Drg: 036.108.00.002 B	BSL	16	105	1680	102.5	1640	40.00
68	Axle Box for 160 Ton Ingot Drg: OH-01-094-74 R2	BSP	03	125	375	76.9	230.7	144.30
69	SIDE FOR 80T BOGIE WITHOUT BRAKE SLAG LADLE CAR 16M3 Drg: CS-88-007-R1/ Attached with 16 Nut Bolt	BSP	02	1250.0	2500	1160.0	2320	180.00
70	Iron Notch Frame. Drg: 320.02.31.X/71814	DSP	01	1320	1320	1270.0	1270	50.00
71	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	02	475	950	435	870	80.00
72	Skid, Stationary Drg: 04.06.001/106 Drg Item No.:AB"	DSP	01	570	570	465	465	105.00
73	Lining Plate, 13 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00021 POSITION NO - 13	ISP	06	75	450	74.7	448.2	1.80
74	Lining Plate, 14 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00022 POSITION NO - 14	ISP	05	60	300	59.7	298.5	1.50
75	Lining Plate, 15 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00022 POSITION NO - 15	ISP	04	60	240	59.7	238.8	1.20
76	Lining Plate, 16 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00023 POSITION NO - 16	ISP	01	70	70	69.7	69.7	0.30
77	Lining Plate, 19 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00024 POSITION NO - 19	ISP	02	32	64	31.7	63.4	0.60
78	Lining Plate, 20 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00025 POSITION NO - 20	ISP	02	65	130	64.7	129.4	0.60
79	Hammer for Coal Crusher. Drg: 001.027.00.057	BSL	36	21	756	18.2	655.2	100.80
80	Iddler Shaft (1635 X 25)	M/C	300	7.9	2370	7.5	2250	120.00
81	Channel Ø16 Drilling (New Crane)	M/C	1	11.00	11	10.2	10.2	0.80
82	Flange Plate Drilling for Water Works	Water Work	5	7.2	36	6.6	33	3.00
83	Shaft for Electric Dept.	EMD	8	1.9	15.2	1.0	8	7.20
84	PB Bushes Bore Turning	EMD	9	3.2	28.8	2.7	24.3	4.50
85	Shaft for Idler Shop pipe cutting machine	Idler	3	13.2	39.6	8.1	24.3	15.30
86	ISP Arbour Sleeve machining	M/C	1	775.14	775.14	757.92	757.92	17.22
87	Gear for 20 T GC Shop Crane under repair	EMD	2	126.1	252.2	93	186	66.20

88	Cu Plug for Tuyere & Cooler	M/C	34	0.85	28.9	0.25	8.5	20.40
89	Gear for 20 T GC Shop Crane under repair	EMD	2	126.1	252.2	93	186	66.20
90	Pinion for 20 T GC Shop Crane under repair	EMD	2	16.65	33.3	12	24	9.30
91	MS Plug for all Tuyere	M/C	78	1.25	97.5	0.75	58.5	39.00
92	MS Cap 2" NPT for RSP Plate Cooler (5X5)	M/C	2	1.75	3.5	0.70	1.4	2.10
93	80T Boggie Nut Bolt -M30 Long Bolts	M/C	16	9.4	150.4	3.3	53.12	97.28
Grand Total					0		0	6079.60
					0		0	
CHECKED : CALCULATION SHEET FOR THE MONTH OF APRIL-2024 (01.04.2024 TO 30.04.2024)					0		0	
1	Top Insert for Bridge Travel, Drg:BF-22-086 R0	BSP	08	20.5	164	12.4	99.2	64.80
2	Bottom Insert for Bridge Travel, Drg:BF-22-085 R0	BSP	08	20.5	164	11.6	92.8	71.20
3	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	02	109	218	80.2	160.4	57.60
4	Bush, Wrist Pin Drg: 02.16.04/12D F, Item K'	DSP	20	13.0	260	6.3	126	134.00
5	BUSH, PN: 3460331601100/08 FOR C/C SHEAR BLADE LEVER, DIA: 100/700X33MM, DRG: 346033160100/08 DRAWING NO: 346033160110007 ITEM NO: 08	DSP	20	3.5	70	0.7	14.4	55.60
6	BUSH, PN: 3460331601100/25 FOR C/C SHEAR BLADE LEVER, DIA: 220/179.4X95MM DRG: 3460331601100/25 DRAWING NO: 346033160110005 ITEM NO: 25	DSP	13	15	195	6.0	78	117.00
7	Bush, for Shear Connecting Rod. Drg:339.71.30.9.12518B Item No. 1 & 2	DSP	2	28	56	18.0	36	20.00
8	Bushing for foot roll assembly of Size 150mm×150mm Drg: ISP- DES-VI-029-001-55-DE-00016	ISP	37	2.0	74	0.40	14.8	59.20
9	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
10	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	02	109	218	80.2	160.4	57.60
11	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50

12	Bush, for Shear Connecting Rod. Drg:339.71.30.9.12518B Item No. 1 & 2	DSP	2	28	56	18.0	36	20.00
13	BUSH, Stand for H-RAKE Drg: 339.71.30.7.13110	DSP	6	14	84	6.0	36	48.00
14	BUSH, Stand for F/V-RAKE Drg: 339.71.30.7.13111 (In Set)	DSP	01	42	42	18.0	18	24.00
15	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	03	315	945	298	894	51.00
16	MOD.CU TUYERE W/HARD FACING. Drg:320.01.31.W/89465	DSP	01	170	170	146	146	24.00
17	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
18	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	02	165	330	160	320	10.00
19	Spindle Support Base Liner 800X80X10 (Drg: MOD-2792)	ISP	1	12.5	12.5	4.7	4.7	7.80
20	Spindle Support Base Liner 800X48X10 (Drg: MOD-2792)	ISP	03	8.5	25.5	2.9	8.7	16.80
21	Spindle Support Base Liner 600X35X10 (Drg: MOD-2159)	ISP	05	5.0	25	1.6	8	17.00
22	Copper Tuyere, Drg: 5017165-0384407	BSL	03	301	903	287	861	42.00
23	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
24	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	02	145	290	130.1	260.2	29.80
25	Modified Liner-2, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-107	BSP	02	160	320	94.10	188.2	131.80
26	BUSH Drg: 339.71.30.9.12519A	DSP	02	57.0	114	32.20	64.4	49.60
27	Bush, for Shear Connecting Rod. Drg:339.71.30.9.12518B Item No. 1 & 2	DSP	2	28	56	18.0	36	20.00
28	Bush, Wrist Pin Drg: 02.16.04/12D F, Item K'	DSP	11	13.0	143	6.3	69.3	73.70
29	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	05	145	725	130.1	650.5	74.50
30	Mod.Wearing Plate Drg: PMM-182 R1	BSP	1	166	166	117.0	117	49.00
31	Bush for Tundish Tilter Drg:4.38.02.01.V/75808	DSP	02	55.0	110	35.0	70	40.00
32	Bush , Foot/Shear Drg:339.71.30.9.12517 (In Set)	DSP	02	30.0	60	18.85	37.7	22.30
33	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
34	Bush, for Shear Connecting Rod. Drg:339.71.30.9.12518B Item No. 1 & 2	DSP	2	28	56	18.0	36	20.00
35	Bush, Wrist Pin Drg: 02.16.04/12D F, Item K'	DSP	04	13.0	52	6.3	25.2	26.80
36	Slipper Liner Spindle Support Stand 06 to 10 Drg: MOD-1356	ISP	01	12.5	12.5	3.5	3.5	9.00

37	Spindle Support Base Liner 600X70X10mm Drg: MOD-2159	ISP	02	9.25	18.5	3.30	6.6	11.90
38	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	07	11.0	77	2.60	18.2	58.80
39	Spindle Support Base Liner 800X80X10mm Drg: MOD-2792/ Item-1	ISP	02	12.5	25	4.7	9.4	15.60
40	Mod.Wearing Plate Drg: PMM-182 R1	BSP	01	166	166	117.0	117	49.00
41	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
42	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
43	Bearing Eccentric Bush for H-RAKE Drg: 339.71.30.7.13107 (4 Halves)	DSP	02	73	146	35.0	70	76.00
44	Bearing Eccentric Bush for V-RAKE, Drg: 339.71.30.7.13106 (2 Halves)	DSP	01	81	81	32.4	32.4	48.60
45	PLATE COOLER, LMS, ROW: 38-40, RS-3.50.048 FOR BLAST FURNACE-1 LOWER MIDDLE STACK ZONE, ROW 38 TO 40, DRG: RS-3.50.048, ROW NO.: 38 TO 40	RSP	01	142	142	129	129	13.00
46	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	10	177	1770	160	1600	170.00
47	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	02	475	950	435	870	80.00
48	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
49	Plate Armour Upper Drg: NM-82-69 R2	BSP	01	955	955	915	915	40.00
50	Plate Armour Middle Drg: NM-83-69 R1	BSP	01	981	981	920	920	61.00
51	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	RSP	03	120	360	65	195	165.00
52	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	04	390	1560	365	1460	100.00
53	Machining of Chain Sprocket 275 dia	NFF	01	13.56	13.56	10.3	10.3	3.26
54	Machining of Chain Sprocket 137 dia	NFF	01	2.56	2.56	1.2	1.2	1.36
55	Track Wheel Modified for Door Extractor No.1,2,3,4 Drg: COB-78-75 R1	BSP	01	240	240	183	183	57.00

56	First Link Dummy bar Chain Assembly 200X1300 for Dummy bar handling Mechanism (Slab Caster), Drg: CC-420-80 R0	BSP	01	300	300	183.2	183.2	116.80
57	Plate ArmourUpper Drg: NM-82-69 R2	BSP	03	955	2865	915	2745	120.00
58	Plate Armour Middle Drg: NM-83-69 R1	BSP	01	981	981	920	920	61.00
59	Plate Armour Lower Drg: NM-84-69 R1	BSP	01	980	980	920	920	60.00
60	Gear Travel Drg: COB-176-69 R1	BSP	02	300	600	207.9	415.8	184.20
61	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	01	475	475	435	435	40.00
62	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D (Rejected due to Blown)	DSP	01	1780	1780	1745	1745	35.00
63	Hammer for Coal Crusher. Drg: 001.027.00.057	BSL	72	36	2592	21	1512	1080.00
64	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	04	390	1560	365	1460	100.00
65	Bearing Block attachment of Blind Cover Drg: 038.067.SM.013A/ Drg: 038.067.00.026 & 038.067.00.015 (2 Set)	BSL	02	200.0	400	139.7	279.4	120.60
66	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	06	18.4	110.4	10.2	61.2	49.20
67	Steel Teeming Ladle Car Assy with Wheel Assemblies and other attachments but without Gearboxes Motor BEARING BLOCK FOR 1000 DIA-WHEEL, Drg: PD/R-208292	RSP	01	170.0	170	121	121	49.00
68	Steel Teeming Ladle Car Assy with Wheel Assemblies and other attachments but without Gearboxes Motor PART 5 BEARING COVER, Drg: N-29800	RSP	04	27.0	108.1	15.7	62.8	45.30
69	First Link Dummy bar Chain Assembly 200X1300 for Dummy bar handling Mechanism (Slab Caster), Drg: CC-420-80 R0	BSP	01	300	300	183.2	183.2	116.80
70	Plate ArmourUpper Drg: NM-82-69 R2	BSP	01	955	955	915	915	40.00
71	Plate Armour Middle Drg: NM-83-69 R1	BSP	01	981	981	920	920	61.00
72	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	01	475	475	435	435	40.00
73	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
74	Pig Mould Drg: 036.108.00.002 B	BSL	32	105	3360	102.5	3280	80.00
75	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	12	120	1440	65	780	660.00

76	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	25	22.8	570	13.8	345.25	224.75
77	Bearing Block attachment of Blind Cover Drg: 038.067.SM.013A/ Drg: 038.067.00.026 & 038.067.00.015 (3 Set)	BSL	02	200.0	400	139.7	279.4	120.60
78	HOUSING, BASE Item No:AB, B & C Drg: 04.06.002/155A	DSP	01	193.20	193.2	161.20	161.2	32.00
79	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	06	115	690	92.1	552.6	137.40
80	Nut-Bolts for CT drive of 20 Ton DC Crane as per drwaing	EMD	06	3.39	20.34	0.57	3.42	16.92
81	ISP Arbour Sleeve machining	M/C	1	775.14	775.14	757.92	757.92	17.22
82	ISP Arbour Sleeve machining	M/C	1	775.14	775.14	757.92	757.92	17.22
83	Drill of Cutting Machine Wheel Cover (Dia: 11mm) Hole Size	NFF	3	15	45	14	42	3.00
84	Sand Mill Machine's Roller machining as per drawing	NFF	4	33.5	134	19.4	77.6	56.40
85	Machining Bore & Keyway for Chain Sprocket 275 dia	NFF	1	13.56	13.56	10.2	10.2	3.36
86	Machining Bore & Keyway for Chain Sprocket 137 dia	NFF	1	2.56	2.56	1.15	1.15	1.41
87	Machining of Forged round 60mm X 830L as per drawing.	F/O	15	25.6	384	17	255	129.00
88	Side Plate L430XW150XT30 mm as per drawing	F/O	20	15.2	304	10.8	216	88.00
89	Cu Plug for Tuyere & Cooler	M/C	14	0.85	11.9	0.25	3.5	8.40
90	MS Plug for all Tuyere	M/C	79	1.25	98.75	0.75	59.25	39.50
91	MS Cap 2" NPT for RSP Plate Cooler (5X5)	M/C	2	1.75	3.5	0.70	1.4	2.10
Grand Total					0		0	6669.60
					0		0	
CHECKED CALCULATION SHEET FOR THE MONTH OF MAY-2024 (01.05.2024 TO 31.05.2024)					0		0	
1	Mod.Wearing Plate Drg: PMM-182 R1	BSP	02	166	332	117.0	234	98.00
2	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
3	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
4	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50

5	Nut, Half, Modified in Pair R/H for Lower Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
6	BUSH, Dia-280/ Dia-250 LG-95 Drg: BF-460-63	BSP	03	17	51	09	27	24.00
7	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	06	177	1062	160	960	102.00
13	Mod.Wearing Plate Drg: PMM-182 R1	BSP	01	166	166	117.0	117	49.00
14	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
15	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
16	BUSH, Dia-280/ Dia-250 LG-95 Drg: BF-460-63	BSP	05	17	85	09	45	40.00
17	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	09	177	1593	160	1440	153.00
18	Bushing for foot roll assembly of Size 150mm×150mm Drg: ISP-DES-VI-029-001-55-DE-00016	ISP	58	2.0	116	0.40	23.2	92.80
19	Insert for Pinion Stand 800 Drg: RSM-20/12A-R-3	BSP	02	108	216	74.8	149.6	66.40
20	Top Insert for Bridge Travel Drg: BF-22-086-R0 (1 Halves) Rejection due to Blown	BSP	01	20.5	20.5	9.83	9.83	10.67
21	Bottom Insert for Bridge Travel, Drg:BF-22-085 R0 (2 Halves) Rejection due to Blown	BSP	01	41	41	37.02	37.02	3.98
22	Mod. Rope Guide bush for OBC Grab Bucket Drg: BF-97-020 Rejection due to Blown	BSP	01	41	41	27.61	27.61	13.39
23	BUSH, Dia-280/ Dia-250 LG-95 Drg: BF-460-63 Rejection due to Blown	BSP	02	17	34	7.62	15.24	18.76
24	Mod.Wearing Plate Drg: PMM-182 R1	BSP	01	166.0	166	117.0	117	49.00
25	Bush for OTC Travel Mechanism, Drg:B-001-107 R1 (TOP)	BSP	02	25.0	50	17.5	35	15.00
26	Bush for OTC Travel Mechanism, Drg:B-001-108 R1 (Bottom)	BSP	02	33.0	66	23.0	46	20.00
27	Bush Cone, For HMB Cone Crusher, Drg:HMB-1204-03-03	BSP	01	668	668	335.8	335.8	332.20
28	Top Insert for Bridge Travel Drg: BF-22-086-R0 (4 Halves)	BSP	02	20.5	41	12.4	24.8	16.20
30	Bottom Insert for Bridge Travel, Drg:BF-22-085 R0 (4 Halves)	BSP	02	20.5	41	11.6	23.2	17.80
31	Slipper Pad Drg: PD/R-13747,Pos-1	RSP	02	330	660	246	492	168.00
32	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	06	177	1062	160	960	102.00
33	Tuyere, Intermediate, Drg: 036.008.00.049C	BSL	06	181	1086	170	1020	66.00

34	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	04	315	1260	298	1192	68.00
35	Bush for OTC Travel Mechanism, Drg:B-001-107 R1 (TOP)	BSP	02	25.0	50	17.5	35	15.00
36	Bush for OTC Travel Mechanism, Drg:B-001-108 R1 (Bottom)	BSP	02	33.0	66	23.0	46	20.00
37	MODIFIED LINER, For Chock Assly of Bottom backup roll C/S Exit (L2) Inner/Barrel Side of 4 High Roughing/Finishing Stand, Drg: PM-17-012	BSP	01	140	140	91.0	91	49.00
38	MODIFIED LINER, For Chock Assly of Bottom backup roll D/S Entry (R1) Inner/Barrel Side of 4 High Roughing/Finishing Stand, Drg: PM-17-016	BSP	01	130	130	77.5	77.5	52.50
39	BUSH, Dia-280/ Dia-250 LG-95 Drg: BF-460-63	BSP	04	17	68	09	36	32.00
40	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
41	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
42	MODIFIED LINER, For Chock Assly of Bottom backup roll D/S Entry (R1) Outer Side of 4 High Roughing/Finishing Stand, Drg: PM-17-017	BSP	01	137	137	82.5	82.5	54.50
43	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
44	Modified Liner-3, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-108	BSP	01	127	127	81.1	81.1	45.90
45	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	04	165	660	160	640	20.00
46	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	04	145	580	130.1	520.4	59.60
47	Copper Tuyere, Drg: 5017165-0384407	BSL	02	301	602	287	574	28.00
48	Plate ArmourUpper Drg: NM-82-69 R2	BSP	01	955	955	915	915	40.00
49	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	02	475	950	435	870	80.00
50	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	09	115	1035	92.1	828.9	206.10
51	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	DSP	01	1780	1780	1500	1500	280.00
52	ISP Arbour Sleeve machining as per drawing	ISP	01	775.14	775.14	757.92	757.92	17.22
53	Support Block for Shear # 1 Chute Drg: 4.565554.T	ISP	01	120	120	57.7	57.7	62.30

54	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	04	390	1560	365	1460	100.00
55	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	01	475	475	435	435	40.00
56	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	02	115	230	92.1	184.2	45.80
57	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
58	Support Block for Shear # 1 Chute Drg: 4.565554.T	ISP	01	120	120	57.7	57.7	62.30
59	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	02	18.4	36.8	10.2	20.4	16.40
60	Axle Box CT Wheel Assly Drg: MSBSL-1 of Item No.02	VISL	01	40	40	20.2	20.2	19.80
61	Bearing Block attachment of Blind Cover Drg: 038.067.SM.013A/ Drg: 038.067.00.026 & 038.067.00.015 (2 Set)	BSL	01	200	200	139.7	139.7	60.30
62	Plate Liner Drg: 03601601200 Pos 1	BSL	03	20	60	18	54	6.00
63	Plate Liner Drg: 03601601200 Pos 2	BSL	03	24	72	22	66	6.00
65	Plate Liner Drg: 03601601200 Pos 3	BSL	03	24	72	22	66	6.00
66	Plate Liner Drg: 03601601200 Pos 4	BSL	03	24	72	22	66	6.00
67	Plate Liner Drg: 03601601200 Pos 5	BSL	03	21	63	19	57	6.00
68	Plate Liner Drg: 03601601200 Pos 6	BSL	03	21	63	19	57	6.00
69	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
70	Support Block for Shear # 1 Chute Drg: 4.565554.T	ISP	01	120	120	57.7	57.7	62.30
71	Plate Armour Middle Drg: NM-83-69 R1	BSP	01	981	981	920	920	61.00
72	Plate Armour Upper Drg: NM-82-69 R2	BSP	02	955	1910	915	1830	80.00
73	Plate II, Armour, Middle for Jaw Crusher, Drg: NM-30-69-R1	BSP	01	490	490	402	402	88.00
74	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	09	115	1035	92.1	828.9	206.10
75	HOUSING, BASE Item No:AB, B & C Drg: 04.06.002/155A	DSP	01	193.20	193.2	161.20	161.2	32.00
76	Machining of Forged round 60mm X 830L as per drawing.	SGW	17	25.6	435.2	17.0	289	146.20
77	10Ton Crane GC Shop	SGW	04	4.158	16.632	4.048	16.192	0.44
78	10Ton Crane GC Shop	SGW	04	1.69	6.76	1.510	6.04	0.72

79	Check Nut for Sinter Car Assly.	SGW	01	1.0	1	0.13	0.13	0.87
80	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	01	475	475	435	435	40.00
81	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D (Rejected due to Blown)	DSP	01	1780	1780	1750	1750	30.00
82	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	01	1600	1600	1300	1300	300.00
83	Hammer for Coal Crusher. Drg: 001.027.00.057	BSL	36	36	1296	21	756	540.00
84	BSL Pin as per Drg	F/O	04	11.0	44	5.5	22	22.00
85	Floating Shaft for Slag Car Drg: PD/R-208697	F/O	02	115.32	230.64	65.0	130	100.64
86	Floating Shaft for Slag Car Drg: PD/R-208696	F/O	02	226.63	453.26	133.0	266	187.26
87	Nut-Bolts for CT drive of 20 Ton DC Crane as per drwaing	SGW	20	0.27	5.4	0.14	2.8	2.60
88	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
89	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	04	1600	6400	1300	5200	1200.00
90	Bearing Block for Slag Car Drg: PD/R-208292 N-29 700/1	F/O	05	214.0	1070	134.6	673	397.00
91	Plate II, Armour, Middle for Jaw Crusher, Drg: NM-30-69-R1	BSP	01	490	490	402	402	88.00
92	SIDE FOR 80T BOGIE WITHOUT BRAKE SLAG LADLE CAR 16M3 Drg: CS-88-007-R1	BSP	02	1250.0	2500	1160.0	2320	180.00
93	Cap Plate 399MM for Transfer Car, Drg: BOF,MOC: IS:2062 A/B	ISP	08	54.4	435.12	19.4	155.2	279.92
94	Cap Plate 349MM for Transfer Car, Drg: BOF,MOC: IS:2062 A/B	ISP	04	32.5	130	12.1	48.4	81.60
95	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	05	22.8	114	13.8	69.05	44.95
96	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	05	120	600	65	325	275.00
97	Cu Plug for Tuyere & Cooler	M/C	32	0.85	27.2	0.25	8	19.20
98	MS Plug for all Tuyere	M/C	102	1.25	127.5	0.75	76.5	51.00
99	MS Cap 2" NPT for RSP Plate Cooler (5X5)	M/C	2	1.75	3.5	0.70	1.4	2.10
100	Fixture for Dummy Bar (620X360X80)	M/C	01	140.27	140.27	61.70	61.7	78.57
101	Fixture for Dummy Bar (600X380X80)	M/C	01	143.29	143.29	88.50	88.5	54.79
102	Fixture for Dummy Bar (350X300X50)	M/C	01	41.24	41.24	16.10	16.1	25.14
103	Fixture for Dummy Bar (560X70X50)	M/C	01	15.39	15.39	8.50	8.5	6.89

104	Fixture for Dummy Bar(100X60X45)	M/C	05	2.12	10.6	0.75	3.75	6.85
105	Fixture for Dummy Bar (350X60X80)	M/C	02	13.19	26.38	6.50	13	13.38
Grand Total					0		0	8086.04
					0		0	
CHECKED CALCULATION SHEET FOR THE MONTH OF JUNE-2024 (01.06.2024 TO 30.06.2024)					0		0	
1	Modified Liner-3, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-108	BSP	03	127	381	81.1	243.3	137.70
2	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	01	165	165	160	160	5.00
3	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
4	Copper Tuyere, Drg: 5017165-0384407	BSL	02	301	602	287	574	28.00
5	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
6	Copper Cooling Plate For Furnace Cooling System 520X550X780L, Drg:ISP-POSCO E&C-V-007-135-40-DE-14942	F/O	01	127	127	112.5	112.5	14.50
7	Copper Cooling Plate For Furnace Cooling System 520X550X850L, Drg:ISP-POSCO E&C-V-007-135-40-DE-14941	ISP	01	155	155	140.9	140.9	14.10
8	LEADING NUT (RIGHT HAND) ATTACHED TWO HALF NUT Drg:-B-603-179 R1 & B-603-180 R2	BSP	01	109	109	80.2	80.2	28.80
9	LEADING NUT (LEFT HAND) ATTACHED TWO HALF NUT Drg:-B-603-196 R1 & B-603-197 R2	BSP	01	109	109	80.2	80.2	28.80
10	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	06	145	870	130.1	780.6	89.40
11	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.00
12	Tuyere, Intermediate, Drg: 036.008.00.049C	BSL	08	181	1448	170	1360	88.00
13	Plate Cooler,Drg: RS-3.50.048,ROW No.:27 to 37	RSP	01	142	142	129.0	129	13.00
14	Copper Tuyere, Drg: 5017165-0384407	BSL	04	301	1204	287	1148	56.00

15	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	08	177	1416	160	1280	136.00
16	Tuyere, Intermediate, Drg: 036.008.00.049C	BSL	04	181	724	170	680	44.00
17	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	06	145	870	130.1	780.6	89.40
18	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	04	70	280	39.5	158	122.00
19	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3	RSP	02	259	518	187	374	144.00
20	MOD. BUSH for Shear Arms of Bar Mill Drg: MOD-2802	ISP	04	17.5	70	6.9	27.6	42.40
21	Skid Moving, CS DWI: B Upper Series along with 2 Nos of plates Item A. 1 No plate item F of Drawing 04.06.001.117A and 8 Nos of 3/4" BSW Bolts X 2-1/2" Long, All in Assembled Condition., Drg: 04.06.001/105 Drawing Item No: B	DSP	02	800	1600	700	1400	200.00
22	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	02	475	950	435	870	80.00
23	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
24	Hammer for Coal Crusher. Drg: 001.027.00.057	BSL	54	36	1944	21	1134	810.00
25	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	01	1600	1600	1300	1300	300.00
26	Plate II, Armour, Middle for Jaw Crusher, Drg: NM-30-69-R1	BSP	02	490	980	402	804	176.00
27	Lining Plate, 15 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00022 POSITION NO-15	ISP	01	60	60	59.7	59.7	0.30
28	Lining Plate, 16 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00023 POSITION NO-16	ISP	01	70	70	69.7	69.7	0.30
29	Lining Plate, 20 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00025 POSITION NO-20	ISP	01	65	65	64.7	64.7	0.30
30	Lining Plate, 27 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00029 POSITION NO-27	ISP	01	60	60	59.7	59.7	0.30
31	Lining Plate, 30 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00038 POSITION NO-30	ISP	01	82	82	80.8	80.8	1.20
32	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	02	1600	3200	1300	2600	600.00
33	Axle Box Housing for 80T Boggie Frame, Drg: 2069.07.25A/036.182.00.045A	RSP	03	140	420	72.8	218.4	201.60

34	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	02	115	230	92.1	184.2	45.80
35	Pig Mould, Drg:A1/02.17.06/120 60	DSP	26	105	2730	102.5	2665	65.00
36	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	DSP	02	580.0	1160	560.0	1120	40.00
37	BUSH (200X160X125) Drg: 62/104620A	.Shop/SGV	02	43.6	87.2	11.1	22.2	65.00
38	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	02	115	230	92.1	184.2	45.80
39	Pig Mould, Drg:A1/02.17.06/120 60	DSP	50	105	5250	102.5	5125	125.00
40	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	DSP	01	1800	1800	1500	1500	300.00
41	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	BSL	02	390	780	365	730	50.00
42	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	02	1600	3200	1300	2600	600.00
43	PLATE ARMOUR EXTREME FOR JAW CRUSHERDRG:-NM-28-69 R1	BSP	01	630	630	590	590	40.00
44	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	04	475	1900	435	1740	160.00
45	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	02	1600	3200	1300	2600	600.00
46	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	03	115	345	92.1	276.3	68.70
47	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	04	1600	6400	1300	5200	1200.00
48	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	01	115	115	92.1	92.1	22.90
49	Pig Mould, Drg:A1/02.17.06/120 60	DSP	38	105	3990	102.5	3895	95.00
50	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	02	475	950	435	870	80.00
51	Shutter Ledge Inter, 2 FT-6 IN Drg: A1/108.02.17/43784 Item NO.21	DSP	04	69.500	278	64.800	259.2	18.80
52	Shutter Ledge Inter, 3 FT-0 IN Drg: A1/108.02.17/43784 Item NO.22	DSP	04	86.500	346	80.900	323.6	22.40
53	Plate II, Armour, Extreme for Jaw Crusher, Drg: NM-31-69-R2	BSP	05	380	1900	340	1700	200.00
54	Cross Beam for 16M3 Slag Ladle Car Drg: CS-88-009- R1	BSP	02	1240	2480	1150	2300	180.00
55	Cu Plug for Tuyere & Cooler & Plate Cooler (ISP)	M/C	18	0.85	15.3	0.25	4.5	10.80
56	MS Plug for all Tuyere	M/C	142	1.25	177.5	0.75	106.5	71.00

57	MS Cap 2" NPT for RSP Plate Cooler (5X5)	M/C	4	1.75	7	0.70	2.8	4.20
Grand Total					0		0	7693.50
					0		0	
CHECKED: CALCULATION SHEET FOR THE MONTH OF JULY-2024 (01.07.2024 TO					0		0	
1	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
2	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	05	177	885	160	800	85.00
3	Tuyere, Intermediate, Drg: 036.008.00.049C	BSL	03	181	543	170	510	33.00
4	Cooler Monkey, Drg:036.005.00.028 B	BSL	01	84	84	71.06	71.06	12.94
5	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	03	145	435	130.1	390.3	44.70
6	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
7	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	100	1.0	100	0.1	10	90.00
8	BUSH CONE Drg:351.24.31.V/70982E	DSP	04	17.0	68	6.9	27.6	40.40
9	MOD. BUSH for Shear Arms of Bar Mill Drg: MOD-2802	ISP	04	17.50	70	6.9	27.6	42.40
10	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	18	11.0	198	2.60	46.8	151.20
11	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	03	177	531	160	480	51.00
12	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	01	315	315	298	298	17.00
13	Bearing Eccentric Bush for H-RAKE Drg: 339.71.30.7.13107	DSP	02	39.5	79	16.2	32.4	46.60
14	Bearing Eccentric Bush for V-RAKE Drg: 339.71.30.7.13106	DSP	06	37.0	222	17.5	105	117.00
19	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	02	11.0	22	2.60	5.2	16.80
20	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	02	11.0	22	2.60	5.2	16.80
21	SPINDLE SUPPORT 800X80X10 Item-1 Drg: MOD-2792	ISP	02	12.5	25	4.7	9.4	15.60
22	SPINDLE SUPPORT 600X70X10 Item-1 Drg: MOD-2159	ISP	08	9.25	74	3.3	26.4	47.60
23	SPINDLE SUPPORT 600X28X10 Item-2 Drg: MOD-2159	ISP	04	4.50	18	1.3	5.2	12.80
24	SLIPPER LINER SPINDLE SUPPORT, STAND 6 TO 10 Drg: MOD-1356	ISP	01	12.5	12.5	3.5	3.5	9.00
25	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31×70552 REV-1	DSP	04	315	1260	298	1192	68.00
26	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	11	145	1595	130.1	1431.1	163.90
27	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
28	Bush for Water Works as per drawing.	SGW	02	30.9	61.8	8.12	16.24	45.56

29	Bush for Carrier Wheel Assly Bottom half Drg: B 502-604 R1=1 NO.	BSP	06	20.5	123	8.70	52.2	70.80
30	Bush for Carrier Wheel Assly Top half Drg: B 502-605 R1=1 NO.	BSP	06	20.5	123	8.70	52.2	70.80
31	Bush, Bearing for Drive Side Drg:3.30.72.30.W/71297 Item No.2	DSP	03	53	159	35.8	107.4	51.60
32	BUSH CONE Drg:351.24.31.V/70982E	DSP	05	17.0	85	6.9	34.5	50.50
33	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	30	1.0	30	0.1	3	27.00
37	Copper Tuyere, Drg: 5017165-0384407	BSL	01	301	301	287	287	14.00
38	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	04	177	708	160	640	68.00
39	SPINDLE SUPPORT 800X80X10 Item-1 Drg: MOD-2792	ISP	09	12.5	112.5	4.7	42.3	70.20
40	SPINDLE SUPPORT 800X48X10 Item-2 Drg: MOD-2792	ISP	04	8.5	34	2.9	11.6	22.40
41	SLIPPER LINER SPINDLE SUPPORT, STAND 1 TO 5 Drg: MOD-1719	ISP	06	18.0	108	6.0	36	72.00
42	SPINDLE SUPPORT 700X80X10 Item-1 Drg: MOD-30527	ISP	05	10.0	50	4.8	24	26.00
43	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	02	11.0	22	2.60	5.2	16.80
44	Bush, Bearing for Drive Side Drg:3.30.72.30.W/71297 Item No.2	DSP	03	53	159	35.8	107.4	51.60
45	BUSH CONE Drg:351.24.31.V/70982E	DSP	05	17.0	85	6.9	34.5	50.50
46	Bush for Carrier Wheel Assly Bottom half Drg: B 502-604 R1=1 NO.	BSP	04	20.5	82	8.70	34.8	47.20
47	Bush for Carrier Wheel Assly Top half Drg: B 502-605 R1=1 NO.	BSP	04	20.5	82	8.70	34.8	47.20
48	Bush for OTC Travel Mechanism, Drg:B-001-107 R1 (TOP)	BSP	02	25.0	50	17.5	35	15.00
49	Bush for OTC Travel Mechanism, Drg:B-001-108 R1 (Bottom)	BSP	02	33.0	66	23.0	46	20.00
50	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.50
51	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair) (Reject due to Blown)	DSP	01	70	70	39.5	39.5	30.50
52	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	DSP	01	1800	1800	1500	1500	300.00
53	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	DSP	02	580.0	1160	560.0	1120	40.00
54	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	02	475	950	435	870	80.00
55	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	02	115	230	92.1	184.2	45.80
56	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	08	22.8	182.4	13.8	110.48	71.92
57	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	40	26.8	1072	20.4	816	256.00
58	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	02	1600	3200	1300	2600	600.00
59	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	04	1600	6400	1300	5200	1200.00

60	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	DSP	01	580.0	580	560.0	560	20.00
61	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	BSL	06	1600	9600	1300	7800	1800.00
62	Pig Mould Drg: 036.108.00.002 B	BSL	90	105	9450	102.5	9225	225.00
63	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	02	18.4	36.8	10.2	20.4	16.40
64	Block for structure shop as per drawing	structure/SG	04	115.01	460.04	89.01	356.04	104.00
65	Sinter Frame Shaft and shrink fitting	DSP	3	39.70	119.1	10.1	30.3	88.80
	Sinter Frame Shaft check Nut	DSP	1	1.97	1.97	0.81	0.81	1.16
66	Steel Teeming Ladle car	RSP	32	38.2	1222.4	17.4	556.8	665.60
67	Blind Cover as per Drg: 038.067.SM.013A	BSL	02	100.1	200.2	33.5	67	133.20
68	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	03	115	345	92.1	276.3	68.70
69	Skid Stationary, Drg:04.06.001/106 AB	DSP	02	570	1140	480	960	180.00
70	HOUSING, BASE Item No:AB, B & C Drg: 04.06.002/155A	DSP	01	193.20	193.2	161.20	161.2	32.00
71	SIDE FOR 80T BOGIE WITHOUT BRAKE SLAG LADLE CAR 16M3 Drg: CS-88-007-R1	BSP	02	1250.0	2500	1160.0	2320	180.00
74	Cu Plug for Tuyere & Cooler	M/C	10	0.85	8.5	0.25	2.5	6.00
75	MS Plug for all Tuyere	M/C	100	1.25	125	0.75	75	50.00
Grand Total					0		0	8202.98
					0		0	
CHECKED: CALCULATION SHEET FOR THE MONTH OF AUGUST-2024 (01.08.2024 TO 31.08.2024)					0		0	
1	Copper Tuyere, Drg: 5017165-0384407	BSL	02	301	602	287	574	28.00
2	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	1130/40/B	01	177	177	160	160	17.00
3	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	1167/10/D	02	170	340	146	292	48.00
4	TUYERE 140X406 mm Drg:PD/R263969 & 263970 (Reject due to leakage)	1148/10/R	01	190	190	184.7	184.7	5.30
5	Slipper Pad for PM Motor Side /R-13747 R3 (2 Pair)	1145/30/R	04	330	1320	246	984	336.00

6	Bush, Bearing for Drive Side Drg:3.30.72.30.W/71297 Item No.2	1165/10/D	04	53	212	35.8	143.2	68.80
7	BUSH CONE Drg:351.24.31.V/70982E	1163/10/D	02	17.0	34	6.9	13.8	20.20
8	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75571, Item No. 1 & 2	1169/30/D	10	5.0	50	3.5	35	15.00
9	Bush for Mould Foot Roll Drg: 540.04.07.V/77934	1169/20/D	32	2.0	64	1.0	32	32.00
10	SPINDLE SUPPORT 800X80X10 Item-1 Drg: MOD-2792	11147/2/IS	01	12.5	12.5	4.7	4.7	7.80
11	SPINDLE SUPPORT 800X48X10 Item-2 Drg: MOD-2792	11147/3/IS	07	8.5	59.5	2.9	20.3	39.20
12	TUYERE 140X406 mm Drg:PD/R263969 & 263970 (Reject due to leakage)	1148/10/R	01	190	190	184.1	184.1	5.90
13	BUSH CONE Drg:351.24.31.V/70982E	1163/10/D	07	17.0	119	6.9	48.3	70.70
14	Bush for Mould Foot Roll Drg: 540.04.07.V/77934	1169/20/D	28	2.0	56	1.0	28	28.00
19	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3	1145/20/R	02	259	518	187	374	144.00
20	Bush Epicycle Gear 120/75X75mm Drg:PD/R-202589 R2, Part-1	1161/30/R	10	06	60	2.1	21	39.00
21	Bush Epicycle Gear 130/105X100mm Drg:PD/R-202583 R2, Part-26	1161/40/R	05	11	52.5	4.5	22.5	30.00
22	Bush 105/80.8X130 mm Drg: PD/R-40705 R8	1161/20/R	08	10.5	84	4.1	32.8	51.20
23	Bush 82/60X130 mm Drg: PD/R-40705 R8	1161/10/R	10	06	60	2.4	24	36.00
24	Bush Epicycle Gear 270/255X60 mm Drg:PD/R-202583 R2, Part-27	1161/50/R	09	11.0	99	4.3	38.7	60.30
25	Slipper Pad, Square Block Drg: CDB/R-18395 POS-1	1145/40/R	04	132	528	115	460	68.00
26	BUSH HALF FOR UNIVERSAL SPINDKLE DIA 150 MM /PMM-940R1	1119/20/B	10	5.5	55	3.0	30	25.00
27	Chill for RSP/BSL Double Chamber as per drawing	NFF	01	12.6	12.6	8.6	8.6	4.00
28	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	1130/40/B	06	177	1062	160	960	102.00
29	Modified Cast Tuyere Assly, Drg: BF-2011-013	1132/10/B	06	145	870	130.1	780.6	89.40

30	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3	1145/20/R	02	259	518	187	374	144.00
31	Bush, for OTC Consisting of Bottom Insert Drg- B 001-110 R1	1140/20/B	10	3.5	35	1.3	13	22.00
32	Bush, for OTC Consisting of Upper Insert Drg- B 001-111 R1	1140/20/B	10	3.5	35	1.3	13	22.00
33	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	1169/10/D	110	1.0	110	0.1	11	99.00
37	Chill for BSL Double Chamber as per drawing	NFF	02	8.0	16	5.3	10.6	5.40
38	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair) (Reject due to blown)	1155/10/D	04	70	280	39.5	158	122.00
39	Bush upper and lower half man brg of sinter m/c of SP no-1, Drg-A3/10239019/90273	1153/10/D	02	275	550	141.8	283.6	266.40
40	Axle Box Housing for 80T Boggie Frame, Drg: 2069.07.25A/036.182.00.045A	F/O/RSP	08	140	1120	72.8	582.4	537.60
41	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	8925/10DS	02	115	230	92.1	184.2	45.80
42	FRAME SIDE LEFT DRG:-SPM.025.001.00.260	8830/10/BS	02	535.0	1070	391.5	783	287.00
43	FRAME SIDE RIGHT DRG:-SPM.025.001.00.261	8830/20/BS	01	535.0	535	391.5	391.5	143.50
44	Pig Mould Drg: 036.108.00.002 B	8935/10/BS	38	105	3990	102.5	3895	95.00
45	Bearing Block for Slag Car Drg: PD/R-208292 N-29 700/1	F/O	05	214.0	1070	134.6	673	397.00
46	Bearing Block attachment of Blind Cover Drg: 038.067.SM.013A/ Drg: 038.067.00.026 & 038.067.00.015 (TOP & BOTTOM)	F/O/BSL	04	200	800	139.7	558.8	241.20
47	Roll Liner for Coke Crusher, Drg: SPM 026.001.00.067B	81/00010/	01	1600	1600	1300	1300	300.00
48	Skid Stationary, CS,PN:106347-AA#/Drg: 004.06.001.0000.107# DWN Item No: AA"	8903/20/DS	02	560	1120	465	930	190.00
49	FRAME SIDE LEFT DRG:-SPM.025.001.00.260 (Side gauge slotting done)	8830/10/BS	02	535.0	1070	533.8	1067.66	2.34
50	FRAME SIDE RIGHT DRG:-SPM.025.001.00.261 (Side gauge welding & grinding done)	8830/20/BS	02	535.0	1070	533.8	1067.66	2.34

51	FRAME SIDE LEFT DRG:-SPM.025.001.00.260	830/10/B	01	535.0	535	391.5	391.5	143.50
52	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	931/10/D	01	475	475	435	435	40.00
53	Blind Cover as per Drg: 038.067.SM.013A	F/O	02	100.1	200.2	33.5	67	133.20
54	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	052/10/R	02	18.4	36.8	10.2	20.4	16.40
55	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	Fab	12	26.8	321.6	20.4	244.8	76.80
56	Cap Plate 349MM for Transfer Car, Drg: BOF,MOC: IS:2062 A/B	3939/3/IS	04	32.5	130	12.1	48.4	81.60
57	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	3883/7/BS	02	120	240	65	130	110.00
58	FRAME SIDE LEFT DRG:-SPM.025.001.00.260 (Side gauge slotting done by TAMKAN)	830/10/B	01	535.0	535	531.0	531	4.00
59	FRAME SIDE RIGHT DRG:-SPM.025.001.00.261 (Side gauge welding & grinding by TAMKAN))	830/20/B	02	535.0	1070	531.0	1062	8.00
61	Pig Mould Drg: 036.108.00.002 B	935/10/B	28	105	2940	102.5	2870	70.00
62	Axle Box Housing for 80T Boggie Frame, Drg: 2069.07.25A/036.182.00.045A	F/O/RSP	05	140	700	122.4	612	88.00
63	Bearing Block for Slag Car Drg: PD/R-208292 N-29 700/1	F/O	05	214.0	1070	134.6	673	397.00
64	Transfer Car Wheel Drg: 412.1409.03.0003	915/10/B	02	250.0	500	170.0	340	160.00
65	Crane Wheel G.C Shop as per drwaing	C.Shop/SC	02	200	400	138.2	276.4	123.60
66	Crane Wheel G.C Shop as per drwaing	C.Shop/SC	02	200	400	130.1	260.2	139.80
67	40 Ton Crane Coupling of GC Shop as per drawing	C.SHOP/SC	01	110.1	110.1	28.7	28.7	81.40
68	20 Ton Crane Coupling of GC Shop as per drawing	C.SHOP/SC	01	37.5	37.5	9.0	9	28.50
69	Nut for Crop Bucket as per drawing	ASP	04	12.6	50.4	2.4	9.6	40.80
70	Bolt for Crop Bucket as per drawing	ASP	04	49.3	197.2	11.6	46.4	150.80
71	Boss for Crop Bucket as per drawing	ASP	04	52.2	208.76	10.8	43.2	165.56
72	TRANSFER CAR WHEEL OF CRM III DRG:- 412.1409.03..0003	915/10/B	06	250	1500	170	1020	480.00
73	Sinter Frame Shaft as per drawing	DSP	01	39.15	39.15	10.1	10.1	29.05
74	Cu Plug for Tuyere & Cooler	M/C	25	0.85	21.25	0.25	6.25	15.00

75	MS Plug for all Tuyere	M/C	65	1.25	81.25	0.75	48.75	32.50
Grand Total					0		0	6908.89
					0		0	
CHECKED: CALCULATION SHEET FOR THE MONTH OF SEPTEMBER-2024 (01.09.2024 TO 30.09.2024)					0		0	
1	Bush Epicycle Gear 120/75X75mm Drg:PD/R-202589 R2, Part-1	RSP	15	06	90	2.1	31.5	58.50
2	Bush Epicycle Gear 130/105X100mm Drg:PD/R-202583 R2, Part-26	RSP	10	11	105	4.5	45	60.00
3	Bush 105/80.8X130 mm Drg: PD/R-40705 R8	RSP	06	10.5	63	4.1	24.6	38.40
4	Bush 82/60X130 mm Drg: PD/R-40705 R8	RSP	21	06	126	2.4	50.4	75.60
5	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	DSP	09	5.0	45	3.5	31.5	13.50
6	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	DSP	09	170	1530	146	1314	216.00
7	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	03	165	495	160	480	15.00
8	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	06	165	990	160	960	30.00
9	Bushing for foot roll assembly of Size 150mm×150mm Drg: ISP-DES-VI-029-001-55-DE-00016	ISP	140	2.0	280	0.40	56	224.00
10	Copper Tuyere, Drg: 5017165-0384407	BSL	03	301	903	287	861	42.00
11	Slipper Pad for PM Motor Side /R-13747 R3 (2 Pair)	RSP	04	330	1320	246	984	336.00
12	BUSH CONE Drg:351.24.31.V/70982E	DSP	09	17.0	153	6.9	62.1	90.90
13	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	DSP	10	5.0	50	3.5	35	15.00
14	SPINDLE SUPPORT 800X40X10 Item-2 Drg: MOD-2792	ISP	12	8.0	96	2.4	28.8	67.20
15	SPINDLE SUPPORT 800X48X10 Item-2 Drg: MOD-2792	ISP	01	8.5	8.5	2.9	2.9	5.60
16	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	04	11.0	44	2.60	10.4	33.60
17	BUSH, Stand for V-RAKE Drg: 339.71.30.7.13111 (In Set)	DSP	04	28	112	12.0	48	64.00
18	Modified Liner-1, For Top Backup Roll (Exit Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-106	BSP	01	160	160	94.10	94.1	65.90
19	Modified Liner-4, For Top Backup Roll (Entry Side) of 4 High Roughing/Finishing Stand, Drg: PM-22-109	BSP	01	127	127	81.10	81.1	45.90
20	Brass Spindle 8" Valve as per drawing	SGW	01	10.4	10.4	6.4	6.4	4.00

21	Brass Spindle 8" Valve as per drawing	SGW	01	7.2	7.2	4.2	4.2	3.00
22	Brass Nut 8" Valve as per drawing	SGW	08	1.6	12.8	0.7	5.6	7.20
23	DC Motor crane of machine shop no.02	SGW	02	3.9	7.8	1.1	2.2	5.60
24	Bearing Block for Slag Car Drg: PD/R-208292 N-29 700/1	F/O	03	214.0	642	134.6	403.8	238.20
25	Skid Stationary, CS,PN:106347-AA#/Drg: 004.06.001.0000.107# DWN Item No: AA"	DSP	01	560	560	465	465	95.00
26	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	04	26.8	107.2	20.4	81.6	25.60
27	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	05	475	2375	435	2175	200.00
28	Skid Moving, CS DWI: B Upper Series along with 2 Nos of plates Item A. 1 No plate item F of Drawing 04.06.001.117A and 8 Nos of 3/4" BSW Bolts X 2-1/2" Long, All in Assembled Condition., Drg: 04.06.001/105 Drawing Item No: B	DSP	01	800	800	700	700	100.00
29	Skid Stationary,Drg: 004.06.001.106# DWN Item No: AB"	DSP	01	570	570	465	465	105.00
30	HOUSING, BASE Item No:AB, B & C Drg: 04.06.002/155A	DSP	01	193.2	193.2	161.2	161.2	32.00
31	Axle Box for LT Wheel Assly Drg:MSBSL-107 Item No.-02	VISL	01	130	130	70.2	70.2	59.80
32	Axle Box for CT Wheel Assly Drg:MSBSL-107 Item No.-02	VISL	01	40	40	20	20.2	19.80
33	Sprocket / Crusher tooth Item No 02 of Single Roll crusher of Sinter Plant#2 Drg:3.29.03.31.W/71114C.	DSP	01	475	475	435	435	40.00
34	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	05	115	575	92.1	460.5	114.50
35	Blind Cover as per Drg: 038.067.SM.013A	F/O	02	100.1	200.2	33.5	67	133.20
36	MS Spindle 6" Valve as per drawing	SGW	03	4.4	13.2	2.6	7.8	5.40
37	MS Spindle 6" Valve as per drawing	SGW	03	5.2	15.6	3.7	11.1	4.50
38	Adapter with 10" hollow pipe as per dimensons for RSP DC	SGW	02	4.1	8.2	3.5	7	1.20
Grand Total					0		0	2691.10
					0		0	
CHECKED: CALCULATION SHEET FOR THE MONTH OF OCTOBER-2024 (01.10.2024 TO 31.10.2024)					0		0	
1	Copper Tuyere, Drg: 5017165-0384407	BSL	05	301	1505	287	1435	70.0
2	Tuyere, Intermediate, Drg: 036.008.00.049C	BSL	06	181	1086	170	1020	66.0
3	Cooler Monkey, Drg:036.005.00.028 B	BSL	01	84	84	71.06	71.06	12.9

4	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3	RSP	02	259	518	187	374	144.0
5	Bush, Drg:02.16.04/12D F/Wrist Pin Item No: K	DSP	20	13	260	6.3	126	134.0
6	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	01	11.0	11	2.60	2.6	8.4
7	SPINDLE SUPPORT 800X80X10 Item-1 Drg: MOD-2792	ISP	01	12.5	12.5	4.7	4.7	7.8
8	SLIPPER LINER SPINDLE SUPPORT, STAND 6 TO 10 Drg: MOD-1356	ISP	01	12.5	12.5	3.5	3.5	9.0
9	SPINDLE SUPPORT 700X80X10 Item-1 Drg: MOD-30527	ISP	02	10.0	20	4.8	9.6	10.4
10	SPINDLE SUPPORT 800X48X10 Item-2 Drg: MOD-2792	ISP	07	8.5	59.5	2.9	20.3	39.2
11	Copper Tuyere, Drg: 5017165-0384407	BSL	05	301	1505	287	1435	70.0
12	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	BSL	02	177	354	160	320	34.0
13	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3	RSP	02	259	518	187	374	144.0
14	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.0
15	Bush, Drg:02.16.04/12D F/Wrist Pin Item No: K	DSP	20	13	260	6.3	126	134.0
16	Bush Epicycle Gear 120/75X75mm Drg:PD/R-202589 R2, Part-1	RSP	10	06	60	2.1	21	39.0
17	SLIPPER LINER, F/SPINDLE SUPPORT FOR SPINDLE SUPPORT OF STAND 1 TO 5, DRG: MOD-1719 (L TYPE)	ISP	14	18.5	259	6.00	84	175.0
18	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	17	11.0	187	2.60	44.2	142.8
19	SLIPPER LINER SPINDLE SUPPORT, STAND 6 TO 10 Drg: MOD-1356	ISP	08	12.5	100	3.5	28	72.0
20	SPINDLE SUPPORT 700X80X10 Item-1 Drg: MOD-30527	ISP	01	10.0	10	4.8	4.8	5.2
21	Spindle Support Base Liner 700X38X10 Drg: MOD-30529	ISP	08	6.0	48	1.30	10.4	37.6

22	ALUMINIUM MATCH PLATE FOR STEEL FOUNDRY MACHINE MOULDING, DRG: AS PER REFERENCE DRAWING	7/PATTE	02	140	280	126.4	252.8	27.2
23	Skid Moving, CS DWI: B Upper Series along with 2 Nos of plates Item A. 1 No plate item F of Drawing 04.06.001.117A and 8 Nos of 3/4" BSW Bolts X 2-1/2" Long, All in Assembled Condition., Drg: 04.06.001/105 Drawing Item No: B	DSP	01	800	800	700	700	100.0
24	Skid Moving,CS DWI: A lower Series along with 2 Nos of Plates Item B. 1 Nos plate item F of Drg 04.06.001.117A and 10 Nos of 3/4" BSW Bolts X2 - 1/2" Long, All in Assembled Condition./ Drg- 04.06.001/105 DWN Item No: A"	DSP	01	560	560	450	450	110.0
25	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	05	115	575	92.1	460.5	114.5
26	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	DSP	5	580.0	2900	560.0	2800	100.0
27	PLATE ARMOUR MIDDLE FOR JAW CRUSHERDRG:-NM-29-69 R1	BSP	02	950	1900	900	1800	100.0
28	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 12 DISC ROLL, Drg: CSP-2011-09	BSP	12	80	960	60	720	240.0
29	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 13 SHAFT, Drg: C-2010-039 R-0	BSP	02	286	572	208	416	156.0
30	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 14 PARALLEL KEY 36x20x92	BSP	12	01	12	0.4	4.8	7.2
31	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 18 END COVER, Drg: C-2010-042.3 R1	BSP	02	07.2	14.4	02.8	5.6	8.8
32	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 19 STOPPER, Drg: C-2010-042.5	BSP	02	0.21	0.42	0.18	0.36	0.1
33	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 20 LOCK NUT M135x2, Drg: IS: 6731-77	BSP	04	05.4	21.6	01.4	5.6	16.0

34	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 22 PARALLEL KEY 36X20X150, Drg: IS: 2048-83	BSP	02	01.5	3	0.8	1.6	1.4
35	LABYRYNTH RING: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 4 LABYRINTH, Drg: C-2010-042.8 R1	BSP	06	6.0	36	1.0	6	30.0
36	END COVER: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 18 END COVER, Drg: C-2010-042.5 R1	BSP	06	7.2	43.2	2.1	12.6	30.6
37	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.7	BSP	04	27.7	110.8	7.8	31.2	79.6
38	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.4	BSP	02	12.6	25.2	5.4	10.8	14.4
39	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.6	BSP	02	5.0	10	2.2	4.4	5.6
Grand Total					0		0	2557.70
					0		0	
CHECKED. CALCULATION SHEET FOR THE MONTH OF NOVEMBER-2024 (01.11.2024 TO 30.11.2024)					0		0	
1	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3	RSP	02	259	518	187	374	144.0
2	Bush Epicycle Gear 120/75X75mm Drg:PD/R-202589 R2, Part-1	RSP	13	06	78	2.1	27.3	50.7
3	Bush Epicycle Gear 130/105X100mm Drg:PD/R-202583 R2, Part-26	RSP	16	11	168	4.5	72	96.0
4	Bush Epicycle Gear 130/105X100mm Drg:PD/R-202583 R2, Part-26 (REJECTED DUE TO FDY DEFECT)	RSP	1	11	10.5	4.5	4.5	6.0
5	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	01	70	70	39.5	39.5	30.5
6	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.30.72.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.0
7	Bush for Shaft W/Lever of Unloader Mechanism Drg:540.01.1.W/85463 (In Set)	DSP	03	20	60	7.0	21	39.0

8	Rope Guide Bush for OBC Grab Bucket Drg:BF-97-020	BSP	01	41	41	30.8	30.8	10.2
9	Mod Wearing Plates Drg: PM-18-104 Serial-1	BSP	01	202	202	132.9	132.9	69.1
10	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1	BSP	01	160	160	117	117	43.0
11	SLIPPER LINER, F/SPINDLE SUPPORT FOR SPINDLE SUPPORT OF STAND 1 TO 5, DRG: MOD-1719 (L TYPE)	ISP	06	18.5	111	6.00	36	75.0
12	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	ISP	02	11.0	22	2.60	5.2	16.8
13	Modified Cast Tuyere Assly, Drg: BF-2011-013	BSP	02	145	290	130.1	260.2	29.8
14	Modified Tuyere Cooler, Drg: BF-2011-012	BSP	04	165	660	160	640	20.0
15	Bush Drg: 04.06.006/247 B Item No. C	DSP	04	46	184	13	52	132.0
16	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.0
17	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.30.72.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.0
18	BEARING ECENTRIC BUSH FOR H RAKE/DRG:339.71.30.7.13107	DSP	01	37	37	17.5	17.5	19.5
19	Bearing Eccentric Bush for V-RAKE, Drg: 339.71.30.7.13106	RSP	03	39.5	118.5	16.2	48.6	69.9
20	Bush Epicycle Gear 270/255X60 mm Drg:PD/R-202583 R2, Part-27	RSP	07	11.0	77	4.3	30.1	46.9
21	SLIPPER PAD FOR PM ROLL SIDE / DRG;-CDB/R-18397 R3 (In Set)	RSP	02	259	518	187	374	144.0
22	Mod Wearing Plates Drg: PM-18-104 Serial-1	BSP	01	202	202	132.9	132.9	69.1
23	Bush for Elevator Counter Weight Drg: B-210102 R1	BSP	10	3.0	30	1.4	14	16.0
24	BUSH for PMTM of OTC One Set Drg: B 001-107 R1=1 NO.	BSP	01	25.0	25	14.5	14.5	10.5
25	BUSH for PMTM of OTC One Set Drg: B-001-108 R1=1 No.	BSP	01	33.0	33	19.0	19	14.0
26	Bushing for foot roll assembly of Size 150mm×150mm Drg: ISP-DES-VI-029-001-55-DE-00016	ISP	94	2.0	188	0.40	37.6	150.4
27	Skid Stationary, CS,PN:106347-AA#/Drg: 004.06.001.0000.107# DWN Item No: AA"	DSP	01	560	560	465	465	95.0
28	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	04	115	460	92.1	368.4	91.6

29	Side Bearer 30mm for 100 Ton Platform Car, Drg: As per reference drawing	F/O	20	26.8	536	20.4	408	128.0
30	Side Bearer 20mm for 100 Ton Platform Car, Drg: As per reference drawing	F/O	08	22.8	182.4	13.8	110.48	71.9
31	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	F/O	14	9.3	130.2	6.2	86.8	43.4
32	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	03	120	360	65	195	165.0
33	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	04	18.4	73.6	10.2	40.8	32.8
34	PLATE ARMOUR MIDDLE FOR JAW CRUSHERDRG:-NM-28-69 R1	BSP	04	630	2520	590	2360	160.0
35	Head, 250 X L1800 mm, for Dummy Bar of CCS Drg: CC-422-76 SLNo.-1	BSP	01	1200	1200	1000	1000	200.0
36	Dummy Bar, Head Size: 200X1280 mm Drg: CC-422-035 R2	BSP	01	660	660	615	615	45.0
37	Bearing Block For Transfer Car Drg: MOC; GR-40CR4M03, IS:4367-1991 BOF	ISP	03	250	750	153.7	461.1	288.9
38	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	05	115	575	92.1	460.5	114.5
39	BSL Pin as per Drg	BSL	06	11.0	66	5.5	33	33.0
40	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	BSL	03	120	360	65	195	165.0
41	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	RSP	03	113	339	89.8	269.4	69.6
42	Car, Platform, 100T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	RSP	08	18.4	147.2	10.2	81.6	65.6
43	Lower Grinding Ring Drg: CDB/R-80103	RSP	01	2080	2080	1870	1870	210.0
44	Head, 250 X L1800 mm, for Dummy Bar of CCS Drg: CC-422-76 SLNo.-1	BSP	01	1200	1200	1000	1000	200.0
45	Dummy Bar, Head Size: 200X1280 mm Drg: CC-422-035 R2	BSP	01	660	660	615	615	45.0
46	LOWER ARMOUR CONE CRUSHER DRG:-MI-9-36(CC/82 R-1	BSP	01	2680	2680	2080	2080	600.0
47	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 12 DISC ROLL, Drg: CSP-2011-09	BSP	12	80	960	60	720	240.0
48	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 13 SHAFT, Drg: C-2010-039 R-0	BSP	02	286	572	208	416	156.0

49	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 14 PARALLEL KEY 36x20x92	BSP	12	01	12	0.4	4.8	7.2
50	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 18 END COVER, Drg: C-2010-042.3 R1	BSP	02	07.2	14.4	02.8	5.6	8.8
51	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 19 STOPPER, Drg: C-2010-042.5	BSP	02	0.21	0.42	0.18	0.36	0.1
52	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 20 LOCK NUT M135x2, Drg: IS: 6731-77	BSP	04	05.4	21.6	01.4	5.6	16.0
53	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 22 PARALLEL KEY 36X20X150, Drg: IS: 2048-83	BSP	02	01.5	3	0.8	1.6	1.4
54	LABYRYNTH RING: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 4 LABYRINTH, Drg: C-2010-042.8 R1	BSP	06	6.0	36	1.0	6	30.0
55	END COVER: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 18 END COVER, Drg: C-2010-042.5 R1	BSP	06	7.2	43.2	2.1	12.6	30.6
56	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.7	BSP	04	27.7	110.8	7.8	31.2	79.6
57	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.4	BSP	02	12.6	25.2	5.4	10.8	14.4
58	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.6	BSP	02	5.0	10	2.2	4.4	5.6
59	Upper Grinding Ring Drg: CDB/R-80103	RSP	01	910	910	825	825	85.0
60	Reversible Frame for Sinter Car Drg:329.03.31.Y/77714- Rectification in Grate Bar Grooves	DSP	09	184.0	1656	166.0	1494	162.0
61	GUIDE as per drg.	ISP	16	560	8960	547.2	8755.2	204.8
Grand Total					0		0	5351.18
					0		0	
CHECKED: CALCULATION SHEET FOR THE MONTH OF DECEMBER-2024 (01.12.2024 TO 31.12.2024)					0		0	
1	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	DSP	05	170	850	146	730	120.0
2	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31x70552 REV-1	DSP	10	315	3150	298	2980	170.0

3	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	DSP	35	5.0	175	3.5	122.5	52.5
4	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	03	70	210	39.5	118.5	91.5
5	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	DSP	02	70	140	39.5	79	61.0
6	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1	BSP	06	160	960	117	702	258.0
7	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1 Reject due to blown hole	BSP	01	160	160	117	117	43.0
8	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	DSP	300	1.0	300	0.1	30	270.0
9	BUSH PB DRG: A1/04.06.004/41980 C Item No.5	DSP	30	9.5	285	4.2	126	159.0
10	Bush for Elevator Counter weight Drg: B-210102 R1	BSP	10	3.0	30	1.4	14	16.0
11	Machined Gun Metal Bush for 500 mm dia wheel 75x105x320	VISL	02	20.5	41	10.7	21.4	19.6
12	Machined Gun Metal Bush for 600 mm dia wheel 85x115x320	VISL	05	23.0	115	11.9	59.5	55.5
13	Skid Stationary, CS,PN:106347-AA#/Drg: 004.06.001.0000.107# DWN Item No: AA"	DSP	01	560	560	465	465	95.0
14	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	DSP	05	115	575	92.1	460.5	114.5
15	Skid Moving,CS DWI: A lower Series along with 2 Nos of Plates Item B. 1 Nos plate item F of Drg 04.06.001.117A and 10 Nos of ¾" BSW Bolts X2 - ½" Long, All in Assembled Condition./ Drg-04.06.001/105 DWN Item No: A"	DSP	02	560	1120	450	900	220.0
16	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	DSP	02	580.0	1160	560.0	1120	40.0
17	Pig Mould Drg: 036.108.00.002 B	DSP	96	105	10080	102.5	9840	240.0
18	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	DSP	03	1800	5400	1500	4500	900.0
19	Bearing Block For Transfer Car Drg: MOC; GR-40CR4M03, IS:4367-1991 BOF	ISP	05	250	1250	153.7	768.5	481.5
20	Dummy Bar, Head Size: 200X1280 mm Drg: CC-422-035 R2	BSP	01	660	660	615	615	45.0

21	Head, 250 X L1800 mm, for Dummy Bar of CCS Drg: CC-422-76 SLNo.-1	BSP	01	1200	1200	1000	1000	200.0
22	Lower Grinding Ring Drg: CDB/R-80103	RSP	01	2080	2080	1870	1870	210.0
23	Jaw Swing Plate Drg:25259	GUA	03	780	2340	600	1800	540.0
24	Sleeve for Tuinion as per drawing	W/O	04	122	488	66.7	266.8	221.2
25	Ring for Tuinion as per drawing	W/O	08	112	896	85	680	216.0
Grand Total					0		0	4402.10
					0		0	
CHECKED : CALCULATION SHEET FOR THE MONTH OF JANUARY-2025 (01.01.2025 TO 31.01.2025)					0		0	
1	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1 (Rejected due to Blown Hole)	1117/40/B	01	160	160	100.4	100.4	59.6
2	Bush for Mould Foot Roll Bar/ Drg: 540.01.017.V/77934 (Rejected due to blown hole)	1181/50/D	12	14	168	10	124.8	43.2
3	Tuyere 140X406 Drg: PD/R263969 & 263970	1148/10/R	02	199	398	184.7	369.4	28.6
4	PLATE COOLER, LMS, ROW: 38-40, RS-3.50.048 FOR BLAST FURNACE-1 LOWER MIDDLE STACK ZONE, ROW 38 TO 40, DRG: RS-3.50.048, ROW NO.: 38 TO 40	1185/10/R	05	142	710	129	645	65.0
5	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	1166/20/D	07	5.0	35	3.5	24.5	10.5
6	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	1181/40/D	240	1.0	240	0.1	24	216.0
7	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	1171/10/D	02	70	140	39.5	79	61.0
8	INSERT FOR COUNTER WEIGHT OF ELEVATOR DRG B- 210103R1	1168/20/B	10	2.0	20	1.0	10	10.0
9	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1	1117/40/B	05	160	800	117	585	215.0

10	WEARING PLATES FOR WORK ROLL WITH CHOCK ASSLY.OF FOUR HIGH STAND DRG; PMM-177 R2 THICK-30 MM	1117/10/B	01	202	202	133	133	69.0
11	Machined Gun Metal Bush for 500 mm dia wheel 75x105x320	1178/1/VI	01	20.5	20.5	10.7	10.7	9.8
12	Machined Gun Metal Bush for 600 mm dia wheel 85x115x320	1178/2/VI	02	23.0	46	11.9	23.8	22.2
	Machined Gun Metal Bush for 600 mm dia wheel 85x115x320 Reject due to blown	1178/2/VI	01	23.0	23	11.9	11.9	11.1
13	Machined Gun Metal Bush for 700 mm dia wheel 95x125x320	1178/3/VI	02	25.0	50	13	26	24.0
14	Machined Gun Metal Bush for 400 mm dia wheel 70x110x250	1178/5/VI	06	20.5	123	11.4	68.4	54.6
	Machined Gun Metal Bush for 400 mm dia wheel 70x110x250 Reject due to blown	1178/5/VI	01	20.5	20.5	11.4	11.4	9.1
15	Machined Gun Metal Bush for 170 mm dia wheel 40x65x170	1178/6/VI	01	6.5	6.5	2.8	2.8	3.7
16	Modified Tuyere Cooler, Drg: BF-2011-012	1170/10/B	02	165	330	160	320	10.0
17	Copper Tuyere, Drg: 5017165-0384407	50/00010/	05	301	1505	287	1435	70.0
18	Tuyere, Intermediate, Drg: 036.008.00.049C	30/00030/	05	181	905	170	850	55.0
19	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	1130/40/B	09	177	1593	160	1440	153.0
20	INSERT FOR ELEVATOR CABIN /DRG:B210104R1	1168/30/B	10	4.4	44	1.8	18	26.0
21	Wearing Plate Drg: A 3/10.05.01/20504A	1176/10/D	12	64.0	768	37.3	447.6	320.4
22	Piston Wearing Plate for Secondary Crusher Drg:IRON/OHP/CR/2/12/72	1177/20/R	02	78.5	157	56.0	112	45.0
23	Bush, for Unloader Shaft and Lateral Transfer Shaft, Drg: 4.38.01.21.W/81429 Halves	1181/30/D	04	9.2	36.8	3.2	12.8	24.0
24	Bearing Eccentric Bush for H-Rake Drg: 339.71.30.7.13107	1169/60/D	04	37.0	148	17.5	70	78.0

25	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	1181/20/D	02	5.0	10	3.5	7	3.0
26	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	1182/10/D	02	170	340	146	292	48.0
27	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	F/O/BSL	01	177	177	160	160	17.0
28	PB Bush for S.Foundry 156X105X80 mm	W/O/SGW	08	14	112	8.33	66.64	45.4
29	PB Bush for S.Foundry 160X65X85 mm	W/O/SGW	08	19	148	9.7	77.6	70.4
30	Bush for Carrier Wheel Assly Bottom half Drg: B 502-604 R1=1 NO. Reject due to blown	1159/10/B	01	20.5	20.5	8.70	8.7	11.8
31	Bush for Carrier Wheel Assly Top half Drg: B 502-605 R1=1 NO. Reject due to Blown	1159/10/B	01	20.5	20.5	8.70	8.7	11.8
32	Bush Epicycle Gear 120/75X75mm Drg:PD/R-202589 R2, Part-1 Reject due to blown	1161/30/R	01	06	6	2.1	2.1	3.9
33	Nut for BLR#3-4 Drg:PD/R-355053	1180/10/R	14	5.3	73.5	2.1	29.4	44.1
34	Nut for BLR#3-4 Drg:PD/R-355053 Reject due to blown	1180/10/R	02	5.3	10.5	2.1	4.2	6.3
35	Upper Grinding Ring Drg: CDB/R-80103	951/20/R	01	910	910	825	825	85.0
36	Front Cover Drg:MI-95-015-3 R1	966/10/B	07	75.0	525	47.0	329	196.0
37	Skid Moving, Drg- 04.06.001/105 DWN Item No: A"	903/10/D	02	560	1120	450	900	220.0
38	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K	9963/2/DS	03	580.0	1740	560.0	1680	60.0
39	Wobbler Coupling Box, Drg: 04.04/-/227F Drawing Item No. K. Reject due to blown	9963/2/DS	01	580.0	580	560.0	560	20.0
40	Grate Bar Drg:329.10.30.V/75774	970/10/D	23	330	7590	320	7360	230.0
	Grate Bar Drg:329.10.30.V/75774 Reject due to blown	970/10/D	01	330	330	320	320	10.0
41	Bearing Block For Transfer Car Drg: MOC; GR-40CR4M03, IS:4367-1991 BOF	9939/10/IS	04	250	1000	153.7	614.8	385.2
42	Bearing Block For Transfer Car Drg: MOC; GR-40CR4M03, IS:4367-1991 BOF. Reject due to blown	9939/10/IS	01	250	250	153.7	153.7	96.3

43	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	3972/1/BS	08	390	3120	365	2920	200.0
44	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	05085	14	26.8	375.2	20.4	285.6	89.6
45	Adaptor Plate as per Drg	05098	02	252	504.2	162	323	181.2
46	Crane Wheel for Steel Foundry as per sample	SGW	08	223.8	1790.4	148.7	1189.6	600.8
47	Crane Wheel Repairing for Steel Foundry as per sample (Re Bushing)	SGW	7	194	1358	184	1288	70.0
48	Jaw Swing Plate Drg:25259	100.00%	03	780	2340	600	1800	540.0
49	Shutter Ledge Inter, 2 FT-6 IN Drg: A1/108.02.17/43784 Item NO.21	100.00%	04	69.5	278	64.8	259.2	18.8
50	Shutter Ledge Inter, 3 FT-0 IN Drg: A1/108.02.17/43784 Item NO.22	100.00%	03	86.5	259.5	80.9	242.7	16.8
51	Ladle Cover Pin as per Drawing	100.00%	09	65.05	585.45	48.56	437.04	148.4
52	Adaptor Plate as per Drg	100.00%	05	252	1260.5	162	807.5	453.0
Grand Total					0		0	5576.57
					0		0	
CHECKED CALCULATION SHEET FOR THE MONTH OF FEBRUARY-2025 (01.02.2025 TO 28.02.2025)					0		0	
1	Main Shaft Step Bearing for Secondary Crusher Drg:IRON/OHP/CR/2/13/72	PB	02	93.5	187	62.4	124.8	62.2
2	INSERT BUSH LOWER OF GUIDE BOX ON OUTPUT SHAFT MIXER SIDE /DRG:-3.30.54.30W./72041B	AB	02	42.0	84	21.4	42.8	41.2
3	INSERT BUSH UPPER OF GUIDE BOX ON OUTPUT SHAFT MIXER SIDE /DRG:-3.30.54.30W./72041	AB	02	42.0	84	21.6	43.2	40.8
4	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	HTB	05	70	350	39.5	197.5	152.5
5	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	HTB	04	70	280	39.5	158	122.0
6	Bush for Mould Foot Roll Drg: 5.40.01.07V/77934	AB	30	2.0	60	1.0	30	30.0
7	Bush Epicycle Gear 270/255X60 mm Drg:PD/R-202583 R2, Part-27	PB	03	11.0	33	4.3	12.9	20.1

8	WEARING PLATES FOR WORK ROLL WITH CHOCK ASSLY.OF FOUR HIGH STAND DRG; PMM-177 R2 THICK-30 MM	AB	04	202	808	133	532	276.0
9	BUSH DIA 280/250X 95 FOR MOV BODY DRG;BF 460-63	CU	02	17.0	34	8.4	16.8	17.2
10	SLIPPER LINER, F/SPINDLE SUPPORT FOR SPINDLE SUPPORT OF STAND 1 TO 5, DRG: MOD-1719 (L TYPE)	HTB	13	18.5	240.5	6.0	78	162.5
11	Slipper Liner, Spindle Support Stands 11 to 16 Drg: MOD-1355	HTB	02	11.0	22	2.6	5.2	16.8
12	Nut, Half, Modified in Pair L/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	HTB	03	70	210	39.5	118.5	91.5
13	Nut, Half, Modified in Pair R/H for Upper Screw of Slag Pot Tilting Car of BOF Part No.: BTS 703529, Drg: 3.3072.30V/70368A (In Pair)	HTB	02	70	140	39.5	79	61.0
14	Bush for Mould Foot Roll Drg: 5.40.01.07V/77934	AB	50	2.0	100	1.0	50	50.0
15	BUSH CONE, DRG: 351.24.31.V/70982E	AB	01	17.0	17	6.9	6.9	10.1
16	Bush Stand for V Rake, Drg- 339.71.30.7.13111	PB	04	24	96	9.2	36.8	59.2
17	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	CU	01	170	170	146	146	24.0
18	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	AB	12	5.0	60	3.5	42	18.0
19	Bush for Mould Foot Roll, Drg: 4.38.01.07.V/81119	AB	20	1.0	20	0.1	2	18.0
20	WEARING PLATES FOR WORK ROLL WITH CHOCK ASSLY.OF FOUR HIGH STAND DRG; PMM-177 R2 THICK-30 MM	AB	02	202	404	133	266	138.0
21	MODI WEARING PLATES FOR BOTTOM WORK ROLL WITH CHOCK ASSLY. OF FOUR HIGH STAND DRG:-PM-18-104 /SL-2	AB	01	210	210	157	157	53.0
22	Bush for Elevator Counter weight Drg: B-210102 R1	AB	13	3.0	39	1.4	18.2	20.8
23	Bush 82/60X130 mm Drg: PD/R-40705 R8	PB	09	06	54	2.4	21.6	32.4
24	Bush 105/80.8X130 mm Drg: PD/R-40705 R8	PB	01	10.5	10.5	4.1	4.1	6.4
25	Bush Epicycle Gear 130/105X100mm Drg:PD/R-202583 R2, Part-26	PB	01	11	10.5	4.5	4.5	6.0
26	ARM BUSH, PN: 16 8 M3 SIZE 7.5X60X90 MM/ DRG:PD/R-20217	AB	07	4.5	31.5	2.2	15.4	16.1
27	PULLY BUSH F/O 8 M3 BUCKET SIZE 90X105X77 MM /DRG:PD/R-42652	AB	04	4.5	18	1.8	7.2	10.8
28	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	ANE CARB	08	390	3120	365	2920	200.0
29	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	ANE CARB	03	120	360	65	195	165.0
30	Pig Mould Drg: 036.108.00.002 B	ANE CARB	96	105	10080	102.5	9840	240.0
31	BSL Pin as per Drg (Platform Car)	LLOY STEEL	02	11	22	5.5	11	11.0

32	SIDE PLATE FOR HAMMAR CRUSHER 1450X1300 /DRG:-CH-21-69R0	LLOY STEEL	01	150	150	140	140	10.0
33	PLATE FOR HAMMAR CRUSHER NO-4/DRG:CH-1-151 R1	LLOY STEEL	01	235	235	200	200	35.0
34	Striker for Hammer Crusher Drg: CH-1-157-R1	LLOY STEEL	01	580	580	460	460	120.0
35	RACK, STATIONARY, IS:1030, GRADE-26/52, Drg: 04.06.024.736A Item No.A	ANE CARB	01	115	115	92.1	92.1	22.9
36	Casting, Side Wall, Inner Drg: A1/05.01.10/58734 Item No.3"	ANE CARB	07	80	560	79	553	7.0
37	Plate, Sand Seal Dipper Casting Drg: 05.01.10/163C Item No.6"	ANE CARB	04	30	120	29	116	4.0
38	Lining Plate, 13 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00021 POSITION NO - 10	LLOY STEEL	01	80	80	79.7	79.7	0.3
39	Lining Plate, 13 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00021 POSITION NO - 13	LLOY STEEL	02	75	150	74.7	149.4	0.6
40	Lining Plate, 15 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00022 POSITION NO - 15	LLOY STEEL	01	60	60	59.7	59.7	0.3
41	Lining Plate, 16 for Coke Bucket of CDCP, Drg: ISP-DES-III-003-020-01-DE-00023 POSITION NO - 16	LLOY STEEL	01	70	70	69.7	69.7	0.3
42	Lining Plate, 12 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00020 POSITION NO - 12	LLOY STEEL	01	55	55	54.7	54.7	0.3
43	Lining Plate, 17 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00023 POSITION NO - 17	LLOY STEEL	01	70	70	69.7	69.7	0.3
44	Lining Plate, 26A for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00026A POSITION NO - 24	LLOY STEEL	01	55.0	55	54.7	54.7	0.3
45	Lining Plate, 26A for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00026A POSITION NO - 23	LLOY STEEL	01	55.0	55	54.7	54.7	0.3
46	Lining Plate, 25 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00027 POSITION NO - 25	LLOY STEEL	02	55.0	110	54.7	109.4	0.6
47	Lining Plate, 26 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00028 POSITION NO - 26 (with welding)	LLOY STEEL	02	95	190	94.7	189.4	0.6
48	Lining Plate, 27 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00029 POSITION NO - 27	LLOY STEEL	01	60	60	59.7	59.7	0.3
49	Lining Plate, 27 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00029 POSITION NO - 28	LLOY STEEL	01	45	45	44.7	44.7	0.3
50	Lining Plate, 29 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00037 POSTION NO - 29- 4 HOLES	LLOY STEEL	01	75	75	73.8	73.8	1.2
51	LINING PLATE ISP-DES-III -003-020-01-DE-00038/POS-30 -4 HOLES	LLOY STEEL	01	82	82	80.8	80.8	1.2

52	Lining Plate, 31 for Coke Bucket of CDCP/Drg: ISP-DES-III-003-020-01-DE-00039 POSTION NO - 31-4 HOLES	LLOY STEEL	01	78	78	76.8	76.8	1.2
53	Spindle for BLR#3-4 Drg: PD/R-355053	MS	09	16.7	150.3	8.6	77.4	72.9
54	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	ANE CARB	02	1780	3560	1500	3000	560.0
55	Pig Mould Drg: 036.108.00.002 B	ANE CARB	34	105	3570	102.5	3485	85.0
56	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	ANE CARB	02	390	780	365	730	50.0
57	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	ANE CARB	05	120	600	65	325	275.0
58	C.S Track Wheel for Door extendor no-1,2,3,4.	ANE CARB	01	240	240	184.2	184.2	55.8
59	C.S track wheel for door extraction No.1 and 4	ANE CARB	01	200	200	156.4	156.4	43.6
60	Plate, Sand Seal Dipper Casting Drg: 05.01.10/163C Item No.6"	ANE CARB	04	30	120	29	116	4.0
61	Jolting Table with PB Bush as per drawing	ANE CARB	01	552.8	552.8	492.8	492.8	60.0
62	Modified Axle Box Top Half Drg: 036.071.00.167	ANE CARB	02	370	740	236	472	268.0
63	Modified Axle Box Bottom Half Drg: 036.071.00.168	ANE CARB	02	420	840	316	632	208.0
64	Adaptor Plate as per Drg	MS	07	254.7	1782.9	181.5	1270.5	512.4
65	Side Bearer 50mm for 100 Ton Platform Car, Drg: As per reference drawing	MS	18	26.5	477	21.3	383.4	93.6
66	Side Bearer 30mm for 100 Ton Platform Car, Drg: As per reference drawing	MS	12	20.5	246	13.1	157.2	88.8
67	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	MS	12	9.3	111.6	6.2	74.4	37.2
68	Sleeve as per drawing (165T LADLE)	MS	04	2.1	8.4	0.9	3.6	4.8
70	ASSLY LAMINATED HOOK ITEM NO. 1, 1A, 1AX, DRG: 6386618600 REV B	MS	02	3460	6920	3398	6796	124.0
71	ASSLY LAMINATED HOOK, WEARING BUSH, ITEM NO.4, Drg:63866618600 Rev B	MS	02	101.6	203.2	47.1	94.2	109.0
72	Ladle Cover Pin as per Drawing (165T LADLE)	MS	01	138.75	138.75	48.56	48.56	90.2
Grand Total					0		0	5121.9
					0		0	
CHECKED CALCULATION SHEET FOR THE MONTH OF MARCH-2025 (01.03.2025 TO 31.03.2025)					0		0	
1	Bush for Mould Foot Roll Drg: 5.40.01.07V/77934	AB	08	2.0	16	1.0	8	8.0

2	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	CU	09	170	1530	146	1314	216.0
3	TUYERE COOLER MOD. CU BLAST DRG: 320.01.31x70552 REV-1	CU	02	315	630	298	596	34.0
4	Bush for Bloom Unloader Fitted with Adjusting screw, Hex. Nut, Lock Nut & 6 mm dia Spring Pin, Drg: 4.38.01.21.V/75 571, Item No. 1 & 2	AB	09	5.0	45	3.5	31.5	13.5
5	Bush for Elevator Counter weight Drg: B-210102 R1	AB	10	3.0	30	1.4	14	16.0
6	INSERT FOR ELEVATOR CABIN /DRG:B210104R1	AB	10	4.4	44	1.8	18	26.0
7	Nut for BLR#3-4 Drg:PD/R-355053	AB	02	5.3	10.5	2.1	4.2	6.3
8	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1	AB	01	160	160	117	117	43.0
9	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	CU	01	177	177	160	160	17.0
10	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 12 DISC ROLL, Drg: CSP-2011-09	LOY STE	12	80	960	60	720	240.0
11	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 13 SHAFT, Drg: C-2010-039 R-0	LOY STE	02	286	572	208	416	156.0
12	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 14 PARALLEL KEY 36x20x92	LOY STE	12	01	12	0.4	4.8	7.2
	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 18 END COVER, Drg: C-2010-042.3 R1	LOY STE	02	07.2	14.4	02.8	5.6	8.8
13	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 19 STOPPER, Drg: C-2010-042.5	LOY STE	02	0.21	0.42	0.18	0.36	0.1
14	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 20 LOCK NUT M135x2, Drg: IS: 6731-77	LOY STE	04	05.4	21.6	01.4	5.6	16.0
	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 22 PARALLEL KEY 36X20X150, Drg: IS: 2048-83	LOY STE	02	01.5	3	0.8	1.6	1.4
15	LABYRYNTH RING: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 4 LABYRINTH, Drg: C-2010-042.8 R1	LOY STE	06	6.0	36	1.0	6	30.0

16	END COVER: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 18 END COVER, Drg: C-2010-042.5 R1	LOY STE	06	7.2	43.2	2.1	12.6	30.6
17	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.7	LOY STE	04	27.7	110.8	7.8	31.2	79.6
19	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.4	LOY STE	02	12.6	25.2	5.4	10.8	14.4
20	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.6	LOY STE	02	5.0	10	2.2	4.4	5.6
21	Shutter Ledge Inter, 3 FT-0 IN Drg: A1/108.02.17/43784 Item NO.22	NE CARE	02	86.5	173	80.9	161.8	11.2
22	Modified Axle Box Top Half Drg: 036.071.00.167	NE CARE	02	370	740	236	472	268.0
23	Modified Axle Box Bottom Half Drg: 036.071.00.168	NE CARE	02	420	840	316	632	208.0
24	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	NE CARE	04	390	1560	365	1460	100.0
25	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	NE CARE	02	120	240	65	130	110.0
26	Pig Mould Drg: 036.108.00.002 B	NE CARE	70	105	7350	102.5	7175	175.0
27	BSL Pin as per Drg	LOY STE	09	11	99	5.5	49.5	49.5
28	Pivot for RSP Platform Car Ref. Drg: 036.182BF 059A	NE CARE	03	120	360	65	195	165.0
29	Car, Platform, 80T, 11M, Pivot Pin, Drg: As per drawing taken from physical measurement	NE CARE	05	18.4	92	10.2	51	41.0
30	165 T Ladle Pin as per drawing	NE CARE	06	65.1	390.3	48.6	291.36	98.9
31	Plate, Sand Seal Dipper Casting Drg: 05.01.10/163C Item No.6"	NE CARE	08	30	240	29	232	8.0
32	Plate, Sand Seal Dipper Casting Drg: 05.01.10/163C Item No.4"	NE CARE	12	30	360	29	348	12.0
33	Casting, Side Wall, Inner Drg: A1/05.01.10/58734 Item No.3"	NE CARE	07	80	560	79	553	7.0
34	MODIFIED HEARTH CASTING (INNER) FOR A FURNACE DRG: A3/005.01.010/59848A DRG. IT 1	NE CARE	04	190	760	146.0	584	176.0
35	Skid Casting/Drg No: A2/04.07.02/90252	LOY STE	08	140	1120	138	1104	16.0

36	Side Bearer 50mm for 100 Ton Platform Car, Drg: As per reference drawing	MS	16	26.5	424	21.3	340.8	83.2
37	Side Bearer 30mm for 100 Ton Platform Car, Drg: As per reference drawing	MS	10	20.5	205	13.1	131	74.0
38	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	MS	12	9.3	111.6	6.2	74.4	37.2
39	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	NE CARB	01	1780	1780	1500	1500	280.0
40	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	LOY STE	02	390	780	365	730	50.0
41	Head, 250 X L1800 mm, for Dummy Bar of CCS Drg: CC-422-76 SLNo.-1	NE CARB	02	1200	2400	1000	2000	400.0
42	LOWER ARMOUR CONE CRUSHER DRG:-MI-9-36(CC/82 R-1	IN STEEL	01	2680	2680	2080	2080	600.0
43	Spindle for BLR#3-4 Drg: PD/R-355053	GED ROU	07	16.7	116.9	8.6	60.2	56.7
44	Plate, Sand Seal Dipper Casting Drg: 05.01.10/163C Item No.4"	NE CARB	04	30	120	29	116	4.0
45	Casting, Side Wall, Inner Drg: A1/05.01.10/58734 Item No.3"	NE CARB	01	80	80	79	79	1.0
46	Skid Casting/Drg No: A2/04.07.02/90252	LOY STE	03	140	420	138	414	6.0
47	Side Bearer for 100 Ton Platform Car, Drg: As per reference drawing	LOY STE	06	9.3	55.8	6.2	37.2	18.6
48	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	NE CARB	04	390	1560	365	1460	100.0
49	Skid Casting/Drg No: A2/04.07.02/90252	LOY STE	03	140	420	138	414	6.0
50	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 12 DISC ROLL, Drg: CSP-2011-09	LOY STE	12	80	960	60	720	240.0
51	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 13 SHAFT, Drg: C-2010-039 R-0	GED ROU	02	286	572	208	416	156.0
52	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 14 PARALLEL KEY 36x20x92	MS	12	01	12	0.4	4.8	7.2

53	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 18 END COVER, Drg: C-2010-042.3 R1	MS	02	07.2	14.4	02.8	5.6	8.8
54	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 19 STOPPER, Drg: C-2010-042.5	MS	02	0.21	0.42	0.18	0.36	0.1
55	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 20 LOCK NUT M135x2, Drg: IS: 6731-77	MS	04	05.4	21.6	01.4	5.6	16.0
56	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 22 PARALLEL KEY 36X20X150, Drg: IS: 2048-83	MS	02	01.5	3	0.8	1.6	1.4
57	LABYRYNTH RING: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 4 LABYRINTH, Drg: C-2010-042.8 R1	MS	06	6.0	36	1.0	6	30.0
58	END COVER: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 18 END COVER, Drg: C-2010-042.5 R1	MS	06	7.2	43.2	2.1	12.6	30.6
59	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.7	MS	04	27.7	110.8	7.8	31.2	79.6
61	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.4	MS	02	12.6	25.2	5.4	10.8	14.4
62	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.6	MS	02	5.0	10	2.2	4.4	5.6
63	Modified Axle Box Top Half Drg: 036.071.00.167	NE CARB	02	370	740	236	472	268.0
64	Modified Axle Box Bottom Half Drg: 036.071.00.168	NE CARB	02	420	840	316	632	208.0
65	Copper Plug for the month of Feb & March	Copper	90	0.9	81	0.5	45	36.0
Grand Total					0		0	5233.5
					0		0	
CHECKED CALCULATION SHEET FOR THE MONTH OF APRIL-2025 (01.04.2025 TO 30.04.2025)					0		0	
1	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	CU	01	177	177	160	160	17.0
2	Modified Cast Tuyere Assly, Drg: BF-2011-013	CU	09	145	1305	130.1	1170.9	134.1

3	MOD.CU TUYERE W/HARD FACING.Drg:320.01.31.W/89465	CU	01	170	170	146	146	24.0
4	MOD.CU TUYERE NOZZLE DRG: BF-2007-021	CU	01	177	177	120	120	57.0
5	Bush for Elevator Counter weight Drg: B-210102 R1	HTB	10	3.0	30	1.4	14	16.0
6	WEARING PLATES FOR WORK ROLL WITH CHOCK ASSLY.OF FOUR HIGH STAND DRG; PMM-177 R2 THICK-30 MM	HTB	02	202	404	133	266	138.0
7	Plate for Top Work Roll Chock of Four High Roughing Stand Drg: PMM-182-1 R1	HTB	01	160	160	100.4	100.4	59.6
8	BUSH DIA 280/250X 95 FOR MOV BODY DRG;BF 460-63	AB	02	17.0	34	8.4	16.8	17.2
9	BUSH BEARING FOR DRIVE SIDE OF SCREW ASEMBLY OF SPTC/Drg: 3.30.72.30.W/71297	AB	07	53.0	371	35.8	250.6	120.4
10	BUSH, PN: 3460331601100/09 FOR C/C SHEAR BLADE HOLDER, DIA: 150/120X60MM, DRG: 346033160100/09 DRAWING NO: 346033160110007 ITEM NO: 09	HTB	08	6.0	48	2.0	16	32.0
11	BUSH, PN: 3460331601100/25 FOR C/C SHEAR BLADE LEVER, DIA: 220/179.4X95MM DRG: 3460331601100/25 DRAWING NO: 346033160110005 ITEM NO: 25	HTB	14	15.0	210	6.2	86.8	123.2
12	Inclined Copper Tuyere dia: 150x562MM, Drg: 036.008.BF.098	CU	03	177	531	160	480	51.0
13	Copper Tuyere, Drg: 5017165-0384407	CU	04	301	1204	287	1148	56.0
14	Tuyere, Intermediate, Drg: 036.008.00.049C	CU	04	181	724	170	680	44.0
15	BUSH DIA 280/250X 95 FOR MOV BODY DRG;BF 460-63	AB	01	17.0	17	8.4	8.4	8.6
16	BUSH BEARING FOR DRIVE SIDE OF SCREW ASEMBLY OF SPTC/Drg: 3.30.72.30.W/71297	HTB	02	53.0	106	35.8	71.6	34.4
17	Bush, PN: 3460331601100/08 for C/C Shear Blade Holder, Dia: 100/70 x 33 mm, Drg: 346033160110007, Drawing Item No. 08	HTB	05	3.0	15	1.4	7	8.0

18	BUSH, PN: 3460331601100/09 FOR C/C SHEAR BLADE HOLDER, DIA: 150/120X60MM, DRG: 346033160100/09 DRAWING NO: 346033160110007 ITEM NO: 09	HTB	12	6.0	72	2.0	24	48.0
19	BUSH, PN: 3460331601100/25 FOR C/C SHEAR BLADE LEVER, DIA: 220/179.4X95MM DRG: 3460331601100/25 DRAWING NO: 346033160110005 ITEM NO: 25	HTB	02	15.0	30	6.2	12.4	17.6
20	PAD BEARING DRG:-04.04.006/28D IT NO-A AND L	PB	04	98.0	392	65.0	260	132.0
21	PAD BEARING DRG:-04.04.006/28D IT NO-A AND L (Reject due to machining defect)	PB	04	98.0	392	42.3	169	223.0
22	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	LOY STE	08	390	3120	365	2920	200.0
23	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	AIN CARE	01	1800	1800	1500	1500	300.0
24	Modified Axle Box Top Half Drg: 036.071.00.167	AIN CARE	03	370	1110	236	708	402.0
25	Modified Axle Box Bottom Half Drg: 036.071.00.168	AIN CARE	03	420	1260	316	948	312.0
26	Clay Gun Nozzle Tip with 2 Nos Locking Pin, Drg:ISP-DES-V-007-323-01-DE-004	AIN CARE	07	45	315	31.3	219.1	95.9
27	Crane Wheel for G.C. Shop as per sample (Rejection for Blown Hole)	AIN STEEL	02	250	500	123.5	247	253.0
28	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 12 DISC ROLL, Drg: CSP-2011-09	LOY STE	12	80	960	60	720	240.0
29	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 13 SHAFT, Drg: C-2010-039 R-0	GED ROU	02	286	572	208	416	156.0
30	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 14 PARALLEL KEY 36x20x92	MS	12	01	12	0.4	4.8	7.2
31	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 18 END COVER, Drg: C-2010-042.3 R1	MS	02	07.2	14.4	02.8	5.6	8.8
32	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 19 STOPPER, Drg: C-2010-042.5	MS	02	0.21	0.42	0.18	0.36	0.1
33	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 20 LOCK NUT M135x2, Drg: IS: 6731-77	MS	04	05.4	21.6	01.4	5.6	16.0

34	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 22 PARALLEL KEY 36X20X150, Drg: IS: 2048-83	MS	02	01.5	3	0.8	1.6	1.4
35	LABYRYNTH RING: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 4 LABYRINTH, Drg: C-2010-042.8 R1	MS	06	6.0	36	1.0	6	30.0
36	END COVER: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 18 END COVER, Drg: C-2010-042.5 R1	MS	06	7.2	43.2	2.1	12.6	30.6
37	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.7	MS	04	27.7	110.8	7.8	31.2	79.6
38	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.4	MS	02	12.6	25.2	5.4	10.8	14.4
39	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.6	MS	02	5.0	10	2.2	4.4	5.6
40	Sprocket Without Hard facing, For Single Roll Crushers, Drg: 0210110102A	LOY STEE	02	390	780	365	730	50.0
41	Spindle, Coupling, Drg: 0404/-/227F Drawing Item No. D	AIN CARE	01	1800	1800	1500	1500	300.0
42	Plate Armour Middle Drg: NM-83-69 R1	AIN STEEL	02	981	1962	920	1840	122.0
43	Plate Armour Lower Drg: NM-84-69 R1	AIN STEEL	01	980	980	920	920	60.0
44	Modified Axle Box Top Half Drg: 036.071.00.167	AIN CARE	01	370	370	236	236	134.0
45	Modified Axle Box Bottom Half Drg: 036.071.00.168	AIN CARE	01	420	420	316	316	104.0
46	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 12 DISC ROLL, Drg: CSP-2011-09	LOY STEE	06	80	480	60	360	120.0
47	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 13 SHAFT, Drg: C-2010-039 R-0	GED ROU	01	286	286	208	208	78.0
48	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 14 PARALLEL KEY 36x20x92	MS	06	01	6	0.4	2.4	3.6
49	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 18 END COVER, Drg: C-2010-042.3 R1	MS	01	7.2	7.2	02.8	2.8	4.4
50	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 19 STOPPER, Drg: C-2010-042.5	MS	01	0.21	0.21	0.18	0.18	0.0

51	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 20 LOCK NUT M135x2, Drg: IS: 6731-77	MS	02	05.4	10.8	01.4	2.8	8.0
52	Roll Assembly-II for Grizzly Screen 1500 x 3300, POS. No. 22 PARALLEL KEY 36X20X150, Drg: IS: 2048-83	MS	01	1.5	1.5	0.8	0.8	0.7
53	LABYRYNTH RING: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 4 LABYRINTH, Drg: C-2010-042.8 R1	MS	03	6.0	18	1.0	3	15.0
54	END COVER: Roll Assembly-II for Grizzly Screen 1500 x 3300, POS.No. 18 END COVER, Drg: C-2010-042.5 R1	MS	03	7.2	21.6	2.1	6.3	15.3
55	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.7	MS	02	27.7	55.4	7.8	15.6	39.8
56	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.4	MS	01	12.6	12.6	5.4	5.4	7.2
57	DISTANCE BUSH: Roll Assembly-II for Grizzly Screen 1500X3300, POS No.97 C-2010-042.6	MS	01	5.0	5	2.2	2.2	2.8

TABLE : PENALTY & BONUS

[illegible]

TENDER DOCUMENT

TENDER NOTICE No.: SGW/CC/25-26/68/OTE/68, DATE: 25/02/2026

**NAME OF WORK: OPERATION OF MACHINES AT MACHINE SHOP
OF SGW KULTI.**



STEEL AUTHORITY OF INDIA LIMITED

SAIL GROWTH WORKS, KULTI

P.O. KULTI, DISTT. BURDWAN (W.B.), PIN-713 343

PHONE: 9477702447/9007910436

FAX: 0341-2515299

**E-mail: burman.sumit@mjunction.in; mkbhangre.sgw@gmail.com;
sumitburman942@gmail.com**

Techno-Commercial Bid (Part - I)

A BRIEF ABOUT SAIL GROWTH WORKS, KULTI

SAIL Growth Works, Kulti, a Unit of Steel Authority of India Limited, is under Burdwan District of West Bengal and is situated at a distance of about 16 kilometers from Asansol on Howrah-New Delhi Grand Chord route. It is also well connected with a number of big cities like Kolkata, Patna, Dhanbad and Jamshedpur by road.

Presently the Operations at Non-Ferrous Foundry, Steel Foundry as well as Machine Shop, and other associated Utilities and Services, are being carried out at SGW, Kulti.

SAIL Growth Works, Kulti has Non ferrous & Steel Foundries, Machine Shops and associated utilities and services, Structural Shops etc.. The Township of Kulti has about 2300 houses and spread over an area of 597 acres with Road network.

CHECK LIST FOR TENDERERS:-

Sl. No.	Documents/Information to be submitted with the Tender	Yes/No
i)	Whether each page of the Tender document signed	
ii)	Whether Techno-Commercial Bid (Part-I) is attached with <u>acknowledgement slip</u> of submission of on line price bid (Refer Mode of submission of tender para) and EMD sealed in separate envelopes and then all these envelopes sealed in a bigger envelope and properly super scribed with tender number and name of work.	
iii)	Whether EMD / Bid-Securing Declaration submitted as prescribed	
iv)	Whether Annexure-3 <u>NON-COLLUSIVE TENDERING CERTIFICATE submitted as prescribed</u>	
v)	Whether GST Registration Number and a Copy of same provided	
vi)	Whether Permanent Account Number and a Copy of PAN Card provided	
vii)	Whether West Bengal Professional Tax Payment Certificate, if applicable, provided	
viii)	Whether details of Bank particulars (Name of the Bank, Branch, Type of Account and the Account No.) provided	
ix)	Whether Copy of Labour License, held under previous Contract, if any provided	
x)	Whether type of Firm has been mentioned and supporting documents provided	
xi)	Whether the Declaration, that the Site has been inspected and fully examined, has been made as per the enclosed Annexure-II (If attached)	
xii)	Whether all the three Declarations, as given in the Tender document, have been filled in and signed	
xiii)	Whether the quoted Rates are inclusive of various Taxes & Duties except GST, as per requirement of the Tender	
xiv)	Whether following documents, in support of Eligibility Criteria as per Clause-4 of 'Special Conditions of Contract' have been provided; i) Documentary evidence in support of work experience self attested ii) Proof of sound financial capability iii) PF Code& ESI Code iv) In respect of PF- copy of ECR	
xv)	One valid email address in their offer for all future correspondences	

NOTE:

- The Tenderers, in their own interest, are advised to check through the above Checklist and make sure that the relevant documents/information, as required, has been provided before submitting the Tender. Call for any clarifications.*
- Submission of physical sealed Price Bid along with the tender document or in any other form/format will lead to rejection of offer.**
- IMPORTANT NOTE: PLEASE NOTE THAT THE TENDERER MUST PROVIDE ONE VALID EMAIL ADDRESS AND MOBILE NO IN THEIR OFFER.**

INVITATION TO TENDER

Sealed tenders are invited from bonafide, experienced and resourceful Contractors for the following Work at SAIL Growth Works, Kulti, PO- Kulti, Distt.- Paschim Burdwan (W.B.), PIN – 713 343. The Scope of Work, Terms & Conditions and other details are given below;

Sl. No.	Item description	Details
1.	Tender Notice No. & date	SGW/CC/25-26/68/OTE/68, DATE: 25/02/2026
2.	Name of Work	OPERATION OF MACHINES AT MACHINE SHOP OF SGW KULTI.
	Mode Of Tender	OTE Followed By Reverse Auction (RA)
3.	Period of Contract	One (01) Year w.e.f 15.06.2026 or from the date of site handover as directed by EA.
4.	Earnest Money Deposit (EMD) / Bid-Securing Declaration	In place of EMD, Bidder may submit a duly signed Bid Securing Declaration (Annexure-I).
5.	<u>DOWNLOADING OF TENDERDOCUMENT</u>	From the website https://www.sailtenders.co.in . (Under the link “search tenders”, Plant/Unit as “SAIL Growth Works (SGW)”). No additional cost is required to be deposited towards the cost of tender document. However, tenderers may opt to collect the complete document from SAIL without any charge if they are unable to download it from the website.
6.	Period of downloading of Tender document	From: 25/02/2026 to 17/03/2026
7.	Date of Pre-bid discussion	Tenderer who may wish to have pre bid discussions may come On any date Before dropping the tender document in the office of GM (CC, MM, Strl & Fab) , SGW, Kulti.
8.	Last date and time of submission of Tender	18/03/2026 by 3-30 P.M.
9.	Tender opening date and time	18/03/2026 at 4-00 P.M.
10.	Paying Authority	Sr. MANAGER (F & A), SGW, Kulti
11.	Validity of Tender	60 days from the date of opening of the price bid/Reverse Auction. Prior withdrawal of the tender after opening shall amount to forfeiture of EMD.
12.	Bidding Basis (for On - Line Reverse Auction)	Rates are to be quoted in Lump sum for (01) Year Excluding GST. Bidders cannot quote below the floor price.
13.	Price Bid (Special Term)	On Line Price Bid is to be submitted on MJ's Byjunction portal at Link https://buyjunction.in and get an acknowledgement slip of on line submission and attach this acknowledgement slip along with the techno commercial offer. Refer “ Submission of Offer “ <i>For normal adequate number of offers received, price will be obtained through On - Line Reverse Auction on the platform of mjunction services limited.</i> <u>Note:</u> RA Conducted as per new RA Guide lines which will be Communicated before RA by mjunction. The same are attached in this tender documents also.

1. MODE OF SUBMISSION OF TENDER:

The Tender consists of the following two Parts:

Part – I : Tender Document (Techno commercial offer) along with **Acknowledgement Slip ***

Part-II: Submission of price bid in buy-junction portal and acknowledgement slip to be submitted along with the Techno-commercial document.

The entire Tender document, except the Price-Bid containing the “Schedule of Rates”, constitutes Part-I of the Tender. All Documents/Information, required as per the tender, are to be submitted along with Part-I of the Tender only.

- The Tenderer should detach the page containing the acknowledgement Slip of Mj Portal where Price Bid was submit in MJ’s Platform at <https://buyjunction.in> Portal and obtain an acknowledgement Slip the same required to be submitted along with Techno-Commercial offer.
- **Note: Submission of physical sealed Price Bid along with the tender document or in any other form/format will lead to rejection of offer.**

Apart from the above, the Earnest Money Deposit (EMD)/Bid Securing shall be enclosed in a separate sealed cover.

There shall therefore be following separately sealed envelopes initially;

- i) Envelope containing Techno-Commercial Offer (Part-I) of the Tender containing the:
 - Entire Tender document as well as all Documents/Information required as per the Tender. The envelope should be marked as **“Techno-Commercial Offer”**.
 - Acknowledgement Slip of Online Price bid submitted in MJ’s Platform at <https://buyjunction.in> Portal.
- ii) Envelope containing the Earnest Money Deposit. The envelope should be marked as **EMD-Earnest Money Deposit/ Bid-Securing Declaration**.

All the above envelopes shall then be put inside a bigger envelope and properly sealed. The Name of the Work, Tender Notice No. and Tender opening date shall be mentioned on the face of the envelope and the same shall be dropped in the **Tender Box** kept in the Office of M-Junction -SAIL GROWTH WORKS-PO: KULTI- DISTT: PASCHIM BARDHMAN-W.B. PIN: 713343) **on or before the last date** of submission of the Tender. The Tenders can also be sent by Registered Post or Courier, however, delayed delivery of Tenders shall not be entertained.

Note:-

- The tender can send the documents either through courier /speed post/by hand or may send scanned copies of all tender documents duly signed and stamped through mail (less then 15 MB file) , in the mail ID mentioned in the tender document.
- A line of confirmation of dropping Tender document in mail be given in SMS in both number 9477702447 & 9007910436)
- Ensure that the price bid is submitted on Link mentioned above for all the mode of submission of tender document.

2. METHOD FOR OPENING OF THE TENDER:

If the prescribed norm of the Company for opening of Tender is met, then the main sealed envelope containing all documents /sealed envelopes, shall be opened as per the given Schedule in presence of the Tenderers or their authorized representatives who may wish to be present during Tender opening. Immediately thereafter **on the same day, the envelope containing Earnest Money Deposit (EMD)/ Bid Securing Declaration (Annexure-I), shall be opened.**

The Part-I of the Tender of only those tenderers, whose Earnest Money Deposit (EMD)/ Bid Securing Declaration (Annexure-I) will be found in order and submitted as prescribed above, shall be opened immediately thereafter, otherwise the offer will be considered as invalid and Techno-Commercial Offer/Bid (Part-I of the Tender) of the same will not be opened.

The Tenders, after opening, shall be evaluated at a later date. Only those tenders which are suitable techno commercially will be eligible for participation in the on-line negotiation mechanism (reverse auction) to be organized on the on-line platform of m-junction services Limited.

3. EXEMPTION FROM SUBMISSION OF 'EMD':

Micro & Small Enterprises (MSEs/PSUs/Government Undertakings and Co-operative Societies/ Start-ups as recognized by the Department for Promotion of Industry and Internal Trade (DPIIT) etc, may be exempted from submission of EMD as per extant Government policy. For MSEs, the exemption will be extended on the basis of:

- 1) Self-certified copy of Udyam Registration Certificate.
- 2) The Concerned MSE is registered in MSME databank
- 3) MSE is registered for given scope of job/Procurement

SSI/NSIC certificate holders are also exempted from submitting EMD.

NOTE:

- 1. Tender document is also available on the SAIL website at www.sailtenders.co.in it shall be responsibility of the persons submitting the tender to ensure that the tenders have been submitted in the formats and as per the terms and conditions prescribed in the SAIL website and no change is made therein before submission of their tender. In the event of any doubt regarding the terms and conditions / formats, the person concerned may seek clarifications from the authorized officer of SAIL. In case of any tampering / unauthorized alteration is noticed in the submitted documents, from the tender document available on the SAIL website, the said tender shall be summarily rejected and the company shall have no liability whatsoever on the matter. However, deviation if any proposed by the tenderer may be separately indicated for acceptance. Such proposed deviation will not be treated as tampering for the purpose of application of the clause”.**
- 2. IMPORTANT NOTE: PLEASE NOTE THAT THE TENDERER MUST PROVIDE ONE VALID EMAIL ADDRESS AND MOBILE NO IN THEIR OFFER.**

(1) Submission of Offer:- Part II

- Techno Commercial Bid:- Bidders have to submit the Techno-Commercial Bid through Email (Email ID : burman.sumit@mjunction.in/mkbbhangre.sgw@gmail.com).
- Price bids to be submitted on MJUNCTION's Platform at <https://buyjunction.in>.
- Methodology for submission of Online Price bid :-

Step 1: Login to [https:// buyjunction.in](https://buyjunction.in).

Step 2: Click on Login Tab

Step 3: Go to Important link and click on “View Live Auctions & Tenders”

Step 4: Click on Tender document.

Step 5: Existing users can login with their user id and password. New users would have to register for a user id & password on the site.

Step 6: Read through the “Bidding Procedure” to proceed further for bidding and place your online sealed price bid.

Step 7: Take a printout of the receipt confirming that the online sealed price bid has been submitted (Acknowledgement).

System generated acknowledgement-slip for online submission of price bid to be submitted along with the techno commercial bid.

MJUNCTION Contact person for assistance:-

Deblina Maity@8336991451 (Primary Contact of Kolkata)

Priyanka Majumdar@9051186491 (primary contact of Kolkata)

Abhishek Chakraborty @ 9163348279 (Burnpur branch)

Pradeep Mohapatra @ 7033297436 (Burnpur branch)

- The timeline for submission of offer is as under :-

Techno-commercial bid - To be submitted up to (client mail id)	18/03/2026 by 03.30 PM
Price Bid - To be submitted at M/s Mjunction's Portal up to https://buyjunction.in	18/03/2026 by 03.30 PM
Due date of opening of Techno-commercial bid	18/03/2026 by 04.00 PM

INSTRUCTIONS TO TENDERERS

- 1) The Tenderers shall Sign and put their Seal on each page of the tender document, as token of acceptance of all the Terms & Conditions of the tender and submit the same.
- 2) The Tender Document is not transferable.
- 3) **The Tenderers shall submit the following Documents/Information along with Techno-Commercial Part (Part-I) of the Tender;**
 - i. GST Registration No. along with a Copy of the same.
 - ii. Permanent Account No. along with a Copy of the PAN Card
 - iii. Valid (Up to date) West Bengal Professional Tax Payment Certificate, if applicable otherwise declaration shall be given in letter head stating reason for non-applicability.
 - iv. PF, ESI, Registration numbers & copies of same. ECR in respect of PF.
 - v. Bank particulars (Name of the Bank, Branch, Type of Account and the Account No.) to which payment is desired.
 - vi. Copy of Labour License, under the Contract Labour (Regulation and Abolition) Act, 1970, held under previous Contract, if any.
 - vii. Type of Firm that is whether Proprietary Firm, Registered Partnership Firm, Private Limited Company, Public Limited Company, Co-operative Society etc. along with the following documents;
 - a. In case of Proprietary Firm, Affidavit of Sole Proprietorship
 - b. In case of Partnership Firm, attested copy of Partnership deed along with amendment, if any and proof of Registration, if any.
 - c. In case of Limited Companies, Memorandum and Articles of association, Certificate of Incorporation, authorized, subscribed and paid up capital
 - d. In case of Cooperative Society, attested copy of the Certificate of Registration from the Registrar of Co-operative Societies.
 - viii. Declaration that the Site has been inspected and fully examined as per Annexure-II
 - ix. The Documents as per Clause-3 (Eligibility Criteria) of 'Special Conditions of Contract' of the Tender, as evidence to prove that the Tenderer fulfills the Eligibility Criteria.
- 4) Rates and quantities quoted in the tender shall be in figure and words. In case of any discrepancy between the two, the Rates quoted in words shall be taken as finally quoted Rates.
- 5) Conditional tenders either in Part-I or Part-II of the tender shall be liable for rejection.
- 6) By submitting the tender for the work, the Tenderer will be deemed to have satisfied himself that the Rates quoted by him in the tender will be adequate to complete such work. Any complaints in this regard on a later date shall not be entertained.
- 7) The Company reserves the right to accept/reject any or all tenders without assigning any reason thereof.
- 8) The Period of Contract, as mentioned in the Tender, can be extended or curtailed at the sole discretion of the Management.
- 9) ***The following three Declarations are to be given by the Tenderer as per the format given below and submitted along with the Techno-Commercial Bid. If these declarations are not enclosed with the Tender, it shall be liable for rejection.***

(I)
DECLARATION

I/we hereby declare that neither I am nor any of Partners/Directors of our Firm/Company is/are made prime accused in a prosecution before any Court of Law or involved in malpractice and/or theft of the property of SAIL.

The above declaration is true to my knowledge and belief and if any part or all found untrue, I/our Firm/Partnership/Company shall be liable for an action for banning of business dealings with SAIL, SGW, Kulti and/or our Tender/Contract shall be liable for cancellation.

Date:

Signature & Seal of the Tenderer

(II)
DECLARATION

I/we hereby declare that I/we (Proprietor or any Partner of the Firm or Director of the Company, as the case may be) have no relation with any employee working in SAIL Growth Works, Kulti/following relatives of mine are working at SAIL Growth Works, Kulti;

Sl.No.

Name of the Employees Relationship
Working at SGW, Kulti

Date:

Signature & Seal of the Tenderer

NOTE: Strike out which is not applicable

(III)
DECLARATION

I/we hereby declare that within the meaning of Section-6 of the Company's Act, 1956, I have no relationship with any Director of SAIL/following Directors of SAIL are related to me;

Sl. No.Name of the Director of SAILRelationship

Date:

Signature & Seal of the Tenderer

NOTE: Strike out which is not applicable

(IV)
DECLARATION

Anti-Bribery Undertaking:

By participating in this tender, the Suppliers/Vendors/Contractors are deemed to have undertaken that they shall not give or take any financial or non-financial bribe, to or from anyone during the tender or during the execution of the contract thereafter and if they notice any such Incident happening, they shall report it to SAIL Vigilance.

Date:

Signature & Seal of the Tenderer

IMPORTANT NOTES

- a. **Clarification:** Acceptance for “time frame of submission of Clarifications/Documents” of all relevant documents required as per tender T&C within 10 days from the date of “First intimation by Contract Cell” after tender opening. If clarification is not submitted within 10 days from date of “First intimation by Contract Cell”, then SGW will evaluate the case as per last document submitted.
- b. Outcome of techno-commercial evaluation (acceptance / rejection of offers) shall be communicated online / through email to the participating bidders. In case of rejection, reasons for rejection shall be mentioned in such communication. Representations thereon if any received within 48 hours shall be considered and replied to expeditiously.
- c. Security Deposit: SD should be deposited within 10 (Ten) days after issuance of W.O. Refer SD clause in the Tender Document.
- d. After final evaluation and assessment of L-1 tender, Work Order will be placed directly.
- e. Validity of contract is meant to complete the work in progress items after the expiry of the period of contract.
- f. The tender can send the documents either through courier /speed post/by hand or may send scanned copies of all tender documents duly signed and stamped through mail , in the mail ID mentioned in the tender document. Hard copies of TD are to be sent to Contract Cell, SGW subsequently.
- g. If L-1 party backs out after tendering process then action will be taken against party, as per company rules.

Minimum Floor Price

The minimum Floor Price is Rs 1,27,68,070.77 (Rupees One Crore Twenty-Seven Lakhs Sixty-Eight Thousand Seventy-One Only)

The Bidders are not allowed to quote below floor price.

Two types of situations may arise after opening of price bid

- 1- Only one party may quote MFP and becomes L-1
- 2- If more than one L-1 bidders quoting MFP, then draw of lots will be carried out with in L-1 Bidder(s)

Methodology of Draw of lot

For Tender sent for RA, where RA could not be conducted for L-1 Bid(s) being equivalent to MFP, involving eligible bidders or the cases where RA is not involved, the bidders who have quoted MFP, draw of Lots will be carried out physically by Tender Committee in presence of L-1 bidder(s) [Quoting MFP] who prefer to be present during draw. Name of the bidders written on a piece of plain white paper will be kept in an envelope, which to be sealed in presence of bidders who are present and on sealed envelope signature will be taken from the members of Tender Opening committee, representative of the tenderer, who will be present. Signed Envelops will be put in a closed container. After shaking the same, the lid of the container will be opened and one of the members of the Tender Committee/Bidders will pick up one envelope, which will be opened & L-1 bidder will be declared.

ELIGIBILITY CRITERIA:

Technical Eligibility	Experience /Documents Required
The Tenderer must have Experience of having successfully Completed or Substantially Completed similar Work during last seven (07) years ending last day of the month previous to the one in which tender is issued should be either of the following : (a) Three similar completed work each costing not less than the amount equal to Rs 51,42,692/- (b) OR (c) Two similar completed work each costing not less than the amount equal to Rs 64,28,365/- (d) OR (e) One similar completed work each costing not less than the amount equal to Rs 1,02,85,384/- Similar work/s is/are defined as: (A) Machining activities done in their Own Premises /Worksfor preparing Engineering sparesfor organizations mentioned in Experience Clause. OR (B) Machining activities done in the premises of organizations mentioned in Experience Clause for preparing DEFINITION OF ENGINEERING SPARE: (i) MACHINING ACTIVITIES OF ENGINEERING MEANS: MACHINING OF CASTINGS/FORGINGS/FABRICATED STRUCTURES TO DRAWING DIMENSIONS. (ii) ONLY MACHINING VALUE WORK ORDERS SHALL BE CONSIDERED. (iii) BIDDER WILL BE ASKED TO GIVE BREAK UP OF MACHINING COST &MATERIAL COSTIFMACHINED SPARE'S WORK ORDER IS WITH MATERIAL COST. (iv) IF BIDDER IS NOT ABLE TO GIVE BREAK UP OF MACHINING COST & MATERIAL COST THEN MACHINING COST SHALL BE CONSIDERED 30 % OF WORK ORDER VALUE.	Experience: Should be from Central/State Govt. Organizations/PSUs/Joint Venture Company where at least one of the partners shall be Central or State Government Organization or PSU/Public Limited Companies: Requisite Documents: • Self- attested copies of work orders • Self- attested copies of Completion Certificate (issued by the concerned work Executing Authority etc.) • Substantially Completed ongoing Work Orders having Duration of contract of more than one year may also be considered for technical evaluation under “ similar work” as per Eligibility criteria, if the party has completed at least 50% of the work as per the work order. In such case already executed value shall be considered for evaluation. • In Case work completion certificate is issued without value of the completed work, ,RA Bills paid or any other document(s) evidencing value of completed work may also be considered. • The work order and completion certificate /Paid Bills so submitted should be in the name of participating tenderer.
Financial Criteria: • Average Annual Financial Turn Over during the last three (03) years ended 31st March 2025 should be at least Rs.38,57,019/- •	Documents Required: 1. If requisite turn over is up to Rs 1 Cr. • Turn over Certificate from a practicing Chartered Accountant (CA) 2. If requisite Turn over > Rs 1 Cr • Audited Profit and Loss (P/L) Accounts Statement. Note: (a) Turn over shall be submitted exclusive of GST. (b) P&L Statement should be in the name of the participating tenderer. (c) Financial documents issued by CA must contain Unique Document Identification Number (UDIN) and Membership Number of CA. (d) Financial Turn over of the bidder during last 3 years , ending 31st March of the Financial Year previous to the one in which NIT is issued , shall be sought as per the existing practice, subject to the following: (i) If tender issue date is in H1 of the FY, last three FYs shall be considered from current FY- 2 Years (ii) If tender issue date is in H2 of the FY, last three FYs shall be considered from current FY- 1 Year

Note:

1. Tenderer should have P.F. Code (Provident Fund Registration No. as provided by the concerned Regional Fund Commissioner) as well as ESI Registration No. and shall submit copy of the same (Original to be produced on demand).
2. In case any document/information submitted by bidder(s) is found to be false or containing any misrepresentation or having fraudulent declaration in it, then in such eventuality legal action (including cancellation of the contract, banning of business dealing, damages, criminal proceedings, etc.) as deemed fit may be initiated by SGW against the bidder.
3. Offers received from any Party whose contracts have been terminated in any agency as per the eligibility criteria would be rejected.
4. All work- orders having similar work where completion date falls within last seven(07) years ending last day of the month previous to the one in which tenders are invited shall be considered for evaluation as per Eligibility Criteria , irrespective of date of work order.
5. Documents to be submitted by the Bidders should be in line with the requisite documents as specified in the tender document.

SPECIAL CONDITIONS OF CONTRACT (SCC)

SCOPE OF WORK

Operation of Machines at Machine Shop of SGW Kulti.

- i) Period of Contract :1 (One) Year
- ii) Machining of Steel and Non-Ferrous Castings/Items at Machine Shop No. 1 & 2 and at Steel Foundry Machine Shop as per Drawings and as per **Monthly Plan.Special Jobs in FERROUS CATEGORY ONLY (WHICH HAVE NOT BEEN DONE IN THE PAST)** which require high degree of skill , accuracy and safety (Such as sleeve and arbor fitting , roll chock refurbishment, cladding and machining , shearing blade machining, 80T/100 Bogie machining and Assembly, Sinter frame assembly, **Hammers, idler shafts** etc. will have to be done as per the SOP given by Executing/Operating Authority. Such Jobs will generate very less swarf, so billing value may be very less than the actual expenses made by contractor in doing the job. Such jobs will require correction in the intricacy factor as per actual time taken to complete the job by machines so that there is balance between bill value and actual expenses made by contractor for doing the job. So for Such Jobs the intricacy factor for Machining Rate (MR) will be fixed and decided by EA/OA which will be binding on the Contractor. (No intricacy will be changed for the regular Jobs).
- iii) Jobs / Activities to be performed in General shift or two Shifts basis as per requirement & instruction of Executing Authority. When felt necessary by the Executing Authority, the jobs are to be carried out on Sundays and Holidays, and also in “Night’ shift. Instruction of EA/OA shall be binding on the contractor.
- iv) Minimum 52 nos. of Man powers (Skilled : 35 heads, Semi Skilled: 9 heads, Unskilled: 8 heads) to be engaged including supervisors. Penalty clause may be referred. Minimum Wages , Govt. of West Bengal circular no: 04/Stat/14/RW/24/2023/LCS/JLC Dated 09/12/2025 under CONSTRUCTION OR MAINTENANCE OF ROADS OR IN BUILDING OPERATIONS to be followed.
- v) All the Critical machines have to be Operated in Two Shifts. List of Critical Machines is attached.
- vi) **Declaration is to be submitted in Tender that all the machine operators to operate the critical machines and operators cum programmers for CNC Machines are available. New CNC Machines and New Conventional Machines will be added in the Machine Shop during the Contract.**
- vii) **List of Manpower to be submitted while submitting the Break up of Rates after RA is done.**
- viii) Expected Swarf Generation in One Year = 100000 Kg
- ix) Payment formula: Swarf calculated (BW-FW) for each Job in (Kg) multiplied by quantity machined multiplied x BR (Rs/Kg) multiplied by intricacy factor). Refer **GUIDE LINES FOR BILLING.**
- x) **The estimated quantity of SWARF GENERATION is Indicative. In case of higher production during the contract period, beyond the work order value, prior approval of the competent authority will be sought as per SGW Delegation of power for processing the additional quantity.**
- xi) **Penalties: Refer Table : Penalty & Bonus**
 - There are 37 nos. Heavy duty Machine and 16 nos. light duty machines which are in working condition and are used for production as per suitability of job/operation required in machining. Please refer annexure List of Machines.
 - **Minimum 25 nos.** of the machines should run every day as per Job Load in consultation with EA/OA.
 - If required remaining machines should also be run on day-to-day basis. Contractor should ensure the Job Rotation of Machine Operators to develop multi skilling.
 - Per day (Basic +DA) Wages of skilled workers will be deducted for no. of machines operated less than 25 nos. machines. (Refer 2nd Column , Row – D&E)
 - **Production loss of swarf generation incurred on day to day basis due to Non operation of Machine will be deducted from Bills. (Refer 2nd Column Row nos. F to N)**

Ex:

- **Total Machines operated in a months : 588 Machines (Sum of all machines operated)**
- **Swarf generated: 9000 Kg**
- **BR : Rs 75/Kg .**
- **Bill Amount: Rs 16,00,000/-**
- **Avg intricacy will be : $16,00,000/(9000 \times 75) = 2.370$**
(Average intricacy arrived from Bill submitted)

xii) Bonus : Refer Table : Penalty & Bonus

- Refer 2ND Column Row- P & Q), even if production is more than targeted monthly swarf generation by operating Less than Avg. 30 nos. Machines, No Bonus shall be paid.
- Refer 2ND Column Row- R) : Bonus of 10 % over BR for increased qty of swarf over targeted monthly swarf generation Will be given only if Avg. 30 nos. or More than 30 nos. machines are operated on an average in the Month.

xiii) Special Note for Penalties & Bonus:

- Penalty will be imposed If all the other machines are not run as per instruction /job load/requirement.
- Monthly utilization of Machines is to be submitted along with Bills.
- Penalty shall not be imposed if there is no Job.
- Penalty shall not be imposed if machine is under break down.
- Contractor has to submit daily production report of Machines , Machine wise.
- Maximum Penalty will be Limited to 10 % of Work Order Value which will be regularized in the Final Bill.
- Final decision lies with EA/OA

(A) Job Details:-

- Handling of black/un machined Castings at Machine Shops for inward or outward movement of material as well as loading & unloading of various machines manually OR by EOT Cranes and must be segregated plant wise on the shop floor. The movement of material may call for late hours even after the working hours of Machine Shop.
The castings incoming from the steel foundry may have to be unloaded at GC Shop/HM Foundry due to lack of storing/unloading space at Machine shop.
- To arrange transportation of in-process castings between Steel Foundry Machine shop , Main Machine Shop and Nonferrous Foundry and any other department as advised by Executing Authority.
- Marking, as per Drawings, on the black Castings/in process materials for machining.
- Machining/Preparing of jigs/fixtures, threading plugs/rings, templates/ gauges etc. if required for setting of the jobs on Machines, may have to be done. This will be considered a job and payment will be made as per intricacy table.
- Machining of the Castings, Forgings etc. as per Drawings & carrying out In-process inspection. In process stage inspection is mandatory. The dimensions of the job must be checked by stage inspector before taking off from the machine.
- The machining rate of copper tuyers has to be matched with the production rate of NFF and no copper casting must be accumulated on the shop floor.
- All jobs of Fitting Section including grinding/finishing, tapping and hydraulic testing, air pressure etc.
- Assembly, if required as per the Drawing.
- To assist in inspection of machined Castings for inspection by the Inspection wing of SGW, Kulti/Customer's Inspector.
- Cleaning of the Machines at the end of the Shift, Housekeeping inside the Shop & surrounding area.
- Removal, segregation and keeping Ferrous and Non-Ferrous Swarf (Copper and copper alloys) separately as per pre-defined grades of copper alloys and colours i.e. P.B, Al. Bronze, High Tensile Brass, Zn Alloy, White Metal, Pb. Brass etc.

- i. The ferrous swarf is to be crushed and packed in the gunny bags/plastic sheets/cloth bundle/potli and sent to designated place as advised by EA.
 - ii. The NF swarf also is to be filled in the good quality bags and is to be weighed in presence of authorized representative of executing authority and handed over to MM department.
 - iii. Swarf generated every month is to be handed over to Machine Shop/Stores packed in good quality gunny bags/plastic bags before submission of the monthly bills. At any point of time, non-availability of the bags will not be put as an excuse. If gunny bags are not brought by the contractor, then the same may be procured by SGW Kulti and the cost of the procurement of such gunny bags shall be recovered from the monthly bills of the contractor.
 - iv. In case, the mixing of the NF swarf is reported by NFF, the cost of the material will be recovered from the bills of the contractor.
 - v. Similarly if the ferrous swarf is scattered on the shop floor, then the cost of scattered swarf will be recovered from the bills of the contractor as per the market value of the swarf and handling charges incurred by some other agency.
 - l) Carrying out Heat treatment operations as per requirement of work. The refractory to be changed if required. Raw materials for refractory change will be provided by SGW. Manpower for doing refractory work is to be arranged by contractor at their cost.(new point)
 - m) Safe operation of Equipment/Machines/EOT Cranes etc. as per Standard Operating Practices of Machine Shop.
 - n) Presently, there are 2 nos. of retrofitted CNC Machine. (one lathe & one horizontal boring). Some more new CNC Machines like CNC VTL , CNC Lathe, PLC Horizontal Boring , Conventional Plano miller and Radial drill may be added during the Contract Period. Apart from this four machines i.e. gear hobbing machine no. 24/2, Floor Types Horizontal boring machine no. 10 , VTL Machine no. 113 and Horizontal Milling machine S-6 are also going to be retrofitted to CNC/PLC (PLC only Gear Hobbing). The Contractor will have to provide duly trained Operators and programmers for the machines.THIS SHOULD BE AGREED WHILE SUBMITTING TENDER**
 - o) Welding of lugs on any job to facilitate clamping of job on the Machine. Welded lugs to be removed after the machining are completed.
 - p) Small modifications, if required in the machines for increasing the productivity, can be done with the approval of the Executing /Operating Authority.
 - q) SGW shall provide the Cutting inserts only. The Contractor has to arrange all the activities like machining of shanks, brazing, grinding etc. required for making the tools. No machining bill/cost will be applicable for manufacturing of the tools.
 - r) Approximately 8-10% of various Castings may require re-machining after repair of the Castings, while the repair will be done by the Foundry; the re-machining is to be undertaken by M/c Shop.
 - s) Small casting defects revealed after machining or during the machining are to be rectified by the contractor at Machine Shop. The welding rods, any kind of adhesives required for this job will be provided by SGW.Painting of any casting if required is also to be done. Paint will be provided by SGW.For these job ,no additional payment will be given**
 - t) The drilling operations in the jobs for fixing/clamping/locating & plugging by Plugs/welding.
 - u) Apart from the monthly schedule there may be a call for any urgent job to be done. The job is to be then taken up immediately as per instruction of the Executing/Operating Authority. If required, any in-process job on the machine may have to be taken off from the machine. No additional price will be given for the job being urgently done nor for the reloading of the taken off job on same machine or on any other machine.
 - v) All the measuring instruments must be used after Calibration using standard gauge available at tools store.
- Important Note:-**
- w) ***Contractor shall have to appoint three (3) nos. of competent Graduate/Diploma Engineers/person having vast working experience. The names of all Three supervisors should be displayed in the machine shop. Before starting the work, the Contractor has to submit documents relating to the qualifications and experience of all the supervisors.***
 - x) These persons so appointed by the Contractor will act as Shop In-charges and Office In Charge & safety supervisor..
 - a. First will look after the whole Production and Inspection and will coordinate among various section of the shop, and will be responsible for day to day Production, Efficiency, Administration/Control.
 - b. Second will Look after planning & generating MIS . Any other computer job/analysis job may be given to him time to time as per requirement.

- c. Third will Look after Safety & Housekeeping.
- y) Regular Functions of supervisors:
- They will be reporting to the Executing Authority on a daily basis.
 - Production In-charge will head the team
 - Other two Supervisors will report to him.
 - They should be well conversant with the Machine shop working/practices and working knowledge on computer (MS Word, Excel, Power point etc.).
 - They are to monitor machine shop operations
 - Do office work for maintaining daily production data (MIS) and making production plan, daily inspection challan etc. Any other job may be assigned time to time by EA/OA
- z) Appointment of Supervisors:
- a) Their appointment MUST be finalized within 7 days from the date of issuance of work order before commencement of actual work.
- b) These three supervisors must continue during the entire tenure of the contract. In case of resignation/ separation from the job, new appointments shall be finalized again within 7 days time from the date of issue of WO/Separation of previous representative with the consultation of Executing Authority/ Operating Authority.
- c) The Contractor has to provide One Desktop Computer to its Supervisors for noting down the records and providing MIS to EA.
- aa) Machine shop office complex to be cleaned on day to day basis to maintain hygiene of the complex for which one person should be dedicatedly working /available in machine shop office complex during working hours. The same person will also look after maintaining office decorum for visitors/officer/guest.

(B) MECHANICAL & ELECTRICAL UPKEEPING OF MACHINES & EQUIPMENTS& CNC MACHINES

- a) Cleaning , Lubrication in Guide ways/moving parts & Top up Gear Oil/Hydraulic Oil to be done by individual Machine Operators.
- b) Maintenance will be carried by another Maintenance Group on Contract.
- c) During maintenance of machine/equipment , the Contractor /machine operator will have to coordinate with Maintenance Group and be present at site if so required.

(C) OBLIGATION OF THE CONTRACTOR:-

- a) The Machining Plan for the next month shall be prepared in consultation with Head of Machine Shops by the last week of the current month based on of the load/urgency. This will be reviewed on fortnightly basis. The Plan shall be jointly signed and shall be adhered to. Any change in the Plan shall be through mutual agreement and covered in writing.
Every Job is important for SAIL, be it small or big. Strictly, no pick and choose for jobs taking up for machining are acceptable.
- b) Deployment of required number of Machinists, CNC Operators, CNC Programmer Crane Operators, Lister/Truck/Tractor Operator, Welders (Electric Welders & Gas Welders), Fitters, Electricians, Tool Grinder, Engineers/Qualified Supervisors each for Tool Room, Non-Ferrous Section, Steel Foundry Machine Shop& Inspection for Operation and optimum utilization of all machines and facilities to fulfill machining targets will be in the scope of the contractor. The manpower deployed can be of Multi Skill type. Each Shift should be headed by qualified Supervisor.
- c) The Contractor shall ensure good housekeeping at Shop Floor as well as shall provide necessary Safety appliances to the workers. No worker must be allowed to work without wearing safety equipment/ helmets. No worker must be allowed to work on the machines with loose clothes, finger rings etc.
- d) The contractor should have back up arrangement to cover the absenteeism of key personnel like crane /truck operator, marker& machine operators including CNC Machine operators etc.
- e) Arranging in process & final inspection of machined castings required by inspection wing of SGW Kulti and customer's inspectors. This must be followed strictly.
- f) The Contractor shall maintain records of Castings received from Foundries for machining with their weights, maintain Forward load cards, Inspection records, Stock of machined Castings with their actual weights, Castings to be sent back to Foundries for rework, Log books, Drawings, Drawing Register etc. including Daily report and all type of Challans in the approved Formats (to be provided by SGW, Kulti).
- g) The job is to be carried out as per the "Scope of work".
- h) The Contractor shall ensure that there is no idle operation of Machines.
- i) In case of break downs and emergency jobs, the work has to be carried out as per requirement. Work may be carried out on Sundays and holidays as per the urgency/requirement of the work. The 4 National holidays i.e. 26th January, 15th

August, 2nd October, 1st May & 24th January (SAIL day) shall be paid holidays for the workmen. The workmen shall be paid for these holidays even if these fall on Sundays.

- j) The Contractor shall offer the Castings for inspection by Inspection cell only after complete machining, finishing by grinding, chipping, filing, etc. and any other required activity such as Hydraulic testing etc. If required, the steel castings are to be painted with one coat of heat resistant primer and one coat of heat resistant paint after the final inspection is done and put to stock.

- k) During Operation of Machines, if the Contractor requests to take any Casting (ferrous or non ferrous) /Tool/Machine part/Equipment part outside the Company's premises, for machining which cannot be performed within the Works in quickest possible time then the contractor may be allowed to do so on giving a returnable gate pass issued by EA/OA. Contractor has to submit BG of equivalent amount of value of item **orequivalent amount of value will be held up from** monthly bill, till the material is brought back.

- Transportation Cost has to be borne by Contractor.
- Such Jobs shall be treated as if done in Machine shop- SGW and payment will be made as per payment formula mentioned at sl no. (ix) in scope of work – Pg-1.
- Cost of Swarf retained by Contractor during machining in outside premises shall be deducted from Bills considering 5 % material Loss.

- The Market rate of Swarf will be taken from MJ which will be binding on the Contractor.

- l) In case any ferrous or non ferrous items which is urgently required by SAIL Plants and cannot be machined in quickest possible time due to non-availability of man power or machine at SGW Machine shop, then the contractor will be allowed to **install their Machines /other facilities like heat treatment /heating etc. inside SGW. Space , Power and Tools , Water ,Compressed air etc. will be provided free of cost.**
- m) Due to unavoidable circumstances which may encounter during the execution period of the contract, SGW, may order to stop any job which is under machining at any point machining stage and may also order to restart the machining of the same job at any point of time (partly done job). The payment for such partly machined job will be done as per the formula of the payment/as per payment conditions.
- n) The Contractor shall ensure that right persons having experience/license are engaged for Jobs of special nature such as Operation of E.O.T Cranes, Electrical jobs, Truck Operation etc.
- o) The Contractor must provide Guarantee towards good workman ship before commencement of the contract.
- p) It will be the responsibility of the Contractor to Hand over and Take over charge on day - to-day basis from 'on duty' CISF personnel deployed at Machine Shops (Main Machine Shop & Steel Foundry Machine Shop) especially for Non-Ferrous items. In case, any item is found missing, the same shall be immediately reported to the Head of the Machine Shops in writing, otherwise the cost of the missing casting will be recovered from the bills.
- q) The Contractor shall be responsible for proper sealing of Steel Foundry Machine Shop, Non-Ferrous Machine Section, Non-Ferrous Cage in Machine Shop No. 1 and 2, Non-Ferrous Godown& Tools Store at the end of the day and hand over to CISF & opening of the same on the next day in presence of CISF personnel.
- r) For new items, the weight of the machined Items will be considered either as appearing in the Drawing or the actual weight as measured on the Company's Weighing Machine. In case the weight of a machined Item has been found to vary from the Drawing weight, then the same will be weighed and verified jointly by the Contractor and SGW, Kulti.
- s) To collect various Materials, Stores & Spares, Oils, Lubricants etc. from the Materials Department of SGW, Kulti.
- t) To provide Personal Protective Equipment (PPE) to his workers at contractor's cost. Use of Helmets, safety helmets, safety boots, flash back arrestor in H/T room while brazing of tools, mask, goggles etc. to be ensured. Non-use of PPEs by the contract employees would attract Penalty as per Safety rules violation clause.
- u) Flash back arrestor in H/T room while brazing of tools, mask, goggles etc. to be ensured at contractor's cost.
- v) **On account of Safety:**
- **Contractor has to deploy skilled Safety Supervisor**
 - **Distribute PPEs (Safety Shoes & Helmets , Hand Gloves of ISI- Make)and other safety appliances as per requirement to Workers twice in a year (first time within 15 days of issue of work order and second time within 15 days after completion of six months from date of work order)**
 - **Contractor has to submit declaration from the workers about receipt of PPEs and submit along with the Bill.**
- w) In case of use of inflammable gas, the gas detector is to be installed in the vicinity of the working area.
- x) Consumable Items like small Cutting tool, HSS drill bit, machine lamps, gunny bags/plastic bags for filling machined swarf ,jute/cotton waste, hand brooms, brushes for soldering, brushes for cleaning of

machines , kerosene, turpentine oil/any other cleaning agent, lime powder, PVC, TEFLONTAPES, Steam joints /packing / sealings etc., electrical insulating tapes, Clamps of battery, distilled water for batteries, acid for batteries, Antirust spray, contactor cleaner, emery paper, small screws up to 6 mm, belt resin, grinding wheel dresser, hack saw blades of ½” , electric bulbs and holders in the machine lamps etc. are to be arranged **at the Contractor's cost** (**This amount is in built in the Estimate and hence shall not be reimbursed**). **The Monthly expenses will be Rs 19,443/- .Monthly expenses more than Rs 19,443/- is allowed subject to maximum limit of Rs 2,33,313/- in a year .(excluding GST) and will have to produce the Bills of purchasing.**

- y) It will be responsibility of the Contractor to send the machined Castings to Non-Ferrous Foundry Repair Shop for any repair/rectification work on account of Foundry/Machining defect with proper documentation i.e. through Challans.

(D) OBLIGATION OF SGW, KULTI:

- a) To provide various Machines, Cranes, Heat treatment furnace and other tackles like wire rope Slings, Sisal rope, flat synthetic slings, Lifting Chain blocks/Pulleys, Welding Machine, Oxygen & DA Cylinders for brazing of tools, brazing rods, welding rods, adhesives like M seal, araldite, ana bond etc. brazing flux, Pneumatic Grinding Machine, Electrical Hand Drills etc.
- b) Hydraulic testing equipment.
- c) Fuel for Heat treatment Furnace.
- d) Equipment for transportation.
- e) Cutting tools including Grinding wheels.
- f) Lubricants & Coolants.
- g) Electrical power and Water
- h) Sheets for Profile templates/Gauges.
- i) Measuring Instruments.
- j) Fitting tools.
- k) Hardness testing equipment.
- l) Weighing Machines (One electronic- 500 Kg cap and one conventional dial type- 500 Kg) of Avery make.
- m) Arrangement for necessary Gate passes for workers and technicians.
- n) Paint /primer for general use. (border marking, painting of any casting other than mentioned in the fixed intricacies).
- o) Necessary electrical power supply for the Machines.
- p) Arrangement of Crane and necessary lifting arrangements.
- q) Paint /primer for general use. (border marking, painting of any casting other than mentioned in the fixed intricacies).
- r) Welding machine with holders, cable and electrodes.
- s) Bearing Joining agents brazing rods, borax powder , zinc chloride etc.
- t) Electric Wires.
- u) Refractory material for furnaces

(E) GUIDE LINES FOR BILLING:

Base Rate, Intricacy Factors & Machining Rates

a. Base Rate (BR):

The Base rate shall be arrived from the Price Quoted as illustrated in the Price Bid Format. The Rates (Over or at Par MFP) quoted by the contractors shall be all-inclusive lump sum. The contractors should take into account all the factors and costs that they may have to incur in Operation of Machines at the Machine Shops as per the “Scope of Work” of the Tender. -

Swarf weight (X) = A – B

Where, A = Actual weight of Black Casting (after necessary cleaning), weighed on Company's Weighing Machine in presence of Work Inspection of SGW, Kulti, and as appearing on the Foundry Challans

B = Actual weight of the Machined Casting, as appearing in the Drawing or weighed on the Company's Weighing Machine on Sample basis, **whichever is Higher.**

The Swarf weight, as calculated above, shall be accepted and binding on both Parties.

b) Intricacy Factor:-

The intricacy factors for the standard items like copper tuyres, coppertuyere coolers, slipper pads (Big size, medium size & very small size), split nuts , Zn alloy nuts & flat wearing strips/Flat liners have been fixed.

For the other items, depending on complexity/difficulty in machining of other Casting, various Intricacy Factors, as per the Intricacy Matrix, have been provided. The **Intricacy Matrix** has been enclosed at **Annexure-I**. Further, **Special Intricacies have been provided at Annexure-II**. The above Matrix as well as Special Intricacies shall be used for calculation of **Machining Rate (MR)** on item-to-item basis in terms of Base Rate and shall remain in force till the completion of the Contracted period.

As may be evident, the Intricacy Factors have been developed for various intricacy levels for different operations/activities and are percentage (%) of Base Rate, in the increasing order with increasing level of complexity/difficulty in machining

The Intricacy Factors with varying Intricacy levels have been provided for the following:-

- i) No. of Machines required in performing various operations during machining of a Casting.
 - ii) The highest required Surface finish on any surface in the Drawing of the Items.
 - iii) Hardness of the Item.
 - iv) Hydraulic pressure testing, if required as per the Customer's Work order.
 - v) Assembly of major parts (excluding nuts, bolts & washers etc.).
 - vi) Special Intricacy multiplying factors as per 1 (a), 1 (b), 1 (c), 2, 3, 4, 5, 6, 7 & 8 of Annexure-II.
- c) Calculation of Machining Rate (MR) on the basis of Base Rate (BR) & Intricacy Factor:**
- i) Based on above operations/activities to be performed, as may be found out from the Drawing/Work order, whenever any Drawing/Work Order is received for the first time for execution, the Machining Rate (MR) shall be worked out for each item before taking up the same for machining.
 - ii) The Machining Rate (MR) thus arrived at shall be jointly signed by the Contractor & SGW, Kulti representatives.
 - iii) The MR so agreed on item-to-item basis will remain firm for the Contracted period.
 - iv) The above Machining Rate (MR) shall be applicable for raising the Bills on monthly basis.

Calculation of Machining Rate (MR):-

Some Illustrations: INDICATIVE EXAMPLES:-

Suppose, the Base Rate (BR) is Rs. 10/-

Illustration:-1

Machining of Split Bearing

- i. Black weight: 22 Kg, Finished Weight: 13 Kg (for one half)
- ii. Amount of Swarf: 9 Kg.
- iii. Qty: 1 no.
- iv. The job requires use of six Machines i.e. (Shaper, Lathe, Vertical Milling, Horizontal Milling, Drilling,)
- v. Surface finish required is triple delta in Bore & on face
- vi. The hardness of the material is below 200 BHN.
- vii. No Hydraulic testing required
- viii. The job requires no assembly

No of Machines (5)	= 2BR = X
Surface Finish (Triple Delta)	= 1.4 (X) = 1.4 (2BR) = 2.8 BR = Y
Hardness (below 200 BHN).	= Z = Y = 2.8 BR
Hydraulic testing (Nil)	= P = Z
Assembly (Nil)	= Q = P
Thus Machining Rate (MR): Billing amount = Intricacy x BR x Swarf x Qty	
= 2.8 x 10 x 9 x 1 = Rs.252/- for one half of Bearing	

Illustration - 2: Blast Tuyere for RSP (Drawing No. PD/R-91919, Item No. 1)

- i. Black weight: 179 Kg, Finished Weight: 158 Kg (for one Casting)
- ii. Amount of Swarf: 21 Kg
- iii. Qty: 1 no
- iv. Fixed intricacy : 2.85 (For fixed intricacy, Refer Annexure –III)

Thus Machining Rate (MR): Billing amount = Intricacy x BR x Swarf x Qty

2.85 x 10 x 21 x 1 = Rs. 598.5/-

Illustration - 3: C.S. Pig Mould for BSP (Drawing No. B 401619 R1)

- i) Black weight : 105 Kg, Finished Weight: 104 Kg (for one Casting)
- ii) Amount of Swarf: 1 Kg.
- iii) Qty: 1 no
- iv) The job requires use of 2 Machines i.e. (Horizontal Boring & Drill)
- v) Surface finish required is one delta
- vi) The hardness of the material is below 200 BHN.
- vii) No Hydraulic testing
- viii) The job requires no assembly

No. of Machines (2) = 1.40 BR = X Surface finish (normal) = Y = X = 1.40 BR Hardness (below 200 BHN) = Z = Y Hydraulic testing (not applicable) = P = Z Assembly (nil) = Q = Z Thus MR = 1.40 BR.

Thus Machining Rate (MR): Billing amount = Intricacy x BR x Swarf x Qty
= 1.4 x 10 x 1x1 = Rs 14/-

Applying condition 1(b) of Annexure-II

Billing amount will be: Rs. 14 x 3= Rs 42/-

**Illustration - 4: P.B. Slipper Pads (Large & Medium size Slipper pads)-
RSP Pad CDB/R-18397 :**

- i) Black weight: 270 Kg, Finished Weight: 190 Kg (for one Casting)
- ii) Amount of Swarf: 80 Kg
- iii) Qty: 1no
- iv) Fixed intricacy : 4.5

Thus Machining Rate (MR): Billing amount = Intricacy x BR x Swarf x Qty
= 4.5 x 10 x 80 x 1 = Rs. 3600/-

Illustration - 5: C.S Roll Coke Liner DrgNo:-000.067A

- i) Black weight : 1600 Kg, Finished Weight: 1330 Kg (for one Casting).
- ii) Amount of Swarf: 270 Kg
- iii) Qty: 1 no
- iv) The job requires use of 2 Machines i.e. (vertical Boring and slotting).
- v) Surface finish required is one delta
- vi) The hardness of the material is above 250 BHN
- vii) No Hydraulic testing.
- viii) The job requires no assembly.
- ix) As per special intricacy, point no: 12, M.R. will be multiplied by 0.90 for swarf more than 100 kg.

No of Machines (2) = 1.40 BR = X

Surface Finish (Normal). = Y = X = 1.40 BR

Hardness (More than 250 BHN- 1.8 Factor) = Z = 1.8 Y

Hydraulic Testing (Not applicable) = P = Z

Assembly (Nil) = Q = Z

Applying condition 12 of Annexure –II

Thus Machining Rate (MR): Billing amount = Intricacy x Hardness x BR x Swarf x 0.9x Qty
= 1.4 x 1.8 x 10 x 270x 0.90 x 1 = Rs 6123.60/-

THE FOLLOWING ARE ATTACHED IN SEPARATE FILES:

1. LIST OF JOBS DONE IN YEAR 2023- 2025 TILL MAY 25-
ANNEXURE-A
2. INTRICACY MATRIX/TABLE – ANNEXURE –B
3. SPECIAL INTRICACY – ANNEXURE –C
4. FIXED INTRICACY – ANNEXURE –D(Pg-1 & 2)
5. MINIMUM MACHINING RATE OF NEW ITEMS BASED ON MFP-
ANNEX-E
6. LIST OF MACHINES - ANNEXURE – F
7. INDICATIVE PENATLTY TABLE-G

GENERAL TERMS & CONDITIONS.

1. EVALUATION OF L-1 TENDER:

The Tenderer shall take into account all costs that he may have to incur in terms of this Contract for 'Operation of Machine Shops' as per the Scope of Work and should quote his all-inclusive Lump sum Rate for the Item of work as per the format provided in the "Schedule of Rates", else the Tender shall be treated as invalid and rejected.

- The Tenderers are required to quote Rates above or at Par the Minimum Floor Price. Rates below MFP are not permitted.
- Party quoting lowest Price will be decided as L-1 Party.
- If the rates are quoted at par by more than one party, then lottery will be taken out to decide L-1 Party.

The quantities indicated in the Scope of work & 'Schedule of Rates' are approximate.. There may be variation in the actual quantities. The Contractor shall have no claim on the quantities mentioned in the 'Schedule of Rates' and payment shall be made on the basis of actual executed quantities only.

The Order shall be placed on one Party.

2. ESCALATION/REVISION OF RATES:

- a) The contractor shall have to pay minimum wages to his workmen as per notification **no.45/Stat/2RW/32/94/LCS/JLC Dated 31-01-2022 or whichever is latest of W.B. Govt.** Minimum wages applicable to "Construction or Maintenance of Roads or in Building operation in W.B (applicable to SGW, Kulti) in three categories – unskilled, semi-skilled and skilled under zone- A".
- b) In case of any increase of such minimum wages / wages related payments (e.g. PF, ESI, Annual Bonus etc.) due to any Govt. Notification / SAIL Notification (e.g. Additional welfare Amenities etc.) issued on or after the base date, during the tenure of the Contract, such difference shall be paid by the Contractor to his workers and subsequently the same shall be reimbursed by the Company based on actual basis on submission of documentary evidence in support of his claim..
- c) Similarly, Any decrease of such wages from base date during the tenure of the contract will be realized from contractor monthly running bill in the reverse mode
- d) Towards payment of AWA by the contractor to his employees (i.e. maximum limit of Rs. 3300 per month) @ Rs. 126.92 per day on actual attendance with ESI, the same will not be linked to wages. Hence, above escalation/de escalation will not be applicable for AWA.
- e) **The base date for the above escalation clause shall be the date of reverse auction / last date of submission of tender documents (whichever is later)**

3. IN THE EVENT OF PARLIAMENTARY/STATE ELECTIONS, IF THERE IS A DECLARED HOLIDAY BY THE GOVERNMENT, THEN PAYMENT OF WAGES FOR THIS DECLARED HOLIDAY WILL BE REIMBURSED TO THE CONTRACTOR ON ACTUALS.

4. INSURANCE:

The Contractor shall provide Group Insurance coverage, as necessary, for any eventuality at work place, to all his workers engaged for the Job.

5. TAXES AND DUTIES:

The Rate quoted by the tenderer should be inclusive of all applicable taxes, duties, levies, Cess and all other statutory obligations/expenses including Contractor's contributions towards P.F. {present rate is 13.6% of minimum wages (including 1.6% as Administrative Charges)}, ESI (present rate is 4.75% of minimum wages), Annual Bonus (present rate is 8.33% of minimum wages), payment against four (04) National Holidays and SAIL day (24th January) even if it falls on Sunday, Additional welfare amenity of maximum Rs. 3300/- per month per worker & ESI contribution thereon as mentioned below, Personal Protective Equipments etc. for the workers engaged for the Work, but excluding GST and shall be firm during the entire period of Contract ((except increase in minimum wages as mentioned under escalation clause).

GST will be reimbursed by the Company at actuals on submission of relevant documents subject to that documentation is done as per GST rule and GST Points as per Annexure-G

In case any Material/Spare part for any Equipment is required to be supplied in emergency, the applicable rates of GST, if any, should be clearly indicated in the bills. These shall be reimbursed by the Company on production of GST Invoice etc. (subject to that documentation is done as per GST rule and GST Points as per Annexure-G) (to be read in conjunction with Scope of work).

6. PAYMENT OF WAGES ETC. BY CONTRACTOR TO HIS WORKERS:

a. The Contractor shall make payment to his workers within seven (07) days of completion of a month, and the wages paid by him to his workers shall not be less than the minimum wages payable as per the Minimum Wages Act as notified by Govt. of West Bengal from time to time for various Categories. The Payment of wages by the Contractor to his workers must be made through Bank by directly depositing the amount payable, to the Bank Account of the Worker. The Contractor will be required to submit the copy of the Bank document in support of the said deposit along with his Bills for necessary verification by the Finance Department.

b. The Contractor shall be fully responsible to comply with all his Statutory obligations/liabilities as Employer in respect of their labour engaged by them for the Job undertaken under the Contract as per applicable Statutory provisions/Law and Government Notification, and will take full liability on this account. The Tenderers should bear this in mind before submitting their Offers and may refer Clause - 8, herein above for guidance. The Contractor shall ensure that the 'Wage Slip' is issued one day in advance to his workers.

Further, the Contractor will ensure that all Half yearly as well as Yearly Returns are submitted in time with the appropriate authority. The Company will not take any liability on this account.

c. The Contractor shall deposit their own Contribution as well as Contribution of their workers, engaged under the Contract, towards P.F. and ESI well in time and submit copies of the Challan with the Bills for verification. Release of payment to the Contractor shall be made after due verification of the actual executed quantities by the Executing Authority and verification of copies of various Challans, as proof of compliance of statutory requirements and payment of all dues to the Contractor's Workmen, by Finance Department.

d. In the event of failure of the Contractor to comply with the above, the Company shall be entitled to recover the amount by deduction from any amount payable to the Contractor under the Contract, including Security Deposit, or as debt payable by the Contractor.

In case the contractor fails to make payment of wages of the workers by 7th of the month, SGW, Kulti being the principal employer, reserves the right to make such payment of wages to the workers and the amount so paid plus 15% as administrative fee will be recovered from any of contractor's bill. In case the contractor fails to pay wages to his worker in time for the second time, action as deemed fit shall be taken against his firm. Timely and regular payment of wages will also be a criterion for evaluation work execution of work performance.

Additional welfare amenity for Contract workers:

The contractor shall pay the additional welfare amenity in the form of allowance (not linked to wages) @ Rs. 126.92 per day of actual attendance, subject to a maximum of Rs. 3300/- per month to each of the contract labours engaged, ESI contributions shall be payable on this allowance by the contractor. Quoted rates shall be considered accordingly and no extra payment shall be made to the contractors on this account.

7. The Tenderer shall be required to carry out the entire work with his own resources including men and any other item required for the work.
8. The job shall be carried out under the direct supervision of the Contractor. Inspection shall be done at site by executing authority or his authorized representative as per scope of work.
9. The Contractor shall be responsible for the safety of his workers and no claim whatsoever shall be entertained by the Company in this regard. The party has to observe all safety rules / precautions and arrange all safety appliances and take necessary instructions from SGW, Kulti executing authority before starting the job.
10. As per terms & conditions, the contractor has to make an action plan for execution of the jobs and the same to be discussed with the executing authority for smooth completion of the job. The contractor has to start the job and mobilize the resources to complete the job in scheduled time.
11. The Contractor will ensure payment of minimum wages through bank to his workers as per State Govt. Notification of West Bengal and shall have to submit True copy of the Payment Sheet for the payment made to his workers as proof of compliance.

12. GENERAL CONDITION OF CONTRACT (GCC):-

The contract shall also be governed by the General Conditions of Contract. Copy of General Conditions of Contract is available at the Contract Cell of SGW, Kulti. The contractors are free to go through the same on any working day during schedule working hours. Special conditions shall prevail in case of conflict with GCC.

13. SPLIT UP OF RATES:-

Order shall be placed on one Party on the basis of L-1 offer and, if required, negotiation will be held with L-1 tenderer only. However, all the tenderers may be required to explain / justify the basis of their quoted price as and when asked for. In case, any tenderer fails to justify his quoted price or refuses to cooperate in this regard, he will not be considered for participating in the retendering, if Order / Contract is not finalized from the present tender.

14. SECURITY DEPOSIT:-

The successful Tenderer shall deposit an amount equivalent to 3% of the total Work value (exclusive of GST) towards Security Deposit. 2% of the work order value has to be deposited before start of the job while 1% shall be recovered from the 1st running/ final bill whichever is earlier. The Security Deposit less the Earnest Money already deposited (if Any) is to be furnished by the successful tenderer **through on line mode /Bank draft /Bank Guarantee/ Banker's Cheque/Fixed Deposit Receipt (FDR)** from any Scheduled Commercial Bank except Co-operative and Gramin Bank but shall not be accepted in cash., in favour of "SAIL Growth Works, Kulti", payable at 'State Bank of India, Kulti', *within ten (10) days from the date of issue of Work Order*. The Security Deposit shall bear no interest and refund will be made as per provision.

In case the contractor delays submission of Security Deposit , a penalty of 0.2% of the original SD per completed day of delay, limited to an amount not exceeding 5% of the original SD shall be applicable. Payment for executed portion of order/contract should only commence on submission of requisite SD

15. REFUND OF SECURITY DEPOSIT:-

The Security Deposit shall be retained by the Company for a period extending one month beyond the validity period of the Contract . *In case the Contract is extended, the validity period of the Security Deposit shall also have to be extended accordingly.* The Security Deposit shall be released after completion of the validity of the Contract Period as above and on submission of “No Dues Certificate” by the Contractor obtained from the Executing Authority and the Operating Authority and compliance of statutory provisions after deduction of any dues from the Contractor arising out of any claim by SGW, Kulti on the Contractor.

16. PERFORMANCE GUARANTEE BOND:-

If a tenderer quotes unworkable Rates, i.e. if the quoted rate is abnormally low, such that it raises material concerns as to the capability of the Bidder to perform the Contract at the offered rate, the Party will be asked to justify the rate quoted. After evaluating the justification provided, if it is felt that the bidder has failed to substantially demonstrate its capability to deliver the contract at the offered rate, the bid/quote may be rejected.

However under compelling circumstances, where the bidder is considered for placement of order despite having quoted abnormally low rates, the Bidder may be required to give Performance Guarantee Bond (PGB) (in addition to the Security Deposit, if applicable) through online mode / Bank Draft / Bank Guarantee/Bankers Cheque / Fixed Deposit Receipt (FDR). The amount of PGB to be decided by SAIL/Plant at the time of placement of order which will be the amount less than lower limit -20% of the Estimate.

In case the Tenderer refuses to give PGB, the following action may be taken against them

If they have submitted Earnest Deposit – their EMD will be forfeited and they will not be considered in re-tendering if order/contract is not finalized from the present tender.

If they have submitted Bid Securing Declaration – action may be taken against them as stipulated in the Bid Securing Declaration.

In case the contractor delays submission of Performance Guarantee, a penalty of 0.2% of the original PG amount per completed day of delay, limited to an amount not exceeding 5% of the original PG amount shall be applicable. Payment for executed portion of order/contract should only commence on submission of requisite PG.

17. RISK & COST:

In case of failure on part of the contractor to complete the entire job within the scheduled completion period as per specification and satisfaction of the Executing & Operating Authorities, the Company will be at liberty to get the unexecuted quantities of the job done through any other agency at the risk & cost of the original Contractor.

Notice for execution of order on risk & cost of any contractor will also be issued during valid period of contract if vendor / contractor either fails to start the work / supply within reasonable time, the progress of the work / supply is poor or for any breach of contract.

The price differential in case of higher cost to SAIL, if any, shall have to be borne by the defaulting Contractor. Moreover, the defaulting Contractor shall have no claim over the quantity, which they failed to execute.

18. LIST OF WORKERS TO BE ENGAGED:-

The Contractor shall submit a Category wise list of workers, to be engaged by him under the Contract, as per the scope of work before start of the work to the Executing department, Personnel department and the Contract cell.

NOTE: WORKERS ABOVE 60 YEARS OF AGE ARE NOT PERMITTED TO WORK

19. FINAL HANDING OVER OF MACHINES/EQUIPMENTS:

During the currency of the Contract, if any Machine or Equipment is damaged or rendered unserviceable due to mishandling or negligence directly attributable to the Contractor, the cost incurred on repair/replacement of the same shall be recovered from the Contractor's Bills. Further, it will be the responsibility of the Contractor to hand over the Machines & Equipments back to SGW, Kulti in good working condition on expiry of the Contracted period. In case, any Machine or Equipment is found damaged during Handing over, the same shall be repaired/rectified by the Contractor within 15 days or else the cost of such repair/replacement shall be recovered from the Contractor's Bills/Security Deposit.

20. VALIDITY PERIOD OF CONTRACT:-

One Year from the date as mentioned in the Work Order or actual date of commencement of Work at Site, whichever is earlier.

21. CONTRACT DOCUMENT: -

The several documents forming the Contract are to be taken as mutually explanatory to one another. Should any condition in 'General Conditions of Contract (GCC)' be contrary to 'Special Conditions of Contract (SCC)', then 'Special Conditions of Contract' shall prevail.

22. SAFETY: -

All Safety precautions on the jobs are to be taken by the Contractor. The Contractor shall ensure provision of all required Personal Protective Equipments, as may be necessary, for the type of job being performed by his workmen. The Contractor shall arrange for Medical check up of his workers at regular interval and shall ensure that the workers working at height are fit for such work.

Important: The contractor will be wholly responsible for taking safety measures and compliance of safety of their workforce. The contractor will have to ensure the compliance of safety rules including arrangement of safety gears / appliances for their workers and supervising of safety measures to be taken from their own resource.

New standard clause on safety violation

- In every contract/ work order/ acceptance of tender, the following clause shall be substituted in place of existing one and if no clause exists earlier, the following clause should be added forming part of the contract.
- “Safety of Persons/ Plant & Equipment being an important matter of the contract, no laxity, omission, violation will be tolerated. A defaulting contractor/ party will be liable to pay a penalty in the form of damages for violation of safety provisions which have been categorised as minor violation/ major violations as enumerated in Annexure I & Annexure II in the following manner.

Category of violation	Penalty per violation (Rs.)
1. Minor Violation	Rs. 3000/- for first violation. For every subsequent violation Rs. 6000/-
2. Major Violation resulting in a Death/ Permanent disability arising out of work due to accident	Up to Rs. 5 lakh
b Injury arising out of work	Rs. 25,000/- subject to maximum of Rs. 1 lakh depending upon the gravity of injury

The contractor shall also be liable for proceeding of banning of business as per prevalent procedure. The decision of the Competent Authority on the safety violation shall be final. The Competent Authority shall take into consideration the report of enquiry committee if constituted to make an enquiry”

Minor Violations

1. Engaging workers without proper safety training
2. Unauthorized entry in hazardous locations
3. Blocking access of emergency equipments or exits
4. Using defective electrical cables, faulty joints in cables
5. Connection taken from board without proper plugs
6. Working without hand gloves/ safety shoes/ safety helmets wherever required
7. Non usage of flash back arrester in cutting sets
8. Rolling of cylinder / dragging on the ground
9. Welding with non-standard holder
10. Taking electrical connections without using ELCB/RCCB
11. Gas hose pipe clamping done by wires
12. Improper use of gas cylinders
13. Wet bags not put on gas cylinders
14. Condition of oxygen/ fuel hose pipes in gas cylinder is not good
15. Not having proper gate pass/ other area passes
16. Use of unsafe slings/ tools/ ropes
17. Use of material handling equipment for other purpose
18. Improper use of hand grinders/ mixture machine
19. Working at height without valid medical certificate
20. Violation of Motor Vehicle Rules
21. Dumping excavated earth on the edge of excavation
22. Dropping/ spillage of material on the road
23. Inadequate road worthiness of mobile equipment used inside the factory
24. Proper ladder/ steps not provided for ascending/ descending
25. Use of improper rope in crane operation

26. Driving of heavy vehicle during restricted hours
27. Parking at unauthorized place

Major Violations

1. Using bamboo or other non-standard material for scaffolding
 2. Railing not provided at working platforms
 3. Unsafe working at height
 4. Workplace left without guarding
 5. Work permits not obtained for hazardous jobs
 6. Not complying with written instructions/ working without proper permission in hazardous area
 7. Absence of supervisor at work in hazardous area
 8. Proper ladder / approach not given for working at height
 9. Inadequate sloping/ shoring of excavated pits
 10. Dismantling of structure without authorized plan
- Tempering with existing operational facilities.

23. PAYMENT TERMS:-

PAYMENT FOR MACHINE SHOP CONTRACTOR – MACHINING CASTINGS AT SGW MACHINE SHOP:

- (a) The Bills shall be raised on the basis of actual quantities executed during the previous month and shall be accompanied with copies of Challans. **The Bills are to be submitted in triplicate within 7 (seven) days of completion of a month.**
- a) Payment shall be made for the actual machined Castings/manufactured, passed by the Inspection wing of SAIL Growth Works, Kulti between 1st day and last day of the previous month based on Challans. No payment shall be made for the Castings rejected on account of defect in machining/manufacture during inspection.
- b) Further, in case of rejection during inspection on account of defect in machining, the cost of machining and the cost of manufacturing of such input material/ Castings shall be recovered from the bills of the Contractor after making adjustments for the weight of the metal of Castings.
- c) **The rates applicable for different items shall be as per the Billing Guide Lines Mentioned earlier .**
- d) In case, the Castings rejected on account of machining/defect in manufacture, but can be salvaged through repair/rectification of the same through approved means (welding/grinding etc.) and are subsequently offered for inspection and are passed, the Contractor shall be paid for such reclaimed Castings after making adjustments for the cost incurred on account of charges for the Company's materials used like electrodes, gas etc.
- e) In case the Castings require rework on account of defect in manufacturing, the same shall be sent back to the respective Foundry through Challans for repair through approved means (welding, caulking, grinding etc.). Such Castings, on receiving back at Machine Shop, shall be re-machined and then re-offered for inspection. If the Casting is passed, the cost of salvaging and re-machining shall be charged back on the Contractor who had manufactured the Casting in the Foundry. The Contractor at the Machine Shop shall be given charges for machining (prior to rework) as well as for re-machining.

- f) In case, the Castings passed by the inspection wing of SGW, Kulti is subsequently found to be rejected at Customer's end, due to defect revealed afterwards, and the rejection is attributed to defect in machining in case of Castings, the cost of manufacturing and cost of machining of such Castings shall be recovered from the subsequent Bills of the Contractor after making adjustments for the metal weight/recovered Items, if any.
- g) On completion of the Contract Period (last day of the Contract); during Handing over/Taking over, only the Semi machined Castings loaded on the Machines under process of manufacturing on that particular day, shall be jointly inspected in presence of EA/OA and all the semi-finished items will be weighed.
 - **Payment to the Outgoing and in coming Contractor** for the semi-finished or work in progress Castings will be made as per following formula:

 (Weight of Black Casting – Weight of complete finished Casting –Drawing Finished weight or actual weight available) x % age Job completed.
 - % age Job completed will be **decided by SGW which shall be final and binding on the outgoing Contractors.**
- h) The Bills shall be supported with all required documents as per the Tender, including copy of the Bank Challan as proof of payment of wages to the workers engaged under the Contract through Bank as well as copies of various other Challans, as proof of compliance of statutory requirements and payment of all dues to the Contractor's Workmen, for verification by the Finance Department.
- i) On verification of the Bills by the Executing and the Operating Authorities for actual quantities executed and subject to any deductions that may have to be made as per the terms of the Contract and provided that all statutory requirements are met, the Bills shall be cleared for payment and the payment shall be made by SGW, Kulti through account payee Cheques within 30 (thirty) days of submission of clear Bills.
- ii) GST and Cess, paid by the Contractor, shall be reimbursed by the Company on production of documentary evidence.
- iii) Income Tax shall be deducted from the Bills of the Contractor as per Income Tax Act, 1961. The Certificate for Tax Deducted at Source (TDS Certificate) shall be provided to the Contractor at the end of the financial year.
- (b) The job shall be carried out under the direct supervision of the Contractor. Inspection shall be done at site by executing authority or his authorized representative as per scope of work.
- (c) The Contractor shall provide tools/ materials, manpower etc as per scope of work.
- (d) As per terms & conditions, the contractor has to make an action plan for execution of the jobs and the same to be discussed with the executing authority for smooth completion of the job. The contractor has to start the job on taking hand over of site and the site clearance and mobilize the resources to complete the job in scheduled time. The contractor has to clean the area after completion of the job.
- (e) The party should work to the complete satisfaction of the executing authority.
- (f) The quality and quantity of work shall be inspected by executing authority.
- (g) Any extra imposition of tax/duty/levy etc. by Govt. authorities after tender / finalization of contract & during the period of contract shall be reimbursed.

- (h) The Contractor will ensure payment of minimum wages through bank to his workers as per State Govt. Notification of West Bengal and shall have to submit True copy of the Payment Sheet for the payment made to his workers as proof of compliance.
24. SGW, Kulti's decision regarding acceptance in whole or part or summary rejection of tenders submitted without assigning any reason shall be final and binding.
25. LABOUR WAGES: The contractor is required to pay to the Workmen, through bank, the minimum wages (**shall not be less than Minimum Wages Act of West Bengal**) and provident fund & ESI notified as per circular/order/notification issued by the Principals/Govt. authorities from time to time and also to comply with the various provisions of the law/circulars applicable to them. In case they default on the payment to the Workmen the Minimum Wage/Provident Fund/ESI or other dues or violate any other provisions of the Law, the Principals reserve the right to pay the Workmen the said amounts on behalf of the Contractor and recover the same from the Contractors bills.
26. ARBITRATION:
In the event of any dispute or differences arising under or out of the Contract, the same shall be referred to the Sole Arbitrator or any person nominated by the Executive Director, SAIL, SGW, Kulti. The award of such Arbitrator shall be final and binding on the Parties to this Agreement.
26. *Employer (Company) reserves its rights to remove from list of approved suppliers / contractors or to ban business dealings if any Agency has been found to have committed misconduct and also to suspend business dealings pending investigation.*
27. *Should any provision in the GTC of contract be contrary to the invitation to Tender, the provisions in the invitation to Tender shall prevail.*
28. **JURISDICTION:** The Asansol Court shall have the exclusive jurisdiction upon any matter arising out of this Contract.

PRICE BID (PART – II)

SCHEDULE OF RATES

Job Description	MFP	Quote Plus or at par. Minus Not Permitted
Operation of Machine Shop- Machining of Items as per the Monthly Plans.	<u>Minimum Floor Price- Rs 1,27,68,070.77</u>	

Note:

- L-1 shall be decided on the basis of Lowest Price Quoted. Below MFP Not Permitted.
 - Lottery will be taken out if Quotes are at Par.
- (a) The total value of the Work shall be Price Quoted as mentioned above.
- (b) Billing formula: $BR (Rs/Kg) \times \text{Swarf Generated (Kg)} \times \text{Intricacy as applicable.}$
- (c) $BR (Rs/Kg) : \text{Price Quoted (Rs)} / (\text{Avg Intricacy } 2.376 \times \text{Swarf in Kg in one year})$

Example :

- Suppose Price Quoted : Rs 1,30,000,00/-
- Targeted Swarf in 1 Year : 1,00,000 Kg
- Avg Intricacy taken for calculation of work order value: 2.376
- BR will be : $(1,30,000,00) / (2.376 \times 100000) = 54.71$

Note:

- **THE AVERAGE LEVEL OF INTRICACY FACTOR OF JOBS IN MACHINE SHOP COMES TO 2.376 . THERE FOR AVERAGE INTRICACY FACTOR OF 2.376 HAS BEEN TAKEN FOR CALCULATION OF BASE RATE WHICH WILL BE MENTIONED IN THE WORK ORDER.**
- **FOR BILLING , ACTUAL INTRICAIES SHALL BE GOVERENED BY INTRICACY TABLES(Annexure-C, D- Pg-1 & Pg-2 & E) AND CORRECTION IN INTRICACY FACTOR FOR NEW/SPECIAL JOBS AS MENTIONED ON FIRST PAGE OF SCOPE OF WORK , A-(ii).**

Bid-Securing Declaration
(In lieu of EMD)

Date:

Bid No.:

Alternative No.:

To

SAIL GROWTH Works Kulti

I/We-----, the undersigned, declare that:

I/We* understand that, according to your conditions, bids must be supported by a Bid-Securing Declaration in lieu of Earnest Money Deposit.

I/We understand that if I/We withdraw or modify our Bids during the period of validity, or if

I/We are awarded the contract and I/We fail to sign the contract, or to submit a performance Security before the deadline defined in the request for bids document, I/We will be suspended for the period of six months from being eligible to submit Bids for all future contracts.

I/We* understand this Bid Securing Declaration shall cease to be valid if I am/we are* not the successful Bidder, upon the earlier of (i) the receipt of your notification of the name of the successful Bidder; or (ii) thirty days after the expiration of the validity of my/our* Bid.

Yours faithfully

[insert signature of person whose name and capacity are shown] In the capacity of
[insert legal capacity of person signing the Bid Securing Declaration]

Name: *[insert complete name of person signing the Bid Securing Declaration]*

Duly authorized to sign the bid for and on behalf of: *[insert complete name of Bidder]*

Dated on _____ day of _____, _____ *[insert date of signing]*

Corporate Seal (where appropriate)

[Note: In case of a Joint Venture, the Bid Securing Declaration must be in the name of all partners to the Joint Venture that submits the bid.]

Annexure-B

(To be submitted by the Tenderer)

DECLARATION ON SITE INSPECTION

This is to state that we have visited the Site of work and the same has been inspected and fully examined by us. This is to confirm that the Rates in the Tender have been quoted by us after considering all aspects.

Signature of the Tenderer with Seal

FORMAT

NON-COLLUSIVE TENDERING CERTIFICATE
(to be signed by an authorised person on the Tenderer's behalf)

To

Plants/ Units
Steel Authority of India Limited
Place

Dear Sir/ Madam,

Non-Collusive Tendering Certificate for {RFQ/ Tender No. () Date ()}

1. We, (name(s) of the tenderer(s)) of (address (es) of the tenderer(s)) refer to the bid/ offer against (the "Tender").
2. **Non-collusion**

We represent and warrant that in relation to the Tender:

- (a) Our bid was developed genuinely, independently and made with the intention to accept the Contract it awarded;
- (b) Our bid was not prepared with any agreement, arrangement, communication, understanding, promise of undertaking with any person (including any other tenderer or competitor) regarding:
 - i) prices;
 - ii) methods, factors or formulas used to calculate prices;
 - iii) an intention or decision to submit a bid;
 - iv) an intention or decision to withdraw a bid;
 - v) the submission of bid that does not conform with the requirements of the tender;
 - vi) the quality, quantity, specifications or delivery particulars of the products or services to which this tender relates; and
 - vii) the terms of the bid, and we undertake that we will not, prior to the award of the Contract, enter into or engage in any of the foregoing.

3. Disclosure of in case of Job/ Project Contracts

We understand that we are required to disclose all intended sub-contracting arrangements relating to the Tender to the Plants/ Units-----, Steel Authority of India Limited, Place -----, including those which are entered into after the Contract is awarded. We warrant that we have duly disclosed and will continue to disclose such

arrangements to the Plants/ Units-----, Steel Authority of India Limited, Place -----
---.

4. Consequences of breach or non-compliance

5. We understand that in the event of any breach or non-compliance with any warranties or undertakings in this certificate, the Plants/ Units-----, Steel Authority of India Limited, Place ----- may, at its discretion, invalidate our bid, exclude us in future tenders, pursue damages or other forms of redress from us (including but not limited to damages for delay, costs and expenses of re-tendering and other costs incurred), and /or (in the event that we are awarded the Contract) terminate the Contract

Signed for and on behalf of the (tenderer)

Signature:

Name:

Position:

Date:

Note:

Para 2(b) is not applicable to Agreements, arrangements, communications, understandings, promises or undertakings with:

- (a) the Plants/ Units, Steel Authority of India Limited, Place;
- (b) a joint venture partner, where joint venture agreements, arrangements, relevant to the bid exist and which are notified to the Plants/ Units-----, Steel Authority of India Limited, Place -----;
- (c) consultants or sub-contractors, provided that the communications are held in strict confidence and limited to the information required to facilitate that particular consultancy arrangement or sub-contract;
- (d) professional advisers, provided that the communications are held in strict confidence and limited to the information required for the adviser to render their professional advice in relation to the Tender;
- (e) insurers or brokers for the purpose of obtaining an insurance quote, provided that the communications are held in strict confidence and limited to the information required to facilitate that particular insurance arrangement; and
- (f) Banks for the purpose of obtaining financing for the contract, provided that the communications are held in strict confidence and limited to the information required to facilitate that financing.

GENERAL CONDITIONS OF CONTRACT (GCC)

1.0 **DEFINITIONS:**

- In the Contract (as hereinafter defined) the following words, definitions and expressions shall have the meanings hereby assigned to them, except where the context otherwise requires.
- 1.1 **"Approved"** means approved in writing, including subsequent written confirmation of previous verbal approvals, and "approval" means approval in writing including as aforesaid.
- 1.2 **Company** means Steel Authority of India Ltd. having its Registered Office at "Ispat Bhawan", Lodi Road, New Delhi –110 003
- 1.3 **Competent Authority** means Head of the Deptt. of the Employer and any other Executive empowered under its Delegation of Powers and also includes its nominee.
- 1.4 **Contract** means the Invitation to Tender, Instruction to Tenderers, General Conditions of Contract, Special Conditions of Contract, Drawing Specifications, Scope of Work showing approximate quantities, Tender submitted by the Tenderer, including his Price offer, Performance Guarantee Bond and Other Bonds, Letter of Intent, Work Order and any communication having the effect of amendment of the Contract.
- 1.5 **Contract Price** means the sum named in the tender that has been accepted, subject to such addition thereto or deduction there from as may be made under the provision of the Contract.
- 1.6 **Contractor** means "the tenderer" who's Tender has been accepted and includes the Contractor's authorized representative, successors or permitted assigns, legal heirs.
- 1.7 **Director** means and includes the Director (Technical), SAIL, New Delhi. and includes the Executive Director (GD & Kulti)/General Manager (SGW), who is exercising the power of the Director in respect of award of Contract, or the Director's designated representative, or other empowered Contract awarding authority. The Director shall make all determination of the Employer/Company with respect to the Contract, except those specifically reserved to the Engineer.
- 1.8 **Duration of Contract** means the period stipulated in the Contract or Work Order and includes any extended period thereof, if any made by written communication.
- 1.9 **Employer** means the **"SAIL Growth Works, Kulti" of Steel Authority of India Ltd.,** a Company registered under the Indian Company's Act, 1956 and includes its legal successors and assigns. Wherever in the Contract, instructions/directions are to be issued by the 'Employer', the Chief Executive of SAIL Growth Works, Kulti or his duly authorized representative shall issue such instructions/directions.
- 1.10 **Engineer** means Head of the Executing Department or any other person appointed from time to time by the Employer and notified in writing to the Contractor to act as Engineer for the purpose of the Contract. Where work must be "Approved" or be performed "As Directed", or "As required" (or words of like import), the approval, direction or requirement will be made by the Engineer.
- 1.11 **Engineer's Representative** mean any assistant of the Engineer appointed from time to time by the Engineer to perform the duties set forth in Clause-2 hereof whose authority shall be notified in writing to the Contractor by the Engineer.
- 1.12 **Equipment** means all tools, instruments, appliances or things of whatsoever nature required in course of execution, of the work or temporary work.
- 1.13 **Notice in writing** or written Notice means a Notice in written, typed or printed characters sent (unless delivered personally or otherwise provided to have been received) by Registered Post/Courier (with POD) to the notified address, or the Registered Office of the addressee, or the Contractor's Site Office, and shall be deemed to be sufficient service, if so sent or left at that address.
- 1.14 **Terms & Conditions** means the 'Special Conditions of Contract' and the 'General Conditions of Contract' herein mentioned and other stipulations incorporated in any part of Tender Documents.
- 1.15 **Tender** means Offer against Enquiry/Advertisement/Downloaded from SAIL Website submitted by Tenderer.
- 1.16 **Tenderer** means and includes the person or firm or Company who have submitted valid tender and also includes its authorized representatives, heirs, executors, administrators, successors and assignees as approved by the employer.
- 1.17 **Work** means all work as given in the 'Scope of Work' in the Tender Document and includes any associated work required for fulfillment of the Scope of Work and as set forth and required by the Specifications, Drawings & Schedules hereto Annexed, and also such additional instructions and drawings as shall from time to time during the progress of work be supplied by the Employer.
- 1.18 Words imparting the singular only shall include the plural and vice-versa. Where the context require, Words imparting person only shall include firms and Corporations, and vice-versa.

2.0 **DUTIES OF ENGINEER'S REPRESENTATIVE:**

- 2.1. The duties of Engineer's Representative are to watch and supervise the work. He shall have no authority to relieve the Contractor of any of his duties or obligations under the Contract except as expressly provided hereunder or elsewhere under the Contract, nor except as expressly provided there under or elsewhere in the Contract, to order any work involving delay or any extra payment by the Company nor to take any variation of or in the works.

- 2.2. The Engineer may from time to time in writing delegate to the Engineer's Representative any of the powers and authorities vested in the Engineer and shall furnish to the Contractor a copy of all such written delegation of power or authorities. Any written instruction or written approval given by Engineer's Representative to the Contractor within the terms of such delegation (but not otherwise) shall bind the Contractor and the Employer provided however:-
- a) Failure of the Engineer's Representative to disapprove any work or materials shall not prejudice the powers of the Engineer thereafter to disapprove such work or materials.
 - b) If the Contractor is dissatisfied by any decision of the Engineer's Representative he may refer the matter to the Engineer who shall thereupon confirm, reverse or vary such decision and the same shall be final and binding on the Contractor.
- 3.0. **ASSIGNMENT/SUB-CONTRACTING:**
- 3.1. The Contractor shall not assign the Contract or any other part thereof or any benefit or interest therein or there under, without the prior written consent of the Employer.
- 3.2. The Contractor shall not Sub-Contract the works, except where otherwise provided by the Contract, without the written consent of the Employer and such consent, if given shall not absolve the Contractor from any responsibility, liability or obligation under the Contract and he shall be responsible for the acts, defaults or neglects of any Sub-Contractor, his agents, servants or workman as fully as if they were the acts, defaults or neglects of the Contractor, his agents, servants or workmen.
Provided always that the provision of labour on a piecework basis shall not be deemed to be subletting under this Clause.
- 4.0 **CONTRACT DOCUMENTS:**
- 4.1 **Documents mutually explanatory:**
The several documents forming the Contract are to be taken as mutually explanatory of one another, and interpreted harmoniously, and in case of ambiguities or discrepancies, the same shall be clarified by the Employer who shall thereupon issue to the Contractor instructions/directions indicating the manner in which the work is to be carried out. However, if there is any conflict regarding any term or expression or instruction used in more than one document, the term or expression or instruction in the later document as arranged seriatim in Article: 1.1.4 shall take precedence and have overriding effect, and modify or replace any such term or expression occurring in an earlier document.
- 4.2 **Further Instructions:** The Engineer shall have full power and authority as delegated to him by the Employer, to issue to the Contractor, from time to time during the progress of the work, such further instructions as shall be necessary for the purpose of the proper and adequate execution of the work and the Contractor (s) shall carry out and be bound by such further instructions.
- 5.0 **GENERAL OBLIGATIONS OF CONTRACTOR:**
- 5.1 **Contract Agreement:**
The Contractor shall enter into and execute a Contract Agreement, if required as per the Contract, in the form as prescribed by the Employer within the time specified in the Letter of Intent and in default thereof the payment to be made to the Contractor under the Contract shall not be made till the Agreement is signed. The cost of the Stamp duty for the Agreement is to be borne and paid by the successful tenderer.
- 5.2 **Site Conditions and Requirement of Work:** By making of Offer/Submitting the tender, it will be deemed that the Tenderer/Contractor had studied the Tender papers carefully. It will be presumed that the Drawings for the work, that were available in the office of the Engineer, have been inspected. It will also be presumed that the Contractor had inspected the Site of the work to acquaint himself with the Site and local conditions, means of access to the Site of work, nature of work and all other matters pertaining thereto. The Contractor has thus made himself aware of the conditions and the requirements under the Contract, and the facts on the ground, before consenting to the Contract. The Company therefore, will not accept any subsequent claim of misunderstanding or misrepresentation of essential information/fact relating to the Work or Contract.
- 5.3 **Sufficiency of tender:**
The Contractor shall be deemed to have satisfied himself before submitting tender as to the correctness and sufficiency of his tender for the work and of the rates stated in the Tender Schedule which shall cover all his obligations, under the Contract and all matters and things necessary for the proper completion and maintenance of the work.

5.4 **Employment of Labour:**

Employment of labour and fixing of their wages and other benefits will be done by the Contractor as per the provisions of the Contract Labour (R&A) Act 1970 and Rules framed there under. The Contractor shall also comply with all Act and rules framed there under by the appropriate Government from time to time including those governing the protection of health, sanitary arrangements, wages, welfare and safety of Contract labour employed. The rules and other statutory obligations with regard to fair wages and all terminal benefits, welfare & safety measures, Maternity leave and Maintenance of necessary registers etc. will be deemed to be part of the Contract and shall be the sole responsibility of the Contractor.

The Contractor will get himself registered with concerned statutory authority as provided in the Act and shall be directly responsible to the authorities there under for compliance with the provisions there under. The Contractor shall keep engaged such number of labors as shall be necessary for the timely execution of the job. The Contractor shall not increase the number of workers without prior permission of the Employer. The retrenchment compensation and other benefits will be payable by the Contractor to their labour as per the provisions of the Industrial Disputes Act. The Contractor shall strictly comply with the statutory rules and regulations in respect of working hours of female labourer at Site and other benefits available to such female labourers.

5.5 **Work to the satisfaction of Engineer:**

The Contractor shall execute the work efficiently and complete it all respects in strict accordance with the Contract terms and conditions and shall comply with and adhere strictly to Engineer's Instructions and Directions on any matter in accordance to the terms of the Contract. The Contractor shall take instructions and directions only from the Engineer or from Engineer's Representative.

5.6 **Contractor's Superintendence:**

As soon as practicable, after the acceptance of his tender, the Contractor shall if required submit to the Engineer for his approval a program showing the order of procedure and method in which he proposes to carry out the work and shall whenever required by the Engineer furnish for his information particulars in writing of the Contractor's arrangements for carrying out the work and of the equipments which the Contractor intends to use for the Contract. The submission to and approval by the Engineer or its authorized representative of such program or the furnishing of such particulars shall not absolve the Contractor or any of his duties or responsibilities under the Contract.

The Contractor shall provide all necessary superintendence during the execution of the work and as long thereafter as Engineer may consider necessary for carrying out the Contract. The Contractor or his competent and authorized representative as approved by the Employer in writing, which approval may, at any time, be withdrawn, shall engage himself continuously on the work and shall make available whole of his time for the superintendence of this Contract work. If such approval is withdrawn by the Employer, the Contractor shall, as soon as, is practicable, having regard to the requirement of replacing him as hereinafter mentioned, after receiving written Notice of such withdrawal, remove his representative from the Site and shall not thereafter employ/engage him again on the Site in any capacity and shall replace him by another representative approved by the Employer. Such authorized representative shall receive on behalf of the Contractor instruction and direction from the Engineer or from the Engineer's representatives subject to Clause-2.0.

If, in the opinion of the Engineer due progress is not made with work in accordance with the Contract and/or any directions given by the Engineer are not properly complied with or attend to, the Engineer may, if he considers it necessary or proper for the due execution of work in accordance with the Contract, of which his Certificate in writing will be sufficient evidence, order the employment of extra supervising staff to supervise the work, and expenses of such employment including the salary of the supervising staff shall be provided by the Contractor, provided that the Engineer shall give to the Contractor not less than 7 days previous Notice in writing of his intention to exercise this power.

5.7 **Contractor's Employees:**

The Contractor shall provide and employ at the Site in connection with the execution of the contracted work, Contractor's employees comprising:

- a) Only such technical assistants and other staff as are skilled and experienced in their respective callings and foreman competent to provide proper supervision to the works.
- b) Such skilled, semi-skilled and unskilled laborer as is necessary for the proper and timely execution of this contracted works. Multi skilling of labour will be permitted, provided the same is to the satisfaction of the Engineer, who in turn will not only examine the skill, but also observe whether the safety norms are being observed by the labour undertaking Multi skill work.

The Engineer shall be at liberty to object to and require the Contractor to remove forthwith from the work any person employed by the Contractor for or in connection with the execution of the work who, in the opinion of the Engineer, has misconduct himself or is incompetent or negligent in the proper performance of his duties or whose employment is otherwise considered by the Engineer to be undesirable and such person shall not be again employed for the work without the written permission of the Engineer. Any person so removed from the work shall be replaced without delay by competent substitute approved by the Engineer.

The Contractor shall submit to the Engineer or the Engineer's representative returns in such forms and at such intervals as the Engineer may prescribe from time to time showing in details the supervisory staff, his permanent employees deployed at Site and the numbers of the several classes of labour engaged by the Contractor at Site and such other information as may be deemed necessary for the purpose of ensuring that the provisions of the Contract, in so far as Contractor's employees are concerned, are being duly complied with by the Contractor.

The Contractor shall not employ labourers below the age as specified in the Factory Act/Industry/Factory Regulation and also shall employ laborers who are medically fit persons.

5.8 Damage to person or property:

The Contractor shall except if and so far as the Contract otherwise provides, indemnify and keep indemnified the Employer against all losses and claims for injuries or damage to any person or property whatsoever which may arise out of and in the course of execution of this contracted work, as a consequence thereof and against all claims, demand, proceedings, damages, costs, charges and expenses whatsoever in respect thereof or in relation thereto.

From commencement of the work to its completion the Contractor shall take full responsibility and care of the work including all temporary works and in case any damage, losses or injury shall happen to the works or to any part thereof or to any temporary works from any cause whatsoever, shall at his own cost repair and make good the same so that at the completion the work shall be in good order and conditions and in conformity in every respect with the requirement of the Contract and the Engineer's instructions.

Before the commencement of the contracted work, the Contractor shall, but without limiting his obligations and responsibility under Clause 5.9 here of, insure against any damages loss or injury which may occur to any property, including that of Employer or to any person including any person/employee of the Employer by or arising out of the execution of contracted work or in carrying out the Contract otherwise than due to 'excepted risks'.

Note: The 'excepted risk' mentioned above are war hostilities (whether war be declared or not), act of foreign enemies, rebellion, revolution, insurrection, or military or usurped, civil war or riot, commotion or disorder (otherwise than among Contractors on Employees) riot, commotion or disorder, or any such operation of the forces of nature or responsible foresight and ability on the part of the Contractor could not foresee or reasonably provide against.

5.9 Accident or Injury to Workmen:

The Employer shall not be liable for or in respect of any damages or compensation payable as per law in respect of and in consequences of any accident or injury to any workmen or any other person in the employment of the Contractor save and except for accident or injury resulting from any act or default of the Employer, his representative or servant. The Contractor shall indemnify and keep indemnified the Employer against all such damages and compensation (save & except as aforesaid) as against all claims, demands, proceedings, costs, charges and expenses whatsoever in respect or in relation thereto.

5.10 Workmen Compensation:

The Contractor shall at all times indemnify the Employer against all claims for Compensation under the provision of the Workmen's Compensation Act, 1923 (VIII of 1923) or any other law from the time being in force, or in respect of any workmen employed by the Contractor in carrying out the Contract and against all costs and expenses for penalties incurred by the Employer in connection therewith and (without prejudice to any other means of recovery) the Employer shall be entitled, to deduct from any money due or to become due to the Contractor (whether under this Contract or any other Contract) all money paid or payable by way of compensation aforesaid or for costs or expenses in connection with any claims thereto and the Contractor abide by the decision of the Employer as to the sum payable by Contractor under the provisions of this Clause.

5.11 Notice & Payments of fees:

The Contractor shall give all Notices and pay all fees required to be given or paid under any Central or State Statute, ordinance or other law or any regulation or by-law of any local or otherwise duly constituted authority in relation to the execution of this contracted work and by the Rules & Regulation of all Public bodies whose property or rights are affected or may be affected in any way by the contracted work.

5.12 **Fossils:**

All fossils, coins, articles of value or antiquity and structures and other remains or things of geological or archaeological interest discovered on Site shall be, as between the Employer and the Contractor, be deemed to be the absolute property of the Employer and the Contractor shall take reasonable precaution to prevent his workmen or any other persons from removing or damaging any such articles or thing and shall, immediately upon discovery thereof deposit the same with the Employer, at the place designated by the Engineer.

5.13 **Patent Right & Royalties:**

The Contractor shall be liable and responsible for any claim made against the Company for any infringement of patents through the use or application of patented tools, articles, appliances, structures, materials, devices, applications, methods, ways, processes or any other use of such patent in the performance of the work or by the use of any process or method connected with the work or by the use of any materials used upon the Work, except to the extent that a claim result from the Contractor's use of a material or product specially required by the Company in the Specifications. The Contractor shall save harmless and indemnify the Company from and against all costs, expenses and damages (including reasonable legal expenses), which any of them shall incur or be obligated to pay, by reason of any such infringement or claim of infringement, and shall, at the election of the Company, defend at the Contractor's sole expense, all such claims in connection with any alleged infringement.

5.14 **Interference with Traffic & Adjoining Properties:**

All operations necessary for the execution of this contracted work shall so far as in compliance with the requirements of the Contract permits, be carried out so as not to interfere unnecessarily or improperly with the public or private convenience or the access to use and occupation of public or private roads and foot-paths or to the properties whether in the possession of the Employer or any other person and the Contractor shall save harmless and indemnify the Employer in respect of all claims, damages proceedings, damages, cost, charges and expenses whatsoever arising out of or in relation to any such matter.

5.15 **Facilities for other Contractors:**

The Contractor shall in accordance with the requirement of the Engineer afford all reasonable facilities to any other Contractors employed by the Employer and their workmen and for the workmen of the Employer and of any other properly authorized authorities or statutory bodies who may employed in the execution on or near the Site of any work not included in the Contract Scope or any Contract which the Employer may enter in to in connection with or ancillary to this contracted work.

5.16 **Clearance of Site on Completion:**

On completion of the Period of Contract, all materials rubbish and temporary structures of any sort or kind used for the purpose of or connected with contracted work including quarters for Contractor's labourers are to be removed by the Contractor and Site handed over in a tidy and workmanlike condition. No final payment in settlement of the accounts for the work shall be held to be due or shall be made to the Contractor till such Site Clearance shall have been affected by him. In the event of the Contractor failing to comply with this provision within 7 days after receiving Notice in writing from the Engineer to that effect, such clearance may be made by the Engineer at the expenses of the Contractor without any further correspondence. The Employer shall under no circumstances be held liable for any loss or damage to such Contractor's property as may be on Site due to removal there from, which removal may be effected by means of public sale of such materials and property or in such way as deemed fit and most convenient to the Engineer. Employer shall not be liable to make any payment in this respect.

5.17 **Force Majeure:**

If, at any time, during the execution of this Contract, the performance in whole or in part, by either party, of any obligation under this Contract, shall be prevented or delayed by reasons of war, hostility, acts of public enemy, civil commotion, sabotage, fires, floods, explosions, epidemic, quarantine restrictions Acts of State or other Acts of God, hereinafter referred to as eventualities, then, provided Notice of the happenings of any such eventualities is given by either party to the other, within 15 days from the date of occurrence thereof, neither party shall by reason of such eventuality be entitled to terminate this Contract, nor shall either party have any claim for damages against the other, in respect of such non-performance or delay in performance, and work under this Contract shall be

resumed as soon as practicable after such eventuality has come to an end or ceased to exist and the decision of the Employer as to whether the work have to be so resumed shall be final and conclusive.

Should one or both parties be prevented from fulfilling their contractual obligations by a state of Force-majeure lasting continuously for a period of at least three months, the two parties should consult each other regarding the further implementation of the Contract, provided always, that if no mutually satisfactory arrangement is arrived at, within a period of a month from the expiry of three months referred to above, the Contract shall be deemed to have expired at the end of the said 4th month referred to above, in which event the liability under the Contract shall be settled taking into consideration the work carried out by the Contractor up to that expiry date without prejudice to the rights of Employer arising prior to that expiry date.

If there are any Judicial Pronouncement/Statutory Notifications/Orders, the Employer reserves to himself the right to withdraw partly or fully any item/items of work as required for compliance of such orders without any financial claim on this account by the other party.

5.18 **Foreclosure of Contract:** The Employer shall have the right to foreclose the Contract, for whatever reasons, by giving three (03) months Notice on the Contractor with out any claim of damages by the Contractor.

5.19 **Supply of Materials:** Except where otherwise specified, the Contractor shall, at his own expense, supply and provide all the construction plant, temporary works, materials both for temporary and for permanent works, labour (including the supervision there of) transport to or from Site and in and about the works and item of any kind required for the contractions, completion and maintenance of the works.

If the Employer has undertaken to supply materials specified in the Special Conditions of Contract or otherwise at terms stated therein, the Contractor shall keep himself in touch with the day to day position regarding the supply of materials from the Engineer and shall so adjust the progress of the work that his labour may not remains idle nor may there be any other claim due to arising from delay in obtaining the materials. No claim, what so ever shall be entertained by the Employer on account of delay in supplying materials or in the event of its inability to such supply at all.

5.20 **Idle labour/Machinery:** No claim on account of idle wages of the labour or idle Charges for machineries engaged shall be entertained under any circumstances.

5.21 **Liquidated Damages:** The work shall throughout the stipulated period of the Contract be proceeded with due diligence. The Contractor is to complete the contracted quantity within the time as stipulated in the Contract. However, if no monthly target is separately specified, the contracted quantity would be prorated monthly and treated as the monthly target.

If the Contractor fails to fulfill the quarterly target during the quarter and/or as required by the Company, Liquidated Damages (L.D) not as a penalty equal to 10% of the value of left out quantity will be recovered. However the total L.D during the entire tenure of the Contract shall be subjected to a maximum of 5% of the Contract Value. Final reconciliation shall be done at the end of the Contract period and adjustment of L.D if any, shall be done. For the purpose of calculation of L.D, quarterly review shall be done.

If the L.D so imposed exceeds the Security Deposit of the Contractor then the Employer may deduct the same from the amount due to the Contractor in any other Contract. This is, however, without prejudice to any other right of recovery that the Employer may have against the Contractor in this regard under the law.

The deduction of such damage as referred above shall not however relieve the Contractor from his obligations to complete the work in time as stipulated in the Contract.

However, if the short fall exceeds 30 % during the quarter, the Company reserves the right to terminate the Contract without any Notice and get the job completed through any other means including engagement of other agencies at the Contractor's risk and cost.

5.22 **Lien:** The Employer shall have a first and foremost lien upon the plant, machineries and materials of the Contractors against its all dues and till the same are paid off by the Contractor, the Employer shall have a exclusive right to retain its all aforesaid goods and materials.

As the Contract includes the Operations and Maintenance of Plant and equipment, the Contractor will be held responsible for loss, destruction or deterioration of the Plant, machineries and materials, because of any alleged inadequacy in maintenance. The Company will be entitled to be reimbursed by the Contractor, any cost incurred towards ensuring the safety and security of the plant, machineries and materials in its custody, when it exercises its security of the plant, machineries and materials in its custody, when it exercises its lien.

6.0 **LABOUR:**

6.1 **Compliance with Labour Statutes:**

The Contractor shall comply with all the obligations under the Contract Labour (Regulation and Abolition) Act, 1970, the Factories Act, the Minimum Wages Act, the Payment of Wages Act, the Employees, Provident Funds and Pension Act, the Payment of Gratuity Act, the Industrial Dispute Act, the Maternity Benefit Act, the Employee's State Insurance Act, the Workmen's Compensation Act, and all other labour statutes for the time being in force, and applicable to the works. The Contractor agrees to keep the Principal Employer/Company indemnified at all times against any demands from the labour or statutory authorities on this account.

- 6.1.1 **Contract Labour (Regulation & Abolition) Act, 1970:** The Contractor is required to produce license as enjoined in the Contract Labour (Regulation & Abolition) Act, 1970, if the Act is applicable to him. The Contractor shall obtain License from the appropriate Licensing Officer of the area before commencement of the work and shall produce a copy thereof of the work along with the original to the Engineer immediately on start of the work. The original will be returned to the Contractor after verification. He shall not be allowed to under take or execute any work through Contract labour except under, and in accordance with, a License issued under the said Act in that behalf by authorized Licensing Officer.

The Contractor shall thoroughly familiarize him self with the provisions of the Contract Labour (Regulation & Abolition) Act, 1970 and rules there under, and submitted Certificate to the Engineer in charge of the work indicating whether the provisions of the said Act and rules are applicable to him. In case the said Act and rules are applicable, the Contractor shall take all steps to comply with their provisions, maintained records and registers as required, submit reports and returns to the prescribed authorities periodically as required, Issue emplacement cards/service certificates and display Notice in accordance with Contract labour rules, with copies thereof to the Company.

The Contractor shall arrange provision of rest room, canteens, latrines and urinals, washing facilities and first aid facilities strictly in accordance with the provision of Contract labour rules. If these facilities are not provided by the Contractor within the stipulated time, the same will be provided by Company and the cost for the same will be debited to the Contractor's account.

The Contractor shall pay wages to his labour in a manner laid down in Contract labour rules, in the presence of the Engineer (Principal Employer), or his authorized representative. He shall at least pay the minimum wages applicable under the Minimum Wages Act. In case the Contractor's labours perform the same or similar kind of work as workmen directly employed by the Principal Employer, the wages rates, holidays, hours of work other conditions of service of the workmen employed by the Principle Employer shall be applicable to the Contractor's labour also. The Contractor shall notify to the Engineer the date and place of payment of wages, who shall nominate a representative to witness the payments. The Contractor shall also submit to the Engineer a copy of the Wage Payment Sheets, which have been receipted by the Contract labour, and counter-signed by the representative of the Engineer. By virtue of default of the Contractor, or otherwise, if the Company is obliged to provide amenities and/or pay wages labour employed by the Contractor directly, or through Sub-Contract (s) under this Contract, then the Contractor shall indemnify the Company fully, the Company shall be entitled to recover from the Contractor the expenditure incurred on providing the said amenities, and/or the wages so paid by deducting it from the Security Deposit or from any sum payable by the Company to the Contractor, either under this or under any other Contract.

Provided that if any dispute arises as to expenditure incurred by the Company on provisions of the said amenities, the decision of the Engineer there of shall be final and binding.

- 6.1.2 **Standing Orders:** The Contractor shall make his own Standing Orders for the employees engaged by him and get the same approved through the concerned Regional Commissioner or other appropriate authority and implement the same in conformity with provisions of Industrial Employment Standing Orders Act. In absence of such Standing Order the Model Standing Order shall be implemented.

6.1.3 **Payment of Wages Act:**

The Contractor shall comply with the provisions of the Payment of Wages Act, 1936 and the Rules there under in respect of all persons employed by him in carrying out this Contract. Any dispute regarding the statutory payment, which may arise, shall be immediately settled by the Contractor with his workmen.

6.1.4 **Minimum Wages Act:**

The Contractor shall comply with the provision of the Minimum Wages Acts 1948 and the Rules made there under by the Central Govt. in respect of all employees employed by him in carrying out this Contract. He shall pay the employees, wages not less than the minimum rate of wages, as notified by the West Bengal Govt. from time to time or settled under any settlement award, which ever is higher for that Category. The Contractor shall make all payments to the labors in the presence of authorized representative of the Employer. The Contractor shall also submit to the Engineer a copy of the Wage Payment Sheets in proof of having paid wages to the various laborers with due counter signature by the representative of the Employer.

The Contractor shall at all times indemnify the Employer against all claims arising out of provisions of the Minimum Wages Act and the Rules framed there under as admissible in respect of any workman employed by the Contractor in carrying out the Contract and against costs and expense incurred by the Employer in connection with and without prejudice to other means of recovery. The Employer shall be entitled to deduct, from any money which is due to or which shall become due to the Contractor whether under this Contract or any other Contract, all money paid or payable by the Employer by way of wages and other dues (including compensations, penalty, if any) imposed for committing breach of any provision of the Act by the Contractor in connection with any claim thereto and the Contractor shall abide by the decision of the Employer as to the sum payable by the Contractor under the provisions of this Clause.

6.1.5 Amenities to Labour:

In respect of labour employed by the Contractor on the work, the Contractor shall provide at his cost reasonable amenities for security, proper working and living conditions such as canteen facilities, safety appliances, first aid facilities etc. The facilities like toilets, drinking water arrangement etc. provided by the Employer to the use of workmen are to be kept in a clean and sanitary conditions satisfactory to the Engineer. Where women labour is employed, urinals and lavatories will be provided separately by the Employer for the infant children of women labourers also in clean and sanitary conditions satisfactory to the Engineer. Labourers engaged shall be provided there under with necessary safety appliances as free of charge required under the provisions of Factory's Act and regulations made by the Contractor.

6.1.6 Medical facilities:

First Aid and Ambulance facilities shall be provided by the Contractor to his labourers in case of accidents/emergency requirements. He must ensure full medical coverage under ESI scheme, wherever statutorily required to do so, or provide medical facilities, where coverage under ESI scheme is not mandated. He must take out insurance coverage of the Contract labour for liability under ESI scheme, and other statutory sickness accident benefit scheme, and provide the Engineer with copies of documentary evidence of the same. Medical facilities at the Company's Hospital/Dispensary, wherever possible, shall be provided on payment of charges applicable to outside parties as per Company's Rules. No free medical treatment shall be provided by the Employer to the Contractor's labourers.

6.1.7 Inspection and Investigations:

The Contractor's establishment will be subject to inspection, investigation etc. by Engineer or such other representatives of the Employer as duly authorized on behalf of him, assuring proper and faithful compliance of the provisions of this Contract by the Contractors particularly with regard to the implementations of Labour Laws and other matter anticipated herein. The Contractor shall abide by the decisions and orders of the Engineer with regard to any such matter and furnish, if required necessary compliance report within the stipulated time.

7.0 OBLIGATION OF THE CONTRACTOR REGARDING OBSERVANCE OF VARIOUS OTHER STATUTES:

The Contractor shall, at his own cost, observe, perform and comply with the provisions of all statutory enactments, rules, regulations and bye-laws framed hereunder as are applicable for control/operation/regulation of Plant, during the execution of work and shall maintain such registers, documents, records etc., as are required under the various statutes, for production of the same before the Employer and/or other statutory authorities prescribed in this behalf, as and when required. Non-compliance of the provisions/stipulations of these Acts/Rules will render the Contractor liable to payment of necessary compensation/penalty, as deemed fit by the Employer/statutory authorities. It will be the sole responsibility of the Contractor to ensure all kinds of statutory payments to his workers and submission of returns etc., to various statutory authorities in time.

In case of Contractor's default in making statutory payment in time, the Employer reserves the right to deduct necessary amount from the Contractor's towards such payments. In case the Contractor fails to supply the required Safety Equipments to his workers then the Company reserves right to supply the same Equipments and deduct cost of such materials and cost of storage from the Contractor's bill. The Contractor should also ensure that his workers used the Safety Equipments during work.

The Contractor shall be fully responsible to comply with all applicable statutory obligations as Employer, e.g., all contributions under P.F. and Pension scheme (EPF) etc. under the Employees Provident Fund and Misc. Provisions Act, 1952, State Insurance Act, 1948, benefits under the Maternity Benefit Act, 1961, payment under Payment of Gratuity Act, Payment of Bonus Act, Industrial Dispute Act, etc., and all other such obligations/liabilities in respect of their labour engaged by them for the job under taken under the Contract as per applicable statutory provisions/law and Govt. Notifications ,and will take full liability on this account. The Company will not take any financial liability on this account. In the event of failure of the Contractor to comply with the above, the Company shall be entitled to recover the amount by deduction from any amount payable to the Contractor under the Contract, including Security Deposit, or as debt payable by the Contractor.

8.0 **WORK MATERIALS & PLANT:**

8.1 **Removal of improper work, equipment & materials:** The Engineer shall during the progress of the work, have power to order in writing from time to time the following:

- a) The removal from Site within such time or times as may be specified in the order, of any materials/equipment which in the opinion of the Engineer are not in accordance with the Contract.
- b) The substitution of suitable equipment and/or materials.
- c) The removal and proper re-execution, notwithstanding any interim payment there against, of any work which, in the opinion of the Engineer is not in accordance with the Contract.

8.2 **Default of Contractor in compliance:**

In case of default on the part of the Contractor in carrying out such Order, the Employer shall be entitled to employ and pay other persons to carry out the same and all expenses consequent thereon or incidental thereto shall be borne by the Contractor and shall be recoverable from him by the Employer or may be deducted by the Employer from any money dues or which may become due to the Contractor.

8.3 **Suspension of work:**

The Contractor shall on the written Order of the Engineer suspend the progress of the work or any parts thereof for such time in such manner as the Engineer may consider necessary and shall during such suspension, properly protect and secure the work as is necessary in the opinion of the Engineer. If the progress of the works or any part thereof is suspended on the written Order of the Engineer for more than ninety days, the Contractor may serve a written Notice on the Employer requiring permission to resume the suspended work with a further period of 30 days from the date such Notice is received by the Employer. If such permission is not granted within that time, the Contractor, by a further written Notice to serve, may elect to treat the suspension as termination of the Contract, without any financial repercussion to either the Employer or to the Contractor.

8.4 **Free Access:** The Employer and/or Engineer and any person authorized by them shall at all times have access to the works at Site and the Contractor shall afford every facility for and every assistance to, in obtaining the right to such access.

9.0 **COMMENCEMENT TIME & DELAYS:**

9.1 **Commencement of work:**

The date of effect of the commencement of the Contract shall be reckoned from the date as specified in the Work Order or the date of taking possession of Site, whichever is earlier, after deposit of Security amount.

9.2 **Possession of Site:**

Save, in so far as the Contract may prescribe, the extent of portions of the Site of which the Contractor is to be given possession from time to time and the order in which portions shall be made available to him and subject to any requirement in the Contract as to the order in which the works shall be executed, the Employer will with the Engineer's written order to commence the work, given to the Contractor possession of so much of the Site as may be required to enable the Contractor to commence and proceed with the work in accordance with such reasonable proposals of the Contractor as he shall by Notice in writing to the Engineer make and will from time to time as the work proceeds, give to the Contractor possession of such further portions of the Site as may be required to enable the Contractor to proceed with the work with due dispatch in accordance with the said proposals.

9.2.1 In case the Contractor delays to take possession of the Site within the time specified in such written order/information/instruction from the Employer or his representative, and the delay subsequently result in further delay/complications, for extension of Contract the sole responsibility of such delay/complications shall be resting with the Contractor.

9.2.2 Without prejudice to the rights of the Employer and without any manner affecting the duties of the Contractor, if in the opinion of the Engineer, the Contractor has failed or is likely to fail to carry out any part of the work and such failure or apprehended failure might result in the delay in supply of the items to be produced as per plan, any loss

or damage to the Company, then the Engineer may cause any part of the work to be carried out departmentally or through other agencies and all expenses incurred by Employer in excess over the Contract rate shall be recoverable from ad-hoc/monthly/fortnightly 'on-account' bills of the Contractor. The decision of the Engineer in regard to the exercise of this provision as well the recovery of additional expenses incurred by the Company is final and binding on the Contractor. Once Contractor takes possession of the Site as per written order of the Employer, the responsibility of security and protection of the Site from any complication shall be the sole responsibility of the Contractor.

- 9.2.3 **Compliance with Statutes:** The Contractor shall at his own cost, observe, perform and comply with the provisions of the following Acts applicable during the execution of the Contract, and Rules/Bye-Laws framed hereunder including but not limiting to the following and shall maintain such registers and documents as are required under the various statutes, for production of the same before the Company and/or other statutory authorities, prescribed in this behalf, as and when required. The Contractor agrees to keep the Employer indemnified at all times against any demands/penalties by statutory authorities, and shall defray to the Employer any costs/suspense's incurred by the Employer in proceedings before the statutory authorities. A list of the relevant laws applicable to the process of execution of Work under the Contract is given below. The list is illustrative and not exhaustive.

- (i) The Contract Labour (Regulation & Abolition) Act, 1970 & Rules framed there under.
- (ii) The Payment of Wages Act, 1936.
- (iii) The Employees' Provident Fund Act, 1952 & Schemes framed there under
- (iv) The Maternity Benefit Act, 1961.
- (v) The Payment of Bonus Act, 1965.
- (vi) The Industrial Disputes Act, 1947
- (vii) The Payment of Workmen's Compensation Act, 1923
- (viii) The Minimum Wages Act, 1948
- (ix) The Payment of Gratuity Act, as and when applicable.
- (x) Air & Water pollution Acts
- (xi) Environment Protection Act, 1986 and Environment protection rules, 1986

- 9.2.4 **Waiver:** The non-enforcement of any of the terms and Conditions of the Contract strictly by any party to the Contract would not amount to waiver of those terms in favour of the other party.

10.0. **REMEDIES AND POWER:**

10.1 **Anticipatory breach and Termination: Risk and Cost:**

If the Contractor shall neglect to execute the work with due diligence and expedition or, shall refuse or neglect to comply with any reasonable orders given to him in writing by the Engineer in connection with work or shall contravene any provision of the Contract, the Company/Employer may give seven (7) days notice in writing, to the Contractor, to make good the failure, neglect or contravention complained of. If the Contractor shall fail to comply with the Notice within the stipulated period from the date of service thereof in the case of a failure, neglect, or contravention capable of the made good within that time, then and in such case the Employer shall be at liberty to employ other workmen, and forthwith perform such work as the Contractor may have neglected to do.

If the Employer shall think fit, it shall be lawful for him to take the work wholly, or in part, out the Contractor's hands and give it to another person on Contract at a reasonable price or prove any other materials, tools tackle or labour for the purpose of completing the work, or any part thereof, and in that event the Employer shall without being responsible to the Contractor for fair wear tear of the same, have the free use of all the materials, tools tackles, or other things which may be on the Site, for use at any time in connection with the work to the exclusion of any right of the Contractor over the same. The Employer shall be entitled to retain and apply any balance, which may be otherwise due on the Contract by him to the payment of the cost of executing such work as aforesaid, and shall be entitled to make a claim for the balance amount, in accordance with law.

10.2 **Bankruptcy and Breach:**

If the Contractor becomes bankrupt or have an order for appointment of any receiver made against him or shall present any position in bankruptcy or shall make an arrangement with or assignment in favour of his Creditors or shall agree to carry out the Contract under Committee of inspection of his Creditors or being a Corporation shall go into liquidation (other than a voluntary liquidation, for the purpose of amalgamation, absorption or reconstruction) or if the Contractor shall assign the Contract without the prior consent in writing of the Employer, or shall have an execution levied on his goods, or if the Engineer shall certify in writing to the Employer that in his opinion, the Contractor;

- a) Has abandoned the Contract or,

- b) Without reasonable excuse has failed to commence the work or has suspended the work for 7 days after receiving from the Engineer written Notice to proceed or,
- c) Is not executing the works in accordance with the Contract or is persistently or flagrantly neglecting to carry out his obligations under the Contract or,
- d) Has to the detriment of good workmanship or in defiance of the Engineer's instructions to the contrary sublet any part of the Contract or,
- e) Otherwise failed to perform his part of the Contract according to the true intent and meaning thereof.

Then the Employer may after fixing 7 days Notice in writing the Contractor, enter upon the Site and expel the Contractor there from, without thereby avoiding the Contractor or releasing the Contractor from any of his obligations or liabilities under the Contract or affecting the rights and powers conferred on the Employer or the Engineer under the Contract or otherwise available under the law, may appoint any other Contractor to complete the work at the cost and risk of the Contractor. However on happening of any eventualities as per above Sub Clause 10.2 (a) to 10.2 (e) the Employer shall be at discretion to terminate the Contract by giving 15 days Notice and the Contract shall stand/terminated w.e.f the 16th day from the date of issue of this Notice. In any of the eventualities mentioned above in (a) to (e), the Employer shall have right to take possession of the plants and machineries of the Contractor and relies the dues by sale of the said plants and machineries.

10.2.1 Valuation in case of forfeiture:

The Engineer shall as soon as practicable after any such entry and expulsion by the Employer fix and determine such investigation or Enquiries as he may think fit to or institute and shall certify what amount (if any) had at the time of such entry and expulsion been reasonably earned by or would reasonably accrue to the Contractor in respect of work actually done by him under the Contract.

10.2.2 Payment after forfeiture:

If the Employer shall enter & expel the Contractor under this Clause, he shall not be liable to pay to the Contractor any money or account of the Contract until the expiry of the period of Contract and thereafter until the costs of completion and damages and all other expenses incurred by the Employer have been ascertained. The Contractor shall then be entitled to receive only such sums (if any) as the Engineer may certify to be due to him upon the completion by him after deducting the said amount payable to Employer. But if such amounts shall exceed the sum which would have been payable to the Contractor on the completion by him then the Contractor shall upon demand pay to the Employer the such excess amount and it shall be deemed a debt due by the Contractor to the Employer and shall be recoverable accordingly.

10.2.3 Illegal gratification, breach of Contract:

The Contract may also be terminated and the Contractor shall be liable to make good any loss or damage resulting from such cancellation (specified under Clause 10.1), If any bribe, gratuity, gift, loan reward or advantage pecuniary or otherwise shall either directly or indirectly be given, promised or offered by the Contractor or any of his servants or agents to any person employed by the Employer in any way relating to his office or employment or if any such person shall become in any way directly or indirectly interested in the Contract or if the Contractor has committed a breach of any of the terms of the Contract and in particular Minimum wages Clause and Labour regulations.

10.3 Retention of Dues:

The Employer will have full liberty to retain and set off all sums due or to become due to the Contractor whether under this Contract or under any other transaction or claim whatsoever, against any sum due or to become due to the Employer under this Contract or any other transaction or claim whatsoever.

11.0 FINAL CERTIFICATE:

The Contract shall not be considered as completed until a Final Certificate has been signed and issued to the Contractor by the Employer stating that the Work has been completed in accordance with the terms of the Contract.

12.0 NOTICE:

12.1 Service of Notice on Contractor:

Any Notice given to the Contractor under the terms of the Contract shall be served by the Engineer or his representative by Registered Post/Courier to, or leaving the same, at the Contractor's notified address or its registered office or at the Contractor's Site office.

12.2 Service of Notice of Employer:

Any Notice to be given to the Employer under the terms of the Contract shall be served by sending the same by Registered Post/Courier at the Office of the Employer at SAIL Growth Works, Kulti.

12.3 **Change of address:** Any change of address of the Contractor shall immediately be notified to SGW, Kulti.

13.0 **MISCELLANEOUS:**

13.1 **Adequate storing to be provided:**

The Contractor shall at his own cost provide adequate storing at all work such that no damage, injury or loss is caused or likely to be caused to any person or property.

13.2 **Provision of shed, stores, houses etc.:**

The Contractor shall at his own cost provide such sheds, stores, and houses as the Employer may consider necessary for storing materials.

13.3 **Work in or around the running plant:**

When work is being carried out in or around a running plant/where the plant must run uninterrupted, the Contractor can only work at specified places and times mutually, arranged. Progress of work in certain areas might have to be expedited as required by the Employer and the Contractor must take this factor into consideration when working out his Rates.

13.4 **Use of intoxicant:**

The Contractor shall prohibit the possession, use and sales of ardent spirits or other intoxicating beverage upon the work or in any of the boarding, houses, encampments or other tenements owned occupied by or within the control of the Contractor or any of his employees. Any Contractor or Sub-Contractor personnel determined by the Company, in its sole discretion, to be in violation of the ban against narcotics, drugs and intoxicants will be prohibited from working on the Contract.

13.5 **Penalty for Accident:** In case any accident involving any Contract work person or any violation reported by any authorized person during the Contractual operation at SGW, Kulti while performing the Contract, a penalty shall be charged on the Contractor as indicated below, over and above the statutory responsibility of the Contractor for payment of Compensation;

Sl. No.	Type of Accident / Violation of Safety Regulation	Penalty of each case		
		% of Contract Value	Minimum (In Rupees)	Maximum (In Rupees)
1.	Fatal Accident	2 %	10,000/-	50,000/-
2.	Serious Accident	0.5%	5,000/-	25,000/-
3.	Reportable Accident	0.1%	1,000/-	5,000/-
4.	Any repeated violation of the safety instruction by the Authorized Person.	0.05%	100/-	500/-

The amount of Penalty as above will be deducted from his running on account bill (in a maximum of 2 subsequent installments) and if not possible, from the Security Deposit.

However, in the subsequent Investigation/Enquiry Report made by the Committee duly approved by Competent Authority, it is proved that the Contractor has taken all reasonable steps to ensure the safety of his workmen and the causes contributing to the accidents is beyond the control of the Contractor, then the amount deducted may be refunded to the Contractor with the approval of Competent Authority with out any interest.

13.6 **Policing of the work:**

Should the general conduct of the works including the premises of the Employer under occupation of the Contractor lead to violation of any of the provisions of the Indian Penal Code either in consequence of riotous or illegal proceedings of the Contractor's labour or Supervising staff or others to such an extent as to necessitate the deployment of Special Police or Magistrate, the cost of such extra forces is to defrayed by the Contractor and not by the Employer.

13.7 **Signature of receipts for amount:**

Every receipt for money which may become payable or for any deposit which may become transferable, notwithstanding to the contrary contained in the partnership deed, if in the name of partnership, signature on receipt by any one of the partners of a Contractor firm be a good any sufficient to the Employer in respect of the money, and in the event of death of any Contractor partners during the currency of the Contract, it is hereby expressly

agreed that every receipt by any one of the surviving Contractor-partners shall, if so signed as aforesaid, be a good and valid discharge, as aforesaid, provided that nothing in this Clause shall be deemed to prejudice or effect any claim which the Employer may hereafter have against the legal representative of any deceased Contractor partner for or in respect or any breach of any of the conditions of Contract, and provided also that nothing in this Clause shall be deemed to prejudice or affect the respective rights or obligations of the Contractor-partners and of the legal representatives of any deceased Contractor-partner's interest.

13.8 Law in force in relation to Contract:

The Contract or amendments thereof entered into between the Employer and the Contractor under the Contract shall be governed and regulated by the relevant laws for the time being in force in the territory of India relating to Contracts.

13.9 Change in Constitution of Firm: If the Contractor is a Partnership firm, the contractual relationship shall be deemed to have terminated if there is any change in the constitution of the firm, whether by retirement, introduction of new partners or otherwise, unless such change has the previous written approval of the Employer, notwithstanding that the Employer may have other dealings with such reconstituted firm. In the event of death of one of the partners, the Contractor shall immediately inform the Employer, giving necessary particulars of the heirs and legal representatives of the deceased partner, and it shall be the option of the Employer either to continue the contractual relationship with the said reconstituted firm, or to treat the Contract as having being terminated by such supervening events, and the decision of the Employer in this regard will be binding on all the partners concerned. No claim for compensation/damages on premature termination will be made or sustainable against the Employer on account of such termination.

13.10 The Contractor shall make his own arrangements for transporting equipment required by him, obtain prior permission for bringing into Plant/Factory or carrying/removing from the Plant/Factory of such equipment. No such permission for removing/carrying of any equipment by the Contractor from the Plant shall be granted unless he has cleared his outstanding to the Employer, nor shall he be entitled for any compensation on this account.

13.11 The Contractor shall not allow any visitor on the Work Sites, without the prior permission of the Engineer.

14.0 ARBITRATION:

14.1 Reference of Disputes to Conciliation/Arbitration: All disputes or differences arising out of the Contract, except disputes or differences for which separate provisions for their resolution have been made in the Contract ('excepted matters'), shall be settled by Conciliation or Arbitration in accordance with the Arbitration and Conciliation Act, 1996, and the provisions made hereinafter in this Article. Such dispute shall first be referred to Conciliation by a Conciliator selected mutually by the parties, who shall also decide the fees/remuneration and the rules of Procedure, which shall be flexible.

14.2 Rules for Arbitration: Rules of Conciliation & Arbitration (2003) framed by SCOPE forum of Conciliation & Arbitration will be applicable in case of this Contract. The expenses, what so ever related to Arbitration will be equally born by both the Parties.

14.3 Work to continue during Conciliation/Arbitration: Work under the Contract shall be continued by the Contractor during the arbitration proceedings and recourse to arbitration shall not be a bar to continuance for the work.

15.0 AMENDMENT:

Except to the extent expressly set forth in the Contract, no change in or modification, termination or discharge of this Contract, in any form whatsoever, shall be valid or enforceable unless it is writing on stamp paper of requisite value, and signed by the party to be charged therewith or its duly authorized representative.

15.1 It shall be the responsibility of the person submitting the tender to ensure that the tender have been submitted in the formats and as per the terms and conditions prescribed in the SAIL Website and no change is made therein before submission of their tender. In the event of any doubt regarding the terms and conditions/formats, the person concerned may seek clarifications from the authorized officer of SAIL Growth Works, Kulti. In case any tampering/unauthorized alteration is noticed in the tender submitted from the tender document available on the SAIL Website, the said tender shall be summarily rejected and the Company shall have no liability whatsoever on the matter. However, deviation if any proposed by the tenderer may be separately indicated for acceptance or otherwise of SAIL Growth Works, Kulti. Such proposed deviation will not be treated as tampering for the purpose of application of this Clause.

16. The terms and conditions in the Tender form are final for all purpose and no other terms and condition contrary or in variation or additions to what are contained therein shall be accepted by the Company unless the same is contained in any formal Agreement duly executed by the Company and the Tender / Contracts.
17. No subsequent correspondence on the Tender initiated by me / us will be entertained by the company.
18. Canvassing in any form is strictly prohibited and that if I / We do resort to canvassing the Tender submitted by me/ us will be liable to rejection.
19. Unless and until a formal "Agreement (in Company's prescribed form) is prepared at my / our cost, the Invitation to Tender, offer of myself/ourselves in this Tender form, the GTC of Contract and the written acceptance of the offer or so much of the same by the Company, shall constitute a binding Contract between the Company and myself / ourselves until the said "Agreement" (in company's prescribed form) is executed.
20. All Tenderers are to note that minimum wages / Basic Plus DA plus VDA payable to labourers as per rules in force, and take this into account in tendering firms rates which will apply during entire tenure of the Contract i.e. no escalation will be paid even on account of change in VDA They make seek clarification, if any, from Contracts Dept& Personnel Dept. & quote accordingly.
21. In case the Tenderer back out or withdraws their Tender offer during the validity within which the Tender are to remain open for consideration, The Tenderer's Earnest Money will be forfeited. This provision for forfeiture shall also apply to the offer that might be made by the Tenderer after negotiations and discussion with IISCO Steel Plant, subsequent to the opening of the Tender. The valid period of the Tender offer shall be considered as that indicated in Invitation to Tender and including also the period as per subsequent written confirmation the Tenderer for further extension of validity, if any, beyond that indicated in the Notice Inviting Tender.
22. In case the L-1 Tenderer backs out he/ they shall be debarred from participation in all types of Tenders for 6 (Six) months.
23. Negotiation: The Company reserves the right to invite / call any Tenderer/s for discussions /negotiations as may be decided after opening of Tenders. Non-compliance by the Tenderer to attend negotiation sub- negotiations/ discussion shall render his /their Tender liable to rejection with forfeiture of Earnest Money without further reference. It is to be noted that holding of discussion / negotiations is the prerogative of Management and there will be no claim on the part of tenderer/s to ask for discussion / negotiations.
24. If my /our Tender is not accepted, I /we shall not be entitled to claim any costs, charges, expenses of and incidental to or incurred by me/us through or in connection with my/ our submission of Tender even through the Company may elect to withdraw the invitation to Tender.
25. Mere sale /issuance of Tender paper does not bind the Company in any way in respect of the proposed Contract with the purchaser of Tender paper.
26. JURISDICTION: The Asansol Court shall have the exclusive jurisdiction upon any matter arising out of this Contract.

Annexure-G

GST TAXES

1. For the purposes of levy and imposition of GST, the expressions shall have the following meanings:
 - (a) GST - means any tax imposed on the supply of goods and/or services under GST Law.
 - (b) Cess – means any applicable cess, existing or future on the supply of Goods and Services as per Goods and Services Tax(Compensation to States) Act, 2017
 - (c) GST Law - means IGST Act 2017, CGST Act 2017, UTGST Act, 2017 and SGST Act, 2017 and all related ancillary Rules and Notifications issued in this regard from time to time.
1. The rates quoted by the Vendor/Supplier/Contractor shall be inclusive of all taxes, duties, levies and Cess except GST. Vendor/Supplier/Contractor has to clearly show the amount of GST separately in the Tax Invoices raised by them. Further, it is the responsibility of the Vendor/Supplier/Contractor to make all possible efforts to make their accounting/IT system GST compliant in order to ensure timely availability of Input Tax Credit (ITC) to SAIL-SGW.
3. SAIL-SGW shall declare in the Tender about value of free issue of material and services, if any, involved in the execution of the contract. The Contractor should consider the same while working out the GST liability, if any. Further in cases where GST is leviable on any facilities provided by SAIL and used by Vendor/Supplier/ Contractors and the consideration for which is recovered by SAILSGW in the form of reduction in the value of invoice raised by Vendor/Supplier/Contractor, then SAIL-SGW will raise GST invoices on such transactions and the same will be borne by Vendor/Supplier/Contractor.
4. Evaluation of L-1 prices shall be done based on landed cost net of Input Tax Credit of GST, if available to SAIL-SGW. SAIL-SGW shall evaluate the offers on the basis of the quoted rates only and any claim subsequently by the Vendor/Supplier/Contractor for additional payment/liability shall not be admitted and has to be borne by the Vendor/Supplier/Contractor. The supplier under Composition Scheme should carefully understand the implication of Input Tax Credit (ITC) before quoting the rates. Composition scheme is available only in respect of intra-state supply of goods, if taxable turnover of vendor is up to Rs.75 Lakhs.
5. For the purpose of this contract, it is agreed between the parties that if any new taxes, duties or levies other than GST is introduced subsequent to the final date of submission of tender or reverse auction by the Central/State Government & Local Authorities and such new taxes, duties or levies become payable, then an equitable adjustment on account of new taxes, duties or levies in the contracted price shall be made which shall be subject to the production of documentary evidence by the Vendor/Supplier/Contractor. This provision shall be applicable only during the original period of contract. However, during the extended period of contract, if any, this provision shall be applicable as follows:-
 - a) If input tax credit (ITC) is available, the adjustment in contract price for such new tax shall be made.
 - b) In other cases (i.e. where tax credit is not available), adjustment in contract price shall be made only if the new tax is enacted during the period of extension arising out of reasons attributable to SGW.Kulti
6. In case of variation (increase/decrease) in the rate of GST after the final date of submission of tender or reverse auction, the said revised rate shall be reimbursed or recovered on production of relevant statutory documentary evidence. This provision shall be applicable only during the original period of contract. However, during the extended period of contract, if any, this provision shall be applicable as follows :
 - bb) If input tax credit (ITC) is available, the said revised rate shall be reimbursed or recovered.
 - cc) In other cases (i.e. where input tax credit is not available), the said revised rate shall be reimbursed or recovered only if the reasons for extension of the contract is attributable to SAIL-SGW. In any case, recovery shall be made in case of a downward variation in the rate of tax.

7. Vendor/Supplier/Contractor agrees to do all things not limited to providing GST compliant Tax Invoices or other documentation as per GST law relating to the supply of goods and/or services covered in the instant contract like raising of and/or acceptance or rejection of credit notes/debit notes as the case may be, payment of taxes, timely filing of valid statutory Returns for the tax period on the Goods and Service Tax Network (GSTN), submission of general information as and when called for by SAIL in the customized format shared by SAIL in order to enable SAIL to update its database etc. that may be necessary to match the invoices on GSTN common portal and enable SAIL-SGW to claim input tax credit in relation to any GST payable under this Contract or in respect of any supply under this Contract.
8. In case Input Tax Credit of GST is denied or demand is recovered from SAIL-SGW by the Central/State Authorities on account of any non-compliance by Vendor/Supplier/Contractor, including non-payment of GST charged and recovered, the Vendor/Supplier/Contractor shall indemnify SAIL-SGW in respect of all claims of tax, penalty and/or interest, loss, damages, costs, expenses and liability that may arise due to such non-compliance. SAIL-SGW, at its discretion, may also withhold/recover such disputed amount from the pending payments of the Vendor/Supplier/Contractor.
9. Vendor/Supplier/Contractor shall maintain high GST Compliance at any given point of time.
10. Vendor/Supplier/Contractor is required to pass on the benefit arising out of introduction of GST, including seamless flow of Input Tax Credit, reduction in Tax Rate on inputs as well as final goods by way of reduction of price as contemplated in the provision relating to Anti-Profiteering Measure as per CGST Act, 2017.
11. Vendor/Supplier/Contractor shall avail the most beneficial Notifications, abatements, exemptions etc., if any, as applicable for the supplies under the Goods and Service Tax Act.

TDS FOR INCOME TAX

Tax deduction at source (TDS) shall be made towards income tax from all the bills of the contractor at applicable rates as per Income Tax Act and Rules.

Annexure to Tender/RFQs

REVERSE AUCTION: ROLE OF SERVICE PROVIDER & BIDDER

Steel Authority of India Limited (All Plants/ units) have engaged the services of its Joint Venture Company M/s mjunction Services Limited (MJ) for conducting the Reverse Auctions of Goods & Services. The purpose of such Reverse Auction would be to meet SAIL's requirement for item(s)/service(s) from the Sellers / Bidders desiring to sell the item(s)/service(s) to SAIL.

The contact details of the mjunction shall be provided by them in the Auction Notice issued by them. mjunction is authorised to give clarifications regarding the reverse auction process to Bidders on behalf of SAIL.

2. Definition of Key Terms –

Reverse Auction

A reverse auction is an auction in which the traditional roles of buyer and seller are reversed. Thus, there is one buyer and many potential sellers and the sellers compete to sell items/ services to the buyer . The sellers are required bid downwards i.e., underbid each other to win the contract.

On-line Reverse Auctions

Reverse Auctions which are conducted through the Internet with simultaneous bidding by the bidders (from one or more locations). In other words, the venue for the auction is on an Internet website/ platform. The mjunction website <https://auction.buyjunction.in> or any other URL assigned by mjunction would constitute the venue for the purpose of the on-line auction.

Award at the Reverse Auction

The bidder quoting the lowest price is declared the L-1 bidder for the item/service to be procured.

Client

SAIL who has contracted with mjunction to conduct the Reverse Auction.

Bidder

Individual/business entity participating in the Reverse Auction, intending to supply the item(s)/service(s) to the Client which has to secure Client approval for participation and provide written acceptance to the General Rules and Regulations.

Auction Engine

The software that encapsulates the entire auction process, processing logic and information flows. mjunction is the sole owner of the auction engine and retains exclusive right over the utilization of the same.

Preview Time

Preview Time refers to the period of time that is provided prior to the commencement of bidding.

Start Time

Start time refers to the time of commencement of the conduct of the On-line auction.

Duration of the Reverse Auction

It refers to the length of time the price discovery process is allowed to continue by accepting bids from competing bidders. The duration of the auction would normally be for a pre-specified period of time. The same may however be subject to bidding rules.

Auto Extension of the Auction Timings

In the event of bids received in the last few minutes of the scheduled bid time, the Bid Timings are automatically extended for a specified period from each such bids placed. Such Auto Extension shall continue until no bids are placed for the specified period (i.e., the auction engine remains inactive for the specified period). The Inactivity Time for Auto Extension purpose is normally 5 minutes. mjunction however retains the right to change the same. The Inactivity time applicable for the particular On-line Bid shall be communicated to the bidder before the bidding process starts, if it is set for less or more than 5 minutes or more.

End of the Reverse Auction

Termination of the bidding event signaling an end to the price discovery process.

Auction Report

Report provided by mjunction containing a summary of the bidding event and outcome..

Start Bid Price

Price decided by the client from where the auction starts which would be put up on the site before commencement of Reverse Auction and also communicated in advance to all the bidders by mjunction. Bidder(s) have to quote a price lesser than the Start Bid Price while participating in the Reverse Auction.

Minimum Decrement

Minimum decrement is the minimum amount a Bidder has to reduce in order to place its bid. It shall be specified in each case prior to start of Reverse Auction.. This minimum decrement shall be pre-decided by Client and will be in-built in the auction engine. **(For cases where foreign bidders are bidding in Landed Cost Net of input Tax credit (LCNITC) & Indian Rupee (INR) (Currency Conversion will be rounded off to the nearest whole number)**

3.0 Process of On-line Reverse Auction :

- 3.1 For tenders issued by SAIL plants/ units , bidders submit their Techno Commercial (TC) bids and the price bids, on or before the due date of tender opening. The price bids shall be submitted by the Bidders online on the SRM platform / EPS Platform or on the buyjunction platform of mjunction as stated in the RFQ terms. In case price is submitted on buyjunction platform of mjunction, a system generated acknowledgement of submission of online sealed price bid submitted prior to TOD on the buyjunction platform of mjunction, which is to be submitted with the TC Bid. The process for submitting online price bid on the buyjunction platform of mjunction is placed at **Annexure A**.
- 3.2 In case SAIL Plants/Units decide to conduct Reverse Auction , SAIL Plants/Units shall inform mjunction about all bidders whose offers are found techno-commercially acceptable. These techno-commercially acceptable bidders shall become eligible for participating in price bidding through On-line Reverse Auction. SAIL Plants/ Units shall intimate the list of Techno-Commercially acceptable Bidders to mjunction.
- 3.3 All eligible bidders having ID & password can after connecting view the bidding on the computer screen and enter their prices any number of times during the duration of RA event.
- 3.4 The L-1 price matching will not be done in the RA event. If Price matching is required to be done, the same will be done by mjunction after finalisation of L-1 price by SAIL Plants / Units.

4.0 Role of Service Provider mjunction

- 4.1 Communication w.r.t Reverse Auction shall be done by mjunction with the authorized representative of Bidders as furnished by Bidders in the Techno commercial bid. The approved bidders will be provided the following information specific to any particular auction:
- i. Preview time & Start Time
 - ii. Duration of the Reverse Auction and the Auto extension of the pre specified auction timings
 - iii. Total number of Items/Markets/Lots etc. for price-discovery.
 - iv. Brief on RA strategy to be employed

- v. Start Bid Price (Market wise in case of multiple Markets), if required
- vi. Minimum bid decrement.
- vii. Loading Factor, if any (w.r.t payment term deviation/any other)
- viii. Special instructions, if any e.g. Exchange rate in case of participation of foreign bidders,
- ix. Issue “Auction Notice” to eligible bidders

4.2 Vendor Enablement- mjunction will:

- i. Familiarize the Techno Commercially acceptable bidders with the Reverse Auction Process (including training to the vendors).
- ii. Explain about the process of participating in Reverse Auction event and also provide training to all the eligible bidders regarding the same. Provide the Login ID and Password to all the Techno Commercially acceptable bidders to participate in the Reverse Auction..
- iii. Enable eligible bidders participation in the RA after obtaining bidders acceptance of “Auction Declaration”
- iv. Launch the screen for the Reverse Auction event at the Scheduled time .
- v. Do the price matching, if required

5.0 Roles and Responsibilities of the Bidder

The role of the bidder is outlined below:

- Get themselves acquainted to their satisfaction with the On-line bidding process by interacting with mjunction.
- Accept “Auction Declaration” to participate in the auction
- Participate in the Reverse Auction with the aim of posting their lowest best price for the auctioned item(s) in the reverse auction.
- Ensure that the User Id and password is not revealed to unauthorized persons.
- Submit their detailed price break-up for the item(s) / service(s) as per RFQ in the event the bidder becomes the Lowest bidder in the Reverse Auction process

6.0 Bidders need not pay any fee towards training or for actual Reverse Auction bidding or any other service with regard to this tender. Bidders need not subscribe to any of the services of mjunction in lieu of participation in the Reverse Auction.

7.0 Cancellation or Reschedule: Mjunction may cancel or reschedule the Reverse Auction in consultation with concerned SAIL Plants/ units, or do so on its own accord in case of any technical issue. The decision of mjunction either made on its own accord or after consultation with SAIL Plants/Units, shall be final.

Annexure A

The process for submitting online price bid on the buyjunction platform of mjunction is as under:

- a. Web-page to be accessed through the URL: <https://www.buyjunction.in>.
- b. There is a link in the above web-page under the section “Open Tender” through which vendors can access the required Request for Quotation(RFQ)/Tender against which sealed price bids are to be accepted.
- c. The above mentioned link would be kept active by mjunction till the due date and time of submission of RFQ/Tender or as per directives/ corrigendum issued by SAIL Plants/Units in case of Open Tender Enquiry(OTE)/Global Tender Enquiry(GTE) or as per instructions in RFQ/tender documents in Limited Tender Enquiry(LTE).
- d. Existing bidder who has earlier registered in mjunction e-platform shall have direct access to the platform. They would be able to do this by using their existing user IDs and passwords after clicking the above link.
- e. New bidder would have to click the link “New User” which would then throw open a pop-up screen showing details on how to (a) register online (b) receive user ID/password on their registered email with which they have registered online and (c) place their bids. With this, new bidders would also become enabled to log in and participate on mjunction e-platform.
- f. After logging in (with respective user IDs/ passwords), bidders shall be directed to a web-page where detailed road map (user manual) on how to place bids would be displayed.
- g. Each bidder can login and place/register a sealed price bid on mjunction e-platform within the stipulated period stated in point ‘c’ above.
- h. Each bidder shall be provided with a system generated acknowledgement of submission of online sealed price bid submitted prior to TOD on the buyjunction platform of mjunction which can be printed.
- i. In case any vendor submits online Price Bid but the Techno-Commercial bid has not been submitted on or before Tender Opening Date(TOD), such online Price Bids will not be considered by SAILPlant/Unit.